

M7 系列模组

用户指令及协议手册

修订记录

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V2.6	1. 增加 NMEATALKER、NMEAFORMAT、NMEAVERSION 指令描述	2025-09

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1. 说明

本章节描述串口、CAN 口、USB 口的相关指令说明，用于指导相关人员设计及验证用户界面的相关需求；本文档适用于 M7 系列模组产品。

1.1. 输入指令格式

所有指令都是以>符开始，其后仅跟指令名称，之后为不定数量的参数，指令名称与各参数之间均以,（逗号）进行分隔，校验位与参数间以*（星号）相隔，指令模式包含校验模式和免校验模式。

免检验模式便于手动便捷输入命令的场景，无需输入*CS。

配置免校验位指令格式：

```
>command,data1,data2,dataN
```

免校验模式指令示例：

```
>CONFIGSIGNAL,DISABLE,GPS,L1CA
```

XOR 校验模式更适应于机器交互场景，对数据完整性有严格的要求，需要输入*（星号）及完整的校验位。

配置带校验位指令格式：

```
>command,data1,data2,dataN*CS
```

校验模式指令示例：

```
>CONFIGSIGNAL,DISABLE,GPS,L1CA*57
```

1、校验位 2 字节,16 进制大写显示。



2、校验算法为 XOR 模式，校验数据从指令头到参数 N，包含中间的,（逗号）。

3、>与*之间的参数间隔符为英文逗号,。

4、参数不区分大小写，整条命令的最大长度为 2048 字节。

1.2. 返回信息格式

所有指令输入后，接收机均会返回执行结果和输入的原始指令信息，通过执行结果可判断配置是否生效。

返回信息格式：

```
>RESPONSE,Status,"Command"*CS
```

校验模式指令示例：

```
>CONFIGSIGNAL,DISABLE,GPS,L1CA*57
```

返回信息示例：

```
>RESPONSE,OK,">CONFIGSIGNAL,DISABLE,GPS,L1CA*57"*46
```


免校验模式指令示例：

```
>CONFIGSIGNAL,DISABLE,GPS,L1CA
```

返回信息示例：

```
>RESPONSE,OK,">CONFIGSIGNAL,DISABLE,GPS,L1CA"*6E
```

表1-1 指令返回信息参数说明

参数项	参数值	参数描述
Status	OK	表示配置成功
	CHECKSUM_ERROR	校验位错误
	INVALID_COMMAND	无效指令，包含指令头错误、不支持指令
	INVALID_PARAMETER	无效参数，包含不符合参数定义以及枚举类型的参数识别到纯数字等等
	PARAMETER_EXCEED_LIMIT	参数超限，包含数字类参数超出定义范围
	PARAMETER_NUMBERS_ERROR	指定的参数个数超限(少于或者多于命令定义的参数个数)或者必选参数个数错误
	UNKNOWN_ERROR	未知错误
Command		与输入的指令信息保持一致

- 1、INVALID_PARAMETER：不符合参数定义，例如参数模式定义为字符串 base，但传入数字类参数 1、0 属于此类错误；
- 

2、PARAMETER_NUMBERS_ERROR：参数个数超限，例如某指令定义参数个数为 2，实际输入参数个数为 3，亦或者定义 3 个必选参数，只输入 2 个参数；
- 3、PARAMETER_EXCEED_LIMIT：数字类参数超出定义范围，例如参数范围为 0-255 的整数，256、245.5 均属于此类；

2. 配置指令

2.1. CONFIGANTENNA TYPE 配置天线参数

适用产品：M720、M722

该指令用于设置与接收器一起使用的天线类型。

配置天线类型指令格式：

```
>CONFIGANTENNA TYPE, AntennaIndex, AntennaType[, RadomeType]*CS
```

指令输入示例：

```
>CONFIGANTENNA TYPE, MASTER, CHCC220GR*7F
```

表2-1 配置天线类型指令参数

参数项	参数值	参数描述
AntennaIndex	见表 AntennaIndex	
AntennaType	见表 AntennaType	
RadomeType	见表 RadomeType	设置为用户自定义天线类型时，此字段默认无效

2.2. CONFIGAUTOBASE 配置自动基站模式

适用产品：M720、M722

此命令可设置接收机自主优化定位坐标，可定制平滑时间、水平精度偏差、垂直精度偏差、基站移动距离偏差。当发送 CONFIGAUTOBASE 命令时，开始位置平滑，位置平滑持续进行，直到达到指定的水平精度、垂直精度、时间到期。位置平滑完成后，计算出的位置保存为基站的 Fix 坐标，然后在向流动站传输差分校正时使用此 Fix 的坐标，可配置是否将此坐标存储 NVM。

配置自动基站模式指令格式：

```
>CONFIGAUTOBASE, Switch[, MaxTime[, MAXHStd[, MAXVStd[, Distance[, SaveNvm]]]]*CS
```

指令输入示例：

```
>CONFIGAUTOBASE, ENABLE, 5*24
```

表2-2 平滑基站坐标参数说明

参数项	参数值	参数描述
Switch	DISABLE	禁用 AUTOBASE 功能,并且结束当前正在进行的 AUTOBASE 任务
	ENABLE	启用 AUTOBASE 功能
MaxTime	1~6000	坐标平滑的最长时间，单位：分钟，默认为 1440 分钟
MAXHStd	0~100	水平标准差，单位：米，默认为 0.1 米
MAXVStd	0~100	垂直标准差，单位：米，默认为 0.1 米

Distance	0~100	比较平滑的位置和保存的 fix 位置之间的最大距离，在 AUTOBASE 过程中，如果平滑出的位置和上次保存的位置之间的距离小于该值，则使用之前保存的位置坐标，单位：米，默认为 0.1 米
SaveNvm	OFF	不将平滑出的坐标保存到 NVM 中
	ON	将平滑出的坐标保存到 NVM 中

2.3. CONFIGBASEANTENNA TYPE 配置基站天线参数

适用产品：M720、M722

该指令用于设置基准站使用的天线类型。

配置基站天线参数指令格式：

```
>CONFIGBASEANTENNA TYPE,AntennaType[,RadomeType]*CS
```

指令输入示例：

```
>CONFIGBASEANTENNA TYPE,CHCC220GR*5A
```

表2-3 配置基站天线参数指令参数说明

参数项	参数值	参数描述
AntennaType	见表 AntennaType	
RadomeType	见表 RadomeType	设置为用户自定义天线类型时，此字段默认无效

2.4. CONFIGCOM 配置串口连接参数

适用产品：M720、M722

本指令对模组的串口以及虚拟串口进行波特率等连接参数配置，所有串口功能相同，但各串口数据输入输出以各自配置进行独立工作，另外，所有串口也可以相互配置。

配置串口连接参数指令格式：

```
>CONFIGCOM[,Port],Baudrate[,Parity[,DataBits[,StopBits]]]*CS
```

指令输入示例：

```
>CONFIGCOM,COM2,115200,NONE,8,1*10
```

表2-4 配置串口连接参数指令说明

参数项	参数值	参数值描述
Port	见端口列表	
Baudrate	9600、19200、115200、460800、921600、1500000、3000000	默认为 115200
Parity	NONE、ODD、EVEN	默认为 NONE
DataBits	5、6、7、8	默认为 8
StopBits	1、2	默认为 1

2.5. CONFIGEVENTMARK 配置 EVENTMARK 功能

适用产品：M720、M722

该指令配置 EventMark 功能及相关参数。EventMark 功能默认为关闭状态。

配置 EVENTMARK 功能指令格式：

```
>CONFIGEVENTMARK,Mark,Switch[,TGuard]*CS
```

指令输入示例：

```
>CONFIGEVENTMARK,MARK1,POSITIVE*7C
```

表2-5 配置 EVENTMARK 功能指令参数说明

参数项	参数值	参数描述
MarkIndex	MARK1	对第几个 EventMark 进行配置
	MARK2	
Switch	DISABLE	关闭 EventMark 功能
	POSITIVE	开启 EventMark 功能，上升沿有效
	NEGATIVE	开启 EventMark 功能，下降沿有效
TGuard	2~3,599,999	两个有效脉冲之间的最短时间要求，单位 ms。若小于 TGuard，则第二个 Event 被忽视。默认值:10。

2.6. CONFIGHEADINGBASELINE 配置双天线基线长度

适用产品：M722

配置双天线的基线长度及标准偏差。

配置双天线基线长度指令格式：

```
>CONFIGHEADINGBASELINE,Length[,LengthStd]*CS
```

指令输入示例：

```
>CONFIGHEADINGBASELINE,1.05*6F
```

表2-6 配置双天线基线长度指令参数说明

参数项	参数值	参数描述
Length	≥ 0	单位 m，保留三位有效位
LengthStd	> 0	单位 m，保留三位有效位

2.7. CONFIGHEADINGOFFSET 配置双天线航向及俯仰角偏差

适用产品：M722

配置双天线航向及俯仰角，双天线航向是指双天线接收机的主天线（ANT1）与从天线（ANT2）之间构成一个基线向量，确定此基线向量逆时针方向与真北的夹角。

配置双天线航向及俯仰角偏差指令格式：

```
>CONFIGHEADINGOFFSET,HeadingOffset,PitchOffset*CS
```

指令输入示例：

```
>CONFIGHEADINGOFFSET,90,45*47
```

表2-7 配置双天线航向及俯仰角偏差参数说明

参数项	参数值	参数描述
HeadingOffset	-180.0~180.0	添加到航向角输出的偏移量；单位：度
PitchOffset	-90.0~90.0	添加到俯仰角输出的偏移量，单位：度

2.8. CONFIGIMUROTATION 配置 IMU 旋转角

适用产品：M720、M722

配置 IMU 坐标系与其他坐标系（如车体系或基线）的旋转偏移。偏移的输入须要是从 IMU 坐标系转向其他坐标系的。旋转顺序为 Z,X,Y。全部旋转均遵从右手系。

配置 IMU 旋转角指令格式：

```
>CONFIGIMUROTATION,INSRotationType,XRotation,YRotation,ZRotation[,XRotationStd[,YRotationStd[,ZRotationStd]]]*CS
```

指令输入示例：

```
>CONFIGIMUROTATION,RBV,180,0,0,5.0,5.0,5.0*37
```

表2-8 配置 IMU 旋转角参数说明

参数项	参数值	参数描述
INSRotationType	见表旋转偏移类型	要设置的旋转偏移方式
XRotation	-180.0~180.0	IMU 坐标系相对于载体坐标系 X 轴旋转角度，单位：度
YRotation	-180.0~180.0	IMU 坐标系相对于载体坐标系 Y 轴旋转角度，单位：度
ZRotation	-180.0~180.0	IMU 坐标系相对于载体坐标系 Z 轴旋转角度，单位：度
XRotationStd	0.25~45	X 轴旋转角度标准偏差，单位：度
YRotationStd	0.25~45	Y 轴旋转角度标准偏差，单位：度
ZRotationStd	0.25~45	Z 轴旋转角度标准偏差，单位：度

表2-9 旋转偏移类型

参数项	参数名	参数描述
INSRotationType	USER	从 IMU 系旋转到用户定义的输出坐标系。
	MARK1	从 IMU 系旋转到 MARK1 期望输出的坐标系。
	MARK2	从 IMU 系旋转到 MARK2 期望输出的坐标系。
	ALIGN	从 IMU 系转到一个对准的双天线结果。
	RBV	从 IMU 系旋转到车辆坐标系

2.9. CONFIGJAMDETECT 配置干扰检测

适用产品：M720、M722

该指令用于对模组抗干扰功能开关及对射频通道选择配置。

配置干扰检测指令格式：

```
>CONFIGJAMDETECT,RFBand*CS
```

指令输入示例：

```
>CONFIGJAMDETECT,NONE*6D
```

表2-10 配置干扰检测指令参数说明

参数项	参数值	参数描述
RFBand	NONE	关闭干扰检测
	ALL	所有射频通道
	L1	只检测 L1 频段
	L2	只检测 L2 频段 *
	L5	只检测 L5 频段

2.10. CONFIGJAMPROGFILT 配置窄带/单音陷波滤波器

适用产品：M720、M722

该指令用于对射频滤波器进行配置。

配置窄带/单音陷波滤波器指令格式：

```
>CONFIGJAMPROGFILT,RFBand,FilterID,Switch[,NotchFreq[,NotchBW]]*CS
```

指令输入示例：

```
>CONFIGJAMPROGFILT,L1,NBIFILT,ENABLE,1575.42,0.25*64
```

表2-11 配置窄带/单音陷波滤波器指令参数说明

参数项	参数值	参数描述
RFBand	L1	L1
	L2	L2
	L5	L5
FilterType	CWIFILT	单音陷波器
	NBIFILT	窄带陷波器
Switch	DISABLE	关闭该类型滤波器

*L2 频段已覆盖 GalileoE6、QZSSL6 和 BDSB3

	ENABLE	打开该类型滤波器
NotchFreq	1170.0~1610.0	干扰所在的频率，单位为 MHz
NotchBW	0.25/0.5/1.0/2.0	干扰带宽，单位为 MHz

2.11. CONFIGJAMSPECTRALANALYSIS 配置频谱分析

适用产品：M720、M722

该指令为控制频谱分析，可通过报文输出监测到的频谱信息。

配置频谱分析指令格式：

```
>CONFIGJAMSPECTRALANALYSIS,Mode[,Frequency[,Period[,FFTsize]]]*CS
```

指令输入示例：

```
>CONFIGJAMSPECTRALANALYSIS,FILTERBEFORE,L1,1000,1k*6F
```

表2-12 配置频谱分析指令参数说明

参数项	参数值	参数描述
Mode	OFF	关闭频谱分析
	FILTERBEFORE	抗干扰前的数据频谱
	FILTERAFTER	经过抗干扰后的数据频谱
Frequency	L1	L1
	L2	L2
	L5	L5
Period	1000/2000/5000/10000	频谱刷新率单位 ms(默认 1000)
FFTsize	1k,2k,4k,8k,16k,32k,64k	默认为 1k

2.12. CONFIGJAMWARNING 配置控制警告等级

适用产品：M720、M722

该指令用于配置抗干扰模式下的告警等级。

配置控制警告等级指令格式：

```
>CONFIGJAMWARNING,WarningLevel*CS
```

指令输入示例：

```
>CONFIGJAMWARNING,3*19
```

表2-13 配置控制警告等级指令参数说明

参数项	参数值	参数描述
WarningLevel	0	0 最敏感的 3 最不敏感的
	1	
	2	

2.13. CONFIGLBANDBEAM 配置 Lband 跟踪

适用产品：M720、M722

使用该指令设置 L-Band 通道需要捕获跟踪的卫星参数，包括频率，波特率，捕获范围以及电文解码模式等参数。

当用 AUTO 选项时，会自动跟踪预定义的波束列表中的所有卫星；

配置 Lband 跟踪指令格式：

```
>CONFIGLBANDBEAM,Option[,Name]*CS
```

指令输入示例：

```
>CONFIGLBANDBEAM,MANUAL,SIO*FF
```

表2-14 配置 Lband 跟踪指令参数说明

参数项	参数值	参数描述
Option	IDLE	不跟踪 LBand 信号
	AUTO	跟踪 LBandBeamTable 里面所有信号
	MANUAL	只跟踪指定 LBand 信号
Name	见表波束名称说明	波束名称

表2-15 波束名称说明

参数项	参数值	参数描述
Name	SIO	位置:东经 83.5°，覆盖区域：印度洋区域

2.14. CONFIGPPS 配置秒脉冲

适用产品：M720、M722

设置接收机输出特定周期、脉宽的 PPS 脉冲信号，并可对 PPS 延迟进行补偿。

配置秒脉冲指令格式：

```
>CONFIGPPS,PPSIndex,Switch[,Polarity[,Width[,Period[,Delay]]]]*CS
```

指令输入示例：

```
>CONFIGPPS,PPS1,ENABLE,POSITIVE,10,1000,40*35
```

表2-16 配置秒脉冲指令参数说明

参数项	参数值	参数描述
PPSIndex	PPS1	
	PPS2	
Switch	DISABLE	关闭(默认)

	ENABLE	打开
Polarity	POSITIVE	上升沿有效
	NEGATIVE	下降沿有效
Width	1~999999	单位微秒，应小于脉冲周期的一半
Period	1,2,5,10,20,50,100,200,250,500,1000,2000,3000,...,20000	单位毫秒，脉冲周期
Delay	-10000~10000	单位纳秒，脉冲输出延迟

2.15. CONFIGSIGNAL 配置卫星系统/频点跟踪

适用产品：M720、M722

对卫星系统、频点是否启用进行配置，可直接对某个系统进行设置，也可用于开关特定的频点。

配置卫星系统/频点跟踪指令格式：

```
>CONFIGSIGNAL,Switch,System[,Signal]*CS
```

指令输入示例：

```
>CONFIGSIGNAL,DISABLE,GPS,L1CA*57
```

表2-17 配置卫星系统/频点跟踪指令参数说明

参数项	参数值	参数描述
Switch	ENABLE	启动
	DISABLE	关闭(不捕获跟踪)
System	见表卫星系统及频点	
Signal	说明	

表2-18 卫星系统及频点说明

卫星系统	频点	备注
GPS	L1CA、L1C、L2C、L2P、L5	
BDS	B1I、B2I、B3I、B1C、B2a、B2b	
GLO	L1、L2、L3、L10C、L20C	
GAL	E1、E5a、E5b、E5、E6	
SBAS	L1CA、L5	
QZSS	L1CA、L1C、L2C、L5、L6	
NAVIC	L5	
ALL		代表所有系统所有频点

2.16. FRESET 恢复出厂设置

适用产品：M720、M722

该指令会将所有的配置信息清除，恢复成出厂设置，使用该指令会清除所有存储在非易失性存储器中的位置信息、卫星星历及用户设置等数据，并强制重启接收机。

恢复出厂设置指令格式：

```
>FRESET*CS
```

指令输入示例：

```
>FRESET*13
```

2.17. HELP 命令用法查询

适用产品：M720、M722

该指令用于显示命令的用法。

命令用法查询指令格式：

```
>HELP,Command*CS
```

指令输入示例：

```
>HELP,CONFIGSIGNAL*29
```

2.18. NMEAFORMAT 自定义 NMEA 输出

适用产品：M720、M722

使用该指令自定义 NMEA 报文中定位结果精度。

命令用法查询指令格式：

```
>NMEAFORMAT,Filed,Precision*CS
```

指令输入示例：

```
>NMEAFORMAT,LATITUDE,4*FF
```

```
>NMEAFORMAT,LONGITUDE,4*FF
```

```
>NMEAFORMAT,ALTITUDE,2*FF
```

表2-19 自定义 NMEA 输出指令参数说明

参数项	参数值	参数描述
Filed	LATITUDE	纬度
	LONGITUDE	经度
	ALTITUDE	海拔高度，参考 MSL（大地水准面）
Message	1~10	坐标需要保留的小数位

2.19. NMEATALKER NMEA talker ID 设置

适用产品：M720、M722

使用该指令自定义 NMEA 报文中定位结果精度。当 NMEATALKER 设置为 AUTO 时，使用单系统进行解算时，GPS 的 Talker 为 GP，GLONASS 的 Talker 为 GL，GALILEO 的 Talker 为 GA，BDS 的 Talker 为 BD(GB)，QZSS 的 Talker 为 GQ，NavIC 的 Talker 为 GI；使用多系统进行解算时，Talker 为 GN（GSV 除外）。

当 NMEATALKER 设置为 GP 时，此时 Talker 不受系统影响，按设定值输出（GSA, GSV 除外）
命令用法查询指令格式：

```
>NMEATALKER,TalkerID*CS
```

指令输入示例：

```
>NMEATALKER,GP*FF
```

```
>NMEATALKER,GB*FF
```

```
>NMEATALKER,BD*FF
```

```
>NMEATALKER,AUTO*FF
```

表2-20 NMEA talker ID 设置指令参数说明

参数项	参数值	参数描述
Talker ID	GP	设置 NMEA 报文 Talker 为 GP
	GB	设置 NMEA 报文中 BDS 相关信息 Talker 为 GB
	BD	设置 NMEA 报文中 BDS 相关信息 Talker 为 BD
	AUTO	设置 NMEA 报文 Talker 根据解决方案定义

2.20. NMEAVERSION NMEA 版本设置

适用产品：M720、M722

命令用法查询指令格式：

```
>NMEAVERSION,Version*CS
```

指令输入示例：

```
>NMEAVERSION,V301*FF
```

```
>NMEAVERSION,V410*FF
```

表2-21 NMEA 版本设置指令参数说明

参数项	参数值	参数描述
Version	V301	设置输出 NMEA 3.01 版本的报文
	V410	设置输出 NMEA 4.10 版本的报文

2.21. OFFMSG 关闭协议输出

适用产品：M720、M722

该指令会将某端口的协议输出全部关闭，如果想只关闭某条协议输出，请在配置协议输出的语句里进行设置。

关闭协议输出指令格式：

```
>OFFMSG[,Port[,Message]]*CS
```

指令输入示例：

```
>OFFMSG,COM1*4A
```

表2-22 关闭协议输出指令参数说明

参数项	参数值	参数描述
Port	见端口列表	可省略，省略时默认为当前端口
Message	例如：GGA	支持 RawMsg、NMEA、RTCM 等报文

表2-23 端口列表

参数项	参数值	参数描述
Port	COM1	COM1-COM3 为 UART 口
	COM2	
	COM3	
	USB1	USB1-USB3 为 USB 虚拟串口
	USB2	
	USB3	

2.22. OUTCONFIG 生效配置查询

适用产品：M720、M722

打印端口配置已生效的指令历史，包含保存的上电指令。

生效配置查询指令格式：

```
>OUTCONFIG[,Type]*CS
```

指令输入示例：

```
>OUTCONFIG*44
```

表2-24 生效配置查询指令参数说明

参数项	参数值	参数描述
Type	缺省	输出配置列表和报文列表
	MSG	输出对应报文配置列表

cmd_name

输出该命令的当前配置

2.23. OUTMSG 配置协议输出

适用产品：M720、M722

本指令对模组的串口以及虚拟串口输出协议进行配置。

配置协议输出指令格式：

```
>OUTMSG[,Port],Message,Period*CS
```

输入指令示例：

```
>OUTMSG,COM1,GGA,1*3B
```

表2-25 配置协议输出指令参数说明

参数项	参数值	参数描述
Port	见端口列表	可省略，省略时默认为当前端口
Message	具体协议名称	
Period	0.05、0.1、0.2、0.5、1、2、5、10、15、20、30、60、120、ONNEW、ONCE、ONMARK	输出信息的频度

2.24. POSECUTOFF 配置解算的卫星高度截止角

适用产品：M720、M722

设置参与解算的卫星高度截止角。高度截止角用于筛选跟踪卫星的观测量，以及对外输出的协议，例如 RANGE、RTCM 等，但不影响内部捕获跟踪的相关策略。

配置解算的卫星高度截止角指令格式：

```
>POSECUTOFF,Angle*CS
```

指令输入示例：

```
>POSECUTOFF,10*29
```

表2-26 解算高度截止角参数说明

参数项	参数值	参数描述
Angle	0~90	整数，单位：度

2.25. RESET 设备重启

适用产品：M720、M722

可用于接收机重启方式的设置，支持热重启、温重启、冷重启。

设备重启指令格式：

```
>RESET[,Mode]*CS
```

指令输入示例：

```
>RESET,HOT*2A
```

表2-27 模块重启指令参数说明

参数项	参数值	参数描述
Mode	HOT	热启动（默认）
	WARM	温启动
	COLD	冷启动

2.26. SAVECONFIG 指令保存

适用产品：M720、M722

本指令将用户输入的指令进行保存，下次上电后配置自动生效。

指令保存指令格式：

```
>SAVECONFIG*CS
```

指令输入示例：

```
>SAVECONFIG*0B
```

2.27. SETALIGNMENTVEL 配置 INS 动态对准的最小速度阈值

适用产品：M720、M722

用速度实现初始对准的时候，最小速度的阈值。

配置 INS 动态对准的最小速度阈值指令格式：

```
>SETALIGNMENTVEL,Velocity*CS
```

指令输入示例：

```
>SETALIGNMENTVEL,5.0*45
```

表2-28 配置 INS 动态对准的最小速度阈值指令参数说明

参数项	参数值	参数描述
Velocity	Minimum:0.2m/s	对准所需的最小速度，默认为 0.2m/s，单位为 m/s

2.28. SETBASEID 配置基站 ID

适用产品：M720、M722

该指令用于配置基准站 ID，需注意不同的差分协议基站 ID 范围的区别。

配置基站 ID 指令格式：

```
>SETBASEID,BaseID*CS
```

指令输入示例：

```
>SETBASEID,2022*74
```

表2-29 配置基站 ID 指令参数说明

参数项	参数值	参数描述
BaseID	0~4095	整数，默认为 0

2.29. SETCMD 配置指令模式

适用产品：M720、M722

指令的校验模式设置，校验算法为 XOR 模式，校验数据从指令头到参数 N，包含中间的间隔符，>与*之间的参数间隔符为英文逗号,；当配置免校验位模式时，设置指令可省略*CS 输入。

配置指令模式指令格式：

```
>SETCMD,VerifyMode*CS
```

指令输入示例：

```
>SETCMD,XOR*61
```

表2-30 配置指令模式参数说明

参数项	参数值	参数描述
ChecksumMode	OFF	输入指令为免校验模式，校验位默认为 FF 或无校验位
	XOR	输入指令为 XOR 校验模式

2.30. SETFIXPOS 配置固定坐标

适用产品：M720、M722

该指令配置成功后，模组就被设备为基准站模式。

配置固定坐标指令格式：

```
>SETFIXPOS,Switch,Latitude,Longitude,Height*CS
```

指令输入示例：

```
>SETFIXPOS,ENABLE,31.12345678901,121.12345678901,30.123456*42
```

表2-31 配置固定坐标指令参数说明

参数项	参数值	参数描述
Switch	DISABLE	删除固定坐标
	ENABLE	设置固定坐标
Latitude	-90~90	为大地坐标系下的纬度坐标，单位：度。（输入有效位 11 位）
Longitude	-180~180	大地坐标系下的经度坐标，单位：度。（输入有效位 11 位）

Height	-10000~10000	大地高，单位：米。（输入有效位 6 位）
--------	--------------	----------------------

2.31. SETIMUAZ 配置 IMU 初始方位角

适用产品：M720、M722

配置 IMU 初始方位角。

配置 IMU 初始方位角指令格式：

>SETIMUAZ,Azimuth,AzimuthStd*CS

指令输入示例：

>SETIMUAZ,180,5*04

表2-32 配置 IMU 方位角指令参数说明

参数项	参数值	参数描述
Azimuth	0~360	输入方位角,单位：度
AzimuthStd	1~25	输入方位标准偏差角,单位：度

2.32. SETIMUOFFSET 配置 IMU 的杆臂补偿

适用产品：M720、M722

配置 IMU 的杆臂补偿参数，常用于 INS 卫惯融合场景。

配置 IMU 的杆臂补偿指令格式：

>SETIMUOFFSET,INSTTranslationType,x,y,z,x_e,y_e,z_e*CS

指令输入示例：

>SETIMUOFFSET,ANT1,1.23,-2.41,0.7,0.01,0.02,0.03*45

表2-33 配置 IMU 的杆臂补偿指令参数说明

参数项	参数值	参数描述
INSTTranslationType	见表杆臂补偿类型	配置的杆臂类型
x	-100~100	以惯导设备中心为原点，定位天线的位置坐标。从左向右分别为 x、y、z。x _e 、y _e 、z _e 为配置误差。从左向右分别为 x _e 、y _e 、z _e 。单位 m，小数有效位 2 位。
y	-100~100	
z	-100~100	
x _e	-100~100	
y _e	-100~100	
z _e	-100~100	

表2-34 杆臂补偿类型

参数项	参数值	参数描述
INSTTranslationType	ANT1	从 IMU 中心到 GNSS 主天线相位中心的杆臂
	ANT2	从 IMU 中心到 GNSS 辅天线相位中心的杆臂

	EXTERNAL	从 IMU 中心到外部定位源位置的杆臂
	USER	从 IMU 中心到用户定义位置的杆臂
	MARK1	从 IMU 中心到 MARK1 输出位置的杆臂
	MARK2	从 IMU 中心到 MARK2 输出位置的杆臂

2.33. SETIMUPROFILE 配置 IMU 载体

适用产品：M720、M722

此命令根据系统安装的载体环境，用于配置 IMU 运动载体。

配置 IMU 载体指令格式：

```
>SETIMUPROFILE,Profile*CS
```

指令输入示例：

```
>SETIMUPROFILE, LAND*73
```

表2-35 IMU 运动载体参数说明

参数项	参数值	参数描述
Profile	Default	默认配置
	LAND	陆地车辆配置，支持动态对准和陆地车辆约束策略
	MARINE	船用配置，禁用静态约束策略
	FIXEDWING	固定翼飞行器配置
	FOOT	步行/背包应用配置，支持低速对准策略
	VTOL	垂直起降载具（无人机、直升机等）配置
	RAIL	铁路载具配置，支持铁路载具约束策略
	AGRICULTURE	农业应用配置，支持低速、高震动场景的应对策略

2.34. SETIONOLEVEL 配置电离层活跃状态

适用产品：M720、M722

使用该指令设置电离层活跃状态。

配置电离层活跃状态指令格式：

```
>SETIONOLEVEL,IonoLevel*CS
```

指令输入示例：

```
>SETIONOLEVEL, 3*0C
```

表2-36 配置电离层活跃状态参数指令说明

参数项	参数值	参数描述
IonoLevel	0	自动识别电离层活跃状态
	1	电离层平静期

	2	电离层常规期
	3	电离层扰动期

2.35. SETPOLEHEIGHT 配置杆高

适用产品：M720、M722

配置天线底部到测量点的天线高度，常用倾斜测量等场景。

配置杆高指令格式：

```
>SETPOLEHEIGHT,Height*CS
```

指令输入示例：

```
>SETPOLEHEIGHT,1.76*72
```

表2-37 配置杆高指令参数说明

参数项	参数值	参数描述
Height	任意数值	天线底部到杆尖点的高度，单位米，小数点有效位数 2 位，默认为 0

2.36. SETPPPCONVERGEDTHRESHOLD 配置 PPP 收敛精度门限

适用产品：M720、M722

使用该指令设置 PPP 的收敛精度。

配置 PPP 收敛精度门限指令格式：

```
>SETPPPCONVERGEDTHRESHOLD,Criteria,Tolerance*CS
```

指令输入示例：

```
>SETPPPCONVERGEDTHRESHOLD,TOTAL_STDDEV,0.5*22
```

表2-38 配置 PPP 收敛精度门限指令参数说明

参数项	参数值	参数描述
Criteria	TOTAL_STDDEV	以三维精度为参考
	HORIZONTAL_STDDEV	以水平精度为参考
Tolerance	任意数	单位：m

2.37. SETPPPSOURCE 配置 PPP 校正数

适用产品：M720、M722

该命令用于指示接收机如何跟踪和使用来自卫星增强系统的 PPP 校正数据。

配置 PPP 校正数指令格式：

```
>SETPPPSOURCE,Source*CS
```

指令输入示例：

>SETPPPSOURCE,B2b*31

表2-39 配置 PPP 校正数指令参数说明

参数项	参数值	参数描述
Source	见表 SourceTypes 说明	选择接收机使用的 PPP 校正数据

表2-40 SourceTypes 说明

参数项	参数值	参数描述
Source	NONE	不使用任何 PPP 校正数
	AUTO	自动选择并使用最佳的一种 PPP 校正数
	FUSION	融合所有的 PPP 校正数，可以增加可用卫星数量，提高定位精度和可靠性
	B2b	只使用 B2b 的 PPP 校正数
	HAS	只使用 HAS 的 PPP 校正数
	CLAS	只使用 QZSSCLAS 的校正数
	MADOCA	只使用 QZSSMADOCA 的 PPP 校正数
	SWAS	只使用华测 SWAS 的 PPP 校正数

2.38. SETPPPTIMEOUT 配置 PPP 最大龄期

适用产品：M720、M722

使用该指令设置 PPP 模式下定位解算最大改正数龄期（秒），如果改正数延迟大于此设置，则无法获取 PPP 定位结果。

配置 PPP 最大龄期指令格式：

>SETPPPTIMEOUT,TimeOut*CS

指令输入示例：

>SETPPPTIMEOUT,30*66

表2-41 PPP 最大龄期配置参数说明

参数项	参数值	参数描述
TimeOut	0~300	默认值为 300，单位：秒，

2.39. SETPSRSMOOTH 配置伪距平滑

适用产品：M720、M722

使用该指令设置伪距平滑参数。

配置伪距平滑指令格式：

>SETPSRSMOOTH,Control*CS

指令输入示例：

```
>SETPSRSMOOTH, ENABLE*3C
```

表2-42 配置伪距平滑指令参数说明

参数项	参数值	参数描述
Control	DISABLE	关闭伪距平滑
	ENABLE	开启伪距平滑
	RESET	重置伪距平滑

2.40. SETRTDTIMEOUT 配置 RTD 最大龄期

适用产品：M720、M722

使用该指令设置 RTD 模式下定位解算最大差分龄期（秒）。

配置 RTD 最大龄期指令格式：

```
>SETRTDTIMEOUT, TimeOut*CS
```

指令输入示例：

```
>SETRTDTIMEOUT, 30*74
```

表2-43 配置 RTD 最大龄期指令参数说明

参数项	参数值	参数描述
TimeOut	0~1800	默认值为 30，单位：秒

2.41. SETRTKRELIABILITY 配置 RTK 安全模式

适用产品：M720、M722

使用该指令设置 RTK 安全模式，在信号被遮挡的区域可使用 EXTRA_SAFE 模式，与 NORMAL 模式相比，此模式下会提供更可靠的结果，但固定速度会有所下降。

配置 RTK 安全模式指令格式：

```
>SETRTKRELIABILITY, Mode*CS
```

指令输入示例：

```
>SETRTKRELIABILITY, NORMAL*60
```

表2-44 配置 RTK 安全模式指令参数说明

参数项	参数值	参数描述
Mode	NORMAL	设置 RTK 为常规模式
	EXTRA_SAFE	设置 RTK 为高可靠模式

2.42. SETRTKTIMEOUT 配置 RTK 最大龄期

适用产品：M720、M722

使用该指令设置 RTK 模式下定位解算最大差分龄期（秒）。

配置 RTK 最大龄期指令格式：

```
>SETRTKTIMEOUT,TimeOut*CS
```

指令输入示例：

```
>SETRTKTIMEOUT,30*7B
```

表2-45 配置 RTK 最大龄期指令参数说明

参数项	参数值	参数描述
TimeOut	0~1800	默认值为 30，单位：秒

2.43. SETSBASSOURCE 配置 SBAS 校正数

适用产品：M720、M722

该命令用于指示接收机如何跟踪和使用来自卫星增强系统的校正数据。

配置 SBAS 校正数指令格式：

```
>SETSBASSOURCE,Source*CS
```

指令输入示例：

```
>SETSBASSOURCE,WAAS*74
```

表2-46 配置 SBAS 校正数指令参数说明

参数项	参数值	参数描述
Source	见表 SystemTypes 说明	选择接收机使用的 SBAS 校正数据

表2-47 SystemTypes 说明

参数项	参数值	参数描述
Source	NONE	不使用任何 SBAS 卫星
	AUTO	自动确定要使用的卫星系统，并防止接收器使用服务区域以外的卫星
	WAAS	只使用 WAAS 卫星
	EGNOS	只使用 EGNOS 卫星
	MSAS	只使用 MSAS 卫星
	GAGAN	只使用 GAGAN 卫星
	SDCM	只使用 SDCM 卫星
	BDSBAS	只使用 BDSBAS 卫星
	KASS	只使用 KASS 卫星
	ASBAS	只使用 ASBAS 卫星
	SPAN	只使用 SPAN 卫星
	QZSS	只使用 QZSSL1S 信号

2.44. SETSBASTIMEOUT 配置 SBAS 最大龄期

适用产品：M720、M722

使用该指令设置 SBAS 模式下定位解算最大差分龄期（秒）。

配置 SBAS 最大龄期指令格式：

```
>SETSBASTIMEOUT,TimeOut*CS
```

指令输入示例：

```
>SETSBASTIMEOUT,30*35
```

表2-48 配置 SBAS 最大龄期指令参数说明

参数项	参数值	参数描述
TimeOut	0~300	默认值为 180，单位：秒

2.45. TRACKECUTOFF 配置跟踪高度截止角

适用产品：M720、M722

设置跟踪的卫星高度截止角。高度截止角用于筛选跟踪卫星的观测量，以及对外输出的协议，例如 RANGE、RTCM 等，但不影响内部捕获跟踪的相关策略。

配置跟踪高度截止角指令格式：

```
>TRACKECUTOFF,System[,Angle]*CS
```

指令输入示例：

```
>TRACKECUTOFF,ALL,10*47
```

表2-49 配置跟踪高度截止角指令参数说明

参数项	参数值	参数描述
System	见表卫星系统及频点说明	
Angle	0~90	整数，单位：度

2.46. UNDULATION 配置高程异常改正值

适用产品：M720、M722

该指令用于高程异常改正值设置，输入特定的大地水准面差距或使用内置大地水准面差距格网值。

配置高程异常改正值指令格式：

```
>UNDULATION,Mode[,Source]*CS
```

指令输入示例：

```
>UNDULATION,EGM96*77
```

表2-50 配置高程异常改正值指令参数说明

参数项	参数值	参数描述
Mode	EGM96	使用全球大地水准面高度模型 EGM96 表格
	USER	用户自定义修正值
Source	±1000.0m	模式为 USER 选项时才生效（默认值=0.000）

3. 报文输出

华测产品支持 NMEA3.01 版本（默认）和 NMEA4.10 版本的报文输出。以下参考 NMEA0183Version4.10 版本，对常用语句进行介绍。

3.1. NMEA 报文输出

当使用华测产品请求 NMEA 消息输出时，只需在参数中输入消息名称即可。如下提供命令输入示例及消息输出说明：

COM1 输出 GGA 报文示例：

```
>OUTMSG,COM1,GGA,1*3B
```

3.1.1. GGA

适用产品：M720、M722

本语句用于输出 GNSS 接收机的位置和定位相关数据。

格式：

```
$--GGA,hhmmss.ss,llll.ll,a,yyyy.yy,a,x,xx,x.x,x.x,M,x.x,M,x.x,xxxx*h  
h<CR><LF>
```

输出示例：

```
$GPGGA,094048.00,3109.9557,N,12117.2813,E,1,38,0.4,31.42,M,10.50,M,,*  
5B
```

表3-1 GGA 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-GGA	log 头		\$GPGGA
2	utc	位置对应的 UTC 时间，hh/mm/s.ss	hhmmss.ss	094048.00
3	lat	纬度,DDmm.mm	llll.ll	3109.9557
4	latdir	纬度方向(N=北, S=南)	a	N
5	lon	经度,DDDmm.mm	yyyy.yy	12117.2813
6	lon dir	经度方向(E=东, W=西)	a	E
7	quality	GPS 质量指示符，不得为空，见表 3-2	x	1
8	#sats	使用中的卫星数。可能与所见数不一致	xx	38
9	hdop	水平精度因子	x.x	0.4
10	alt	海拔高	x.x	31.42
11	a-units	大地高单位(M=m)	M	M
12	undulation	大地水准面差距	x.x	10.50
13	u-units	大地水准面差距单位(M=m)	M	M

14	age	差分卫星导航系统数据龄期，单位为秒	x.x	（没有差分数据时为空）
15	stnID	差分基站 ID,0000-1023	xxxx	（没有差分数据时为空）
16	*xx	校验和	*hh	*
17	<CR><LF>	语句结束符		<CR><LF>

表3-2 GPS 质量指示符

标识符	说明
0	定位不可用或无效
1	GPSSPS(StandardPointingService)模式,SPP,定位有效
2	差分 GPSSPS 模式，定位有效(包含 DGNSS 和 SBAS 以及收敛至分米级的 PPP 状态)
4	RTK 固定解
5	RTK 浮点解，系统处于 RTK 模式中，整周模糊度是浮动的
6	INS 或是无 INS 情况下的纯航位推算
7	固定模式（基站固定位置模式）
8	模拟器模式

3.1.2. GSA

适用产品：M720、M722

本语句用于输出 GNSS 接收机工作模式,GGA 或 GNS 语句报告的导航解算中用到的卫星以及精度因子(DOP)的值。

格式：

\$--GSA,a,x,xx,xx,xx,xx,xx,xx,xx,xx,xx,xx,xx,x.x,x.x,x.x,h*hh<CR><LF>

输出示例：

\$GPGSA,M,3,32,24,05,18,23,12,15,10,13,,0.8,0.4,0.6,1*27

表3-3 GSA 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-GSA	log 头		\$GPGSA
2	ModeMA	模式： M—手动，强制用于 2D 或 3D 模式 A—自动，允许 2D/3D 自动变换	a	M
3	Mode123	模式： 1—定位不可用 2=2D 3=3D	x	3

4	prn	解算中用到的卫星号	xx,xx,.....	32,24,05,18,23,12,15,10,13,,,
5	pdop	位置精度因子	x.x	0.8
6	hdop	水平精度因子	x.x	0.4
7	vdop	垂直精度因子	x.x	0.6
8	systemID	GNSS 系统标识符, 不得为空, 见表 3-4	h	1
9	*xx	校验和	*hh	*27
10	<CR><LF>	语句结束符		<CR><LF>

表3-4 GNSS ID

系统	系统 ID	卫星 ID	信号 ID	信号通道
GPS	1(GP)	1-99	0	All Signals
		1-32 GPS	1	L1 C/A
		33-64 GPS SBAS	2	L1 P(Y)
			3	L1 M
			4	L2 P(Y)
			5	L2C-M
			6	L2C-L
			7	L5-I
			8	L5-Q
			9 - F	Reserved
GLONASS	2(GL)	1-99	0	All Signals
		33-64 SBAS	1	G1 C/A
		65-99 GL	2	G1 P
			3	G2 C/A
			4	GLONASS (M) G2 P
			5 - F	Reserved
GALILEO	3(GA)	1-99	0	All Signals
		1-36 GA	1	E5a
		37-64 GA SBAS	2	E5b
			3	E5a+b
			4	E6-A
			5	E6-BC
			6	L1-A
			7	L1-BC
			8 - F	Reserved

BDS	4(GB/BD)	1-99	0	All Signals
		1-64 BD	1	B1I
			2	B1Q
			3	B1C
			4	B1A
			5	B2-a
			6	B2-b
			7	B2 a+b
			8	B3I
			9	B3Q
			A	B3A
			B	B2I
			C	B2Q
			D - F	Reserved
QZSS	5(GQ)	1-99	0	All Signals
		1-10 QZ	1	L1 C/A
		55-63 QZ SBAS	2	L1 C(D)
			3	L1 C(P)
			4	LIS
			5	L2C-M
			6	L2C-L
			7	L5-I
			8	L5-Q
			9	L6D
			A	L6E
			B - F	Reserved
NavIC	6(GI)	1-99	0	All Signals
		1-15 GI	1	L5-SPS
		33-64 SBAS	2	S-SPS
			3	L5-RS
			4	S-RS
			5	L1-SPS
			6 - F	Reserved
Reserved	7 - F			

3.1.3. GSV

适用产品：M720、M722

本语句用于输出可视的卫星数、卫星标识号、仰角、方位角、载噪比。每次传送最多为 4 颗卫星，传送的语句总数和传送的语句号在前两个字段中显示。

格式：

```
$--GSV,x,x,xx,xx,xx,xxx,xx.....xx,xx,xxx,xx,h*hh<CR><LF>
```

输出示例：

```
$GPGSV,3,1,09,24,81,119,55,18,58,233,52,23,50,328,51,15,47,044,52,1*6
```

1

表3-5 GSV 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-GSV	log 头		\$GPGSV
2	#msgs	语句总数 (1-9)	x	3
3	msg#	语句号 (1-9)	x	1
4	#sats	可视的卫星总数	xx	09
5	prn	卫星号	xx	24
6	elev	仰角, 90° (最大值)	xx	81
7	azimuth	真方位角, 000~359°	xxx	119
8	snr	C/N0,00-99dB-Hz,未跟踪时为空格	xx	55
...	...	下一颗卫星的卫星号,仰角,真方位角,SNR,		18,58,233,52,23,50,328,51,15,47,044,52
...	...	...		
...	...	最后一颗卫星的卫星号,仰角,真方位角,SNR		
9	signalID	GNSS 信号标识符, 该字段不能为空,见表 3-4	h	1
10	*xx	校验和	*hh	*61

3.1.4. GLL

适用产品：M720、M722

本语句用于输出载体的纬度与经度、定位时间与状态。

格式：

```
$--GLL,1111.11,a,yyyyy.yy,a,hmmss.ss,A,a*hh<CR><LF>
```

输出示例：

```
$GPGLL,3109.9556822,N,12117.2813212,E,094048.00,A,A*69
```

表3-6 GLL 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-GLL	log 头		\$GPGLL
2	lat	纬度,DDmm.mm	IIII.II	3109.9556822
3	latdir	纬度方向(N=北, S=南)	a	N
4	lon	经度,DDDmm.mm	yyyyy.yy	12117.2813212
5	londir	经度方向(E=东, W=西)	a	E
6	utc	位置对应的 UTC 时间,hh/mm/s.ss	hhmmss.ss	094048.00
7	status	状态, 不得为空, 见表 3-7	A	A
8	modeind	模式指示, 不得为空, 见表 3-8	a	A
9	*xx	校验和	*hh	*69
10	<CR><LF>	语句结束符		<CR><LF>

表3-7 数据状态

标识符	说明
V	数据无效, 除 A 和 D 以外对所有模式指示, 其状态字段应该设置成 V
A	数据有效

表3-8 定位系统模式说明

标识符	说明
A	自主模式
D	差分模式
E	估算（航位推算）模式
M	手动输入模式
S	模拟器模式
N	数据无效

3.1.5. GST

适用产品：M720、M722

本语句用于输出伪距误差信息。

格式：

\$--GST,hhmmss.ss,x.x,x.x,x.x,x.x,x.x,x.x,x.x*hh<CR><LF>

输出示例：

\$GPGST,094048.00,3.53,0.80,0.72,12.2267,0.79,0.72,1.61*51

表3-9 GST 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-GST	log 头		\$GPGST
2	utc	与这条语句有关的 GGA 或 GNS 语句的 UTC 时间	hhmmss.ss	094048.00
3	rms	伪距、DGNS 修正值的标准偏差的均方根	x.x	3.53
4	smjstd	误差椭圆半长轴的标准偏差，单位为米	x.x	0.80
5	smnrstd	误差椭圆半短轴的标准偏差，单位为米	x.x	0.72
6	orient	误差椭圆的半长轴方向（单位为度，与真北夹角）	x.x	12.2267
7	latstd	纬度的标准偏差，单位为米	x.x	0.79
8	lonstd	经度的标准偏差，单位为米	x.x	0.72
9	altstd	高度的标准偏差，单位为米	x.x	1.61
10	*xx	校验和	*hh	*51
11	<CR><LF>	语句结束符		<CR><LF>

3.1.6. KSXT

适用产品：M720、M722

KSXT 输出包含 GNSS 接收机的时间、位置、定位和定向相关数据。

输出示例：

```
$KSXT,20250804115550.00,121.17808532,31.15968682,34.2463,130.33,-80.7
2,82.38,0.005,0.00,1,3,50,51,,,,0.018,0.002,0.023,,*22
```

表3-10 GST 报文格式说明

ID	字段	数据描述	符号	示例
1	\$KSXT	log 头		\$KSXT
2	Utc	UTC 时间	yyyymmddhhmmss.ss	20250804115550.00
3	Lon	经度（单位，度），保留小数点后 8 位有效数字	DDD.DDDDDDDD	121.17808532
4	Lat	纬度（单位，度），保留小数点后 8 位有效数字	DD.DDDDDDDD	31.15968682
5	Height	海拔高（单位，米），保留小数点后 4 位有效数字	xxxxx.xxxx	34.2463
6	Heading	方位角，保留小数点后 2 位有效数字	xxx.xx	130.33

7	Pitch	俯仰角，保留小数点后 2 位有效数字	xxx.xx	-80.72
8	Track true	速度角，保留小数点后 2 位有效数字	xxx.xx	82.38
9	Vel	水平速度，单位 km/h，保留小数点后 3 位有效数字	xxx.xxx	0.005
10	Roll	横滚角，保留小数点后 2 位有效数字	xxx.xx	0.00
11	Pos qual	定位质量标识符： 0 = 定位不可用或无效 1 = 单点定位 2 = RTK 浮点解 3 = RTK 固定解	X	1
12	Heading qual	定向质量标识符： 0 = 定位不可用或无效 1 = 单点定位 2 = RTK 浮点解 3 = RTK 固定解	X	3
13	#hsolnSVs	从天线当前参与解算的卫星数量	xx	50
14	#msolnSVs	主天线当前参与解算的卫星数量	xx	51
15	East	东向位置坐标：以基准站为原点的地理坐标系下的东向位置，单位：米，小数点后 3 位	xxx.xxx	
16	North	北向位置坐标：以基准站为原点的地理坐标系下的北向位置，单位：米，小数点后 3 位	xxx.xxx	
17	Up	天向位置坐标：以基准站为原点的地理坐标系下的天向位置，单位：米，小数点后 3 位	xxx.xxx	
18	EastVel	东向速度：地理坐标系下的东向速度，小数点后 3 位，单位： Km/h(如无为空)	xxx.xxx	0.018
19	NorthVel	北向速度：地理坐标系下的北向速度，小数点后 3 位，单位： Km/h(如无为空)	xxx.xxx	0.002
20	UpVel	天向速度：地理坐标系下的天顶向速度，小数点后 3 位，单位： Km/h(如无为空)	xxx.xxx	0.023
21	Reserved	保留		
22	Reserved	保留		
23	*xx	校验和	*hh	22
24	<CR><LF>	语句结束符		<CR><LF>

3.1.7. VTG

适用产品：M720、M722

本语句用于输出相对于地面的实际航向和速度。

格式：

```
$--VTG,x.x,T,x.x,M,x.x,N,x.x,K,a*hh<CR><LF>
```

输出示例：

```
$GPVTG,15.473,T,15.473,M,0.025,N,0.046,K,A*26
```

表3-11 VTG 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-VTG	log 头		\$GPVTG
2	tracktruth	对地航向，单位为度（真北）	x.x	15.473
3	T	真北航迹角标识	T	T
4	trackmag	对地航向，单位为度（磁北）	x.x	15.473
5	M	磁北航迹角标识	M	M
6	speedKn	对地速度，单位为节	x.x	0.025
7	N	速度单位，节	N	N
8	speedKm	对地速度，单位为千米/小时	x.x	0.046
9	K	速度单位，千米/小时	K	K
10	modeind	模式指示（不应该为空字段），见表 3-8	a	A
11	*xx	校验和	*hh	*26
12	<CR><LF>	语句结束符		<CR><LF>

3.1.8. ZDA

适用产品：M720、M722

本语句用于输出 UTC 时间，日、月、年及本地时区。

格式：

```
$--ZDA,hhmmss.ss,xx,xx,xxxx,xx,xx*hh<CR><LF>
```

输出示例：

```
$GPZDA,094048.00,02,08,2022,,*6F
```

表3-12 ZDA 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-ZDA	log 头		\$GPZDA
2	utc	UTC 时间	hhmmss.ss	094048.00

3	day	日, 01-31	xx	02
4	month	月, 01-12	xx	08
5	year	年	xxxx	2022
6	localzonehours	本时区小时(0-±13h)	xx	
7	localzoneminutes	本时区分钟 (00-59)	xx	
8	*xx	校验和	*hh	*
9	<CR><LF>	语句结束符		<CR><LF>

3.1.9. RMC

适用产品：M720、M722

本语句用于输出 GNSS 导航接收机提供的时间、日期、位置、航迹向和速度数据。

格式：

\$--RMC,hhmmss.ss,A,llll.ll,a,yyyy.yy,a,x.x,x.x,xxxxxx,x.x,a,a,*hh<CR><LF>

输出示例：

\$GPRMC,094048.00,A,3109.9556822,N,12117.2813212,E,0.025,15.5,020822,0.0,E,A,V*73

表3-13 RMC 报文格式说明

ID	字段	数据描述	符号	示例
1	\$-RMC	log 头		\$GPRMC
2	utc	定位的 UTC 时间	hhmmss.ss	094048.00
3	posstatus	数据状态, 见表 3-7	A	A
4	lat	纬度,DDmm.mm	llll.ll	3109.9556822
5	latdir	纬度方向(N=北, S=南)	a	N
6	lon	经度,DDDmm.mm	yyyyy.yy	12117.2813212
7	londir	经度方向(E=东, W=西)	a	E
8	speedKn	对地速度, 单位为节	x.x	0.025
9	tracktrue	对地真航向, 单位为度	x.x	15.5
10	date	日期:ddmmyy	xxxxxx	020822
11	magvar	磁偏角	x.x	0.0
12	vardir	磁偏角方向(E=东, W=西)	a	E

13	modeind	模式指示, 见表 3-13	a	A
14	navstatus	导航状态, 见表 3-14	a	V
15	*xx	校验和	*hh	*73
16	<CR><LF>	语句结束符		<CR><LF>

表3-14 RMC 定位系统模式指示

标识符	说明
A	自主模式
D	差分模式
E	估算（航位推算）模式
F	RTK 浮点解
M	手动输入模式
N	数据无效
P	高精度。
R	RTK 固定解
S	模拟器模式

表3-15 导航状态

标识符	说明
S	若估计定位精度(95%置信度)在所选取的与实际导航模式对应的精度水平之内, 完好性可用且满足实际导航模式的要求, 且有效位置的计算时间对传统飞机小于 1s, 对高速飞机小于 0.5s
C	完好性不可用
U	当估计定位精度(95%置信度)低于相应的实际导航模式选取的精度水平, 或完好性可用, 但超出了实际导航模式的要求, 或新的有效位置不能在特定时间内计算完成 (传统飞行器为 1s, 高速飞行器为 0.5s)
V	导航状态无效, 设备不能提供导航状态指示

3.1.10. HDT

适用产品：M720、M722

本语句用于输出 GNSS 接收机相对于真北方向的航向角, 单位为度。

格式:

\$--HDT,x.x,T*hh<CR><LF>

输出示例:

\$GPHDT,78.7236,T*15

表3-16 HDT 报文格式说明

ID	字段	数据描述	符号	示例
----	----	------	----	----

1	\$-HDT	log 头		\$GPHDT
2	heading	航向角, 单位为度	x.x	78.7236
3	true	真北	T	T
4	*xx	校验和	*hh	*15
5	<CR><LF>	语句结束符		<CR><LF>

3.2. GMF 主天线报文输出

GMF 数据格式支持 ASCII 和二进制格式输出。二进制信息是一种有严格约定的机器可读格式，适合用于包含大量数据传输的应用。由于固有的压缩格式，二进制信息与 ASCII 相比数据量要小得多，因此接收机的通讯端口能够发送或接收更多的数据。ASCII 数据格式以“#”开头。ASCII 数据格式中的“#”不参与 CRC 校验。GMF 数据格式结构定义如下：

基本格式：

Header（头） 3 个同步字节，一共 24 个头信息字节。请务必检查头的长度。

Data（数据） 变量

CRC（校验） 4 个字节

表3-17 Binary 格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	Sync word	0xAA	Uchar	1	0
2	Sync word	0x44	Uchar	1	1
3	Sync word	0x72	Uchar	1	2
4	CPU Idle	CPU 空闲值	Uchar	1	3
5	Message ID	报文 ID	Ushort	2	4
6	Sequence number	同一报文序列号，倒序排列，值的范围 N-1 到 0	Uchar	1	6
7	Message attribute	报文属性：Bits 0-2 = Measurement source;	Uchar	1	7
8	Reserved	保留字段，值为 0	Ushort	2	8
9	Body length	报文体长度，不包含 CRC	Ushort	2	10
10	Port	端口，参见 Port ID	Uchar	1	12
11	Time status	GPS 时间状态，参见 Time Status	Uchar	1	13
12	Week Number	GPS 周	Ushort	2	14
13	TOW	GPS 周内秒，单位 ms	Ulong	4	16
14	Reserved	保留字段	Ulong	4	20
15	Receiver status	接收机状态	Ulong	4	24

表3-18 Binary 格式说明

Field	Field type	Description	Format
1	Sync word	ASCII 报文开头同步字符，保持为'#'	char
2	Message Name	报文名	char[]

Field	Field type	Description	Format
3	CPU Idle	CPU空闲值	float
4	Sequence number	同一报文序列号，倒序排列，值的范围N-1到0	unsigned char
5	Reserved	填充0	unsigned short
6	Port	端口名，参见Port ID	char[]
7	Time status	GPS时间状态，参见Time Status	Enum
8	Week Number	GPS周	unsigned short
9	TOW	GPS周内秒，单位ms	unsigned long
10	Reserved	保留字段	unsigned long
11	Receiver status	接收机状态，hex格式打印	unsigned long
12	;	ASCII header结束标志	char

表3-19 Time Status 说明

Time Status(ASCII)	值	描述
UNKNOWN	0	时间有效性未知 Time validity is unknown
APPROXIMATE	1	近似时间 Time is set approximately
Reserved	2	
COARSE	3	达到粗略精度 This time is valid to coarse precision
COARSESTEERING	4	时间是粗调设置且正在被调优 Time is coarse set and is being steered
Reserved	5	
Reserved	6	
FINE	7	时间达到高精度 Time has fine precision
Reserved	8	
FINESTEERING	9	时间已被正确配置且正在调优 Time is fine set and is being steered
SATTIME	10	卫星时间，只用于星历和历书的报文中。 Time from satellite. Only used in logs containing satellite data such as ephemeris and almanac

3.2.1. ANTENNASTATUS 天线状态信息

适用产品：M720、M722

支持天线馈电工作状态查询，天线馈电工作状态主要有三种：正常、开路、短路 1)；可以通过报文系统定期查询。

三种工作状态说明如下：

- 1. 有源天线工作正常，板卡给天线馈电，正常工作电流形成回路，接收机可以实时查询到其状态；
- 2. 天线开路，当接收机未连接天线，或者射频线缆损坏、断开等原因导致接收机收到不到卫星信号；
- 3. 天线短路，由于天线故障、连接接收机的射频线缆短路等原因，接收机与天线连接链路短路，导致接收机不能正常工作。

- Message ID: 23909
- Log Type: ASynch

输出示例：

```
#ANTENNASTATUSA,40.5,0,0,COM2,FINESTEERING,2381,393996000,3d7cea15,00008000;2,OK,0,0,OPEN,0,0*3e75689
```

表3-20 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	ANTENNASTATUS	报文头。 Log header.For information about the log headers, see ASCII, Abbreviated ASCII or Binary.		H	0
2	#antenna	对应的天线个数。 Number of antennas with information to follow	Ulong	4	H
3	AntennaStatus	天线馈电状态 Antenna Power Supply Status	ENUM	4	H+4
4	Reserved1	单一天线功耗估计，单位mW Antenna Power Comsumption Measured in unit of mW	Ulong	4	H+8

Field	Field type	Description	Format	Binary Bytes	Binary Offset
5	Reserved2	单一天线功耗异常事件计数	Ulong	4	H+12
6...	Next antenna offset = H + 4 + (#antenna x 12)				
variable	xxxx	32 位CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+4+(#antenna x 12)
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-21 天线馈电状态说明

二进制数值	天线状态	描述
0	UNKNOWN	未知
1	OK	馈电开启并正常工作
2	OFF	馈电处于关闭状态 ((暂不支持))
3	OPEN	开路（断路）
4	SHORT	短路
5	UNSTABLE	馈电不稳定，通常是馈电缆线或接头连接不稳定 ((暂不支持))

3.2.2. ANTISPOOFSTATUS 欺骗检测状态

适用产品：M720、M722

该协议用于输出欺骗检测状态，包括是否存在欺骗，OSNMA 状态信息和欺骗的置信度。

- Message ID: 23911
- Log Type: Async

输出示例：

```
#ANTISPOOFSTATUSA,81.5,0,0,COM2,FINESTEERING,2368,259758000,cba257c1,00008000;00000000,0,0,0,0,0,0,0,0,0000000000000000,0000000000000000,0000000000000000,0000000000000000,00000000*7b573bc8
```

表3-22 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	ANTISPOOFSTATUS	ANTISPOOFSTATUS 报文头。 Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.		H	0
2	OSNMAStatus	OSNMA 状态标志	HEX	4	H
3	Reserved1	Ulong	Ulong	4	H+4
4	SpoofExistFlag	是否存在诱骗信号	Ushort	2	H+8
5	ConfidenceRatio	诱骗信号存在的置信度 (0 ~ 100), 默认检测门限为 80	Ushort	2	H+10
6	SpoofDetectMode	欺骗检测模式 0:No 1:Baseband 2:OSNMA 3:Both	Ushort	2	H+12
7	FirstDetectedWeek	首次检测出欺骗的时间信息, 周计数	Ushort	2	H+14
8	FirstDetectedTOW	首次检测出欺骗的时间信息, 周内秒计数	Ulong	4	H+16
9	DetectSatNum	检测总卫星数	Ulong	4	H+20
10	SpoofSatNum	检测出受到欺骗攻击的总卫星数	Ulong	4	H+24
11	GPSSpoofSatMask	GPS 系统检测出欺骗攻击的 SatMask	HEX	8	H+28
12	GLOSpoofSatMask	GLO 系统检测出欺骗攻击的 SatMask	HEX	8	H+36
13	GALSpoofSatMask	GAL 系统检测出欺骗攻击的 SatMask	HEX	8	H+44
14	BDSSpoofSatMask	BDS 系统检测出欺骗攻击的 SatMask	HEX	8	H+52
15	Reserved2	reserved for QZS Spoof SatMask	HEX	4	H+60
16	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
17	[CR][LF]	语句结束符。 Sentence Terminator (ASCII only)	-	-	-

表3-23 OSNMA 状态标志

二进制位	状态	描述
	Authentication Status	
	0: NOT_AVAILABLE	不可用
	1: CANNOT_AUTHENTICATE	OSNMA 数据未收齐
	2: KEY_NOT_AUTHENTICATED	密钥验证失败
	3: MSG_NOT_AUTHENTICATED	导航信息验证失败
	4: AUTHENTICATED	导航信息验证通过
Bit 3	Public Key Source	
	0: GSC server	EGNSS GNSS Service Centre
	1: SIS	Signal In Space, 空间信号
Bit 4 ~ 5	Start Mode	
	0: Cold Start	每次设备启动时，公钥和根密钥都 需要进行验证
	1: Warm Start	保存有效的公钥，在设备启动时，仅对根密钥进行验证
	2: Hot Start	保存有效的公钥和根密钥，在设备 启动时，直接对链密钥进行验证
Bit 6	Merkle Tree Renewal	
	0: No	
	1: Yes	
Bit 7 ~ 32	Reserved	for future use

3.2.3. BASEOBS 基站的原始观测数据信息

适用产品：M720、M722

BASEOBS 报文传输基站的原始观测数据信息，基本格式与内容与 Range 报文 range 一致，MsgBody 中增加 BaseID 来标识数据来源。

- Message ID: 23001
- Log Type: Asynch

该报文 **novatel** 隐藏的 **baserange** 并不相同，增加了一个 **baseID** 来区分不同的数据流



- BASEOBS Msg Header 的时间，取当前BaseObs的时间。不取当前接收机的时间。
- BASEOBS 的 Phase 观测值是 RTCM 的原始观测值。从 MSM 解析到的，则对齐到参考信号；从 1001 等解析到的，根据 rtcm 协议 1/4 周补偿的表格中确认 phase shift 的问题。
- BASEOBS 格式更新，2024/12/05，ulong 类型的 stationID 更改成 ushort 类型的 stationID + ushort 类型的 vendorID，注意 msg 的解析。虽然部分板卡支持 4 个字符串数据的 StationID 设置（例如 Novatel 板卡），但是 Novatel 设置基站 ID 命令（DGPSTXID）是针对不同输出报文的，针对 RTCMV3 报文其基站 ID 只能按 RTCMV3 的 12bit 大小设置，而 RTCM 目前是 BASEOBS 的 StationID 的唯一输入源，故修改成 ushort 类型满足需求。

输出示例：

```
#BASEOBSA,47.0,0,0,COM2,FINE,2381,378436000,6e54712f,00008000;654,SWAS_VR
S,126,1,0,37435153.885,0.000,-194934008.412493,0.000,0.000,44.0,524.288,0
0041c00,1,0,37435154.600,0.000,-158399591.586404,0.000,0.000,43.0,524.288
,00441c00,1,0,37435153.707,0.000,-150735137.391689,0.000,0.000,46.0,524.2
88,00241c00,2,0,38027255.612,0.000,-198017811.809873,0.000,0.000,39.0,524
.288,00041c00,2,0,38027254.683,0.000,-160905749.569438,0.000,0.000,41.0,5
24.288,00441c00,2,0,38027253.146,0.000,-153119968.266801,0.000,0.000,45.0
,524.288,00241c00,3,0,36682707.662,0.000,-191015795.911016,0.000,0.000,45
.0,524.288,00041c00,3,0,36682705.804,0.000,-155215783.218836,0.000,0.000,
45.0,524.288,00441c00,3,0,36682705.089,0.000,-147705391.703414,0.000,0.00
0,47.0,524.288,00241c00,4,0,38058433.483,0.000,-198180152.675997,0.000,0.
000,40.0,524.288,00041c00,4,0,38058432.357,0.000,-161037621.005261,0.000,
0.000,41.0,524.288,00441c00,4,0,38058432.697,0.000,-153245476.787503,0.00
0,0.000,42.0,524.288,00241c00,6,0,36526677.033,0.000,-190204176.307043,0.
000,0.000,45.0,524.288,00041c00,6,0,36526672.173,0.000,-154556510.566623,
0.000,0.000,46.0,524.288,00441c00,6,0,36526673.138,0.000,-147077980.88068
```

6,0.000,0.000,47.0,524.288,00241c00,8,0,36909052.235,0.000,-192195186.148
182,0.000,0.000,44.0,524.288,00041c00,8,0,36909047.267,0.000,-156174349.6
50560,0.000,0.000,46.0,524.288,00441c00,8,0,36909048.858,0.000,-148617529
.907159,0.000,0.000,48.0,524.288,00241c00,9,0,37746749.673,0.000,-1965574
81.483788,0.000,0.000,43.0,524.288,00041c00,9,0,37746745.920,0.000,-15971
9098.575127,0.000,0.000,44.0,524.288,00441c00,9,0,37746748.476,0.000,-151
990770.034543,0.000,0.000,46.0,524.288,00241c00,13,0,36554055.074,0.000,-
190346633.321989,0.000,0.000,47.0,524.288,00041c00,13,0,36554056.593,0.00
0,-154672245.801905,0.000,0.000,46.0,524.288,00441c00,13,0,36554059.059,0
.000,-147188109.413426,0.000,0.000,47.0,524.288,00241c00,14,0,22021691.35
9,0.000,-114672859.820562,0.000,0.000,47.0,524.288,00041c00,14,0,22021687
.106,0.000,-93181138.368841,0.000,0.000,52.0,524.288,00441c00,14,0,220216
88.822,0.000,-88672381.663090,0.000,0.000,52.0,524.288,00241c00,16,0,3704
7868.347,0.000,-192918134.271502,0.000,0.000,47.0,524.288,00041c00,16,0,3
7047866.560,0.000,-156761821.855927,0.000,0.000,45.0,524.288,00441c00,16,
0,37047869.580,0.000,-149176582.111200,0.000,0.000,48.0,524.288,00241c00,
21,0,24545158.606,0.000,-127813075.367836,0.000,0.000,42.0,524.288,00041c
00,21,0,24545159.071,0.000,-103858578.325208,0.000,0.000,45.0,524.288,004
41c00,21,0,24545163.163,0.000,-96320444.136910,0.000,0.000,31.0,524.288,0
1241c00,21,0,24545163.538,0.000,-98833169.931916,0.000,0.000,45.0,524.288
,01641c00,21,0,24545154.925,0.000,-128985665.831068,0.000,0.000,40.0,524.
288,01441c00,24,0,21868426.801,0.000,-113875001.062576,0.000,0.000,51.0,5
24.288,00041c00,24,0,21868424.300,0.000,-92532843.855244,0.000,0.000,52.0
,524.288,00441c00,24,0,21868422.531,0.000,-85816727.384704,0.000,0.000,39
.0,524.288,01241c00,24,0,21868423.603,0.000,-88055470.078215,0.000,0.000,
52.0,524.288,01641c00,24,0,21868423.674,0.000,-114919726.940451,0.000,0.0
00,49.0,524.288,01441c00,26,0,22556495.384,0.000,-117457916.962083,0.000,
0.000,48.0,524.288,00041c00,26,0,22556495.098,0.000,-95444245.554093,0.00
0,0.000,50.0,524.288,00441c00,26,0,22556493.865,0.000,-88516747.583104,0.
000,0.000,35.0,524.288,01241c00,26,0,22556495.206,0.000,-90826070.743088,

0.000,0.000,49.0,524.288,01641c00,26,0,22556492.364,0.000,-118535498.9763
25,0.000,0.000,46.0,524.288,01441c00,33,0,23715323.788,0.000,-123492008.0
72740,0.000,0.000,46.0,524.288,00041c00,33,0,23715336.315,0.000,-10034739
1.941575,0.000,0.000,47.0,524.288,00441c00,33,0,23715348.019,0.000,-93064
069.604477,0.000,0.000,33.0,524.288,01241c00,33,0,23715349.145,0.000,-954
91879.600663,0.000,0.000,46.0,524.288,01641c00,33,0,23715319.964,0.000,-1
24624961.782255,0.000,0.000,44.0,524.288,01441c00,38,0,37487239.478,0.000
, -195205959.872754,0.000,0.000,45.0,524.288,00041c00,38,0,37487239.318,0.
000, -158620861.001605,0.000,0.000,45.0,524.288,00441c00,38,0,37487240.425
,0.000, -147107926.340199,0.000,0.000,32.0,524.288,01241c00,38,0,37487241.
283,0.000, -150945662.238820,0.000,0.000,46.0,524.288,01641c00,38,0,374872
36.566,0.000, -196996841.016188,0.000,0.000,43.0,524.288,01441c00,39,0,365
15966.019,0.000, -190148349.613031,0.000,0.000,49.0,524.288,00041c00,39,0,
36515962.106,0.000, -154511137.275596,0.000,0.000,50.0,524.288,00441c00,39
,0,36515961.462,0.000, -143296602.733825,0.000,0.000,36.0,524.288,01241c00
,39,0,36515963.106,0.000, -147034801.079335,0.000,0.000,49.0,524.288,01641
c00,39,0,36515962.713,0.000, -191892828.094696,0.000,0.000,47.0,524.288,01
441c00,42,0,21892943.226,0.000, -114002446.793782,0.000,0.000,51.0,524.288
,00041c00,42,0,21892951.303,0.000, -92636420.998995,0.000,0.000,52.0,524.2
88,00441c00,42,0,21892952.679,0.000, -85912700.689994,0.000,0.000,40.0,524
.288,01241c00,42,0,21892954.001,0.000, -88153962.560657,0.000,0.000,53.0,5
24.288,01641c00,42,0,21892940.456,0.000, -115048338.908345,0.000,0.000,50.
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47.0,524.288,00041c00,59,0,37101027.225,0.000, -156986581.596076,0.000,0.0
00,48.0,524.288,00441c00,59,0,37101029.120,0.000, -149390472.489154,0.000,
0.000,49.0,524.288,01641c00,60,0,38281145.141,0.000, -199340774.186584,0.0
00,0.000,44.0,524.288,00041c00,60,0,38281132.114,0.000, -161980638.482682,
0.000,0.000,43.0,524.288,00441c00,60,0,38281135.778,0.000, -154142786.6437
65,0.000,0.000,45.0,524.288,01641c00,40,12,20417740.196,0.000, -109297951.
167792,0.000,0.000,51.0,524.288,00011c00,40,12,20417740.750,0.000, -850095

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-80303856.036514,0.000,0.000,51.0,524.288,00211c00,42,8,22189919.133,0.00
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.000,-92258364.583907,0.000,0.000,44.0,524.288,00211c00,51,0,19562032.767
,0.000,-104276928.987313,0.000,0.000,55.0,524.288,00011c00,51,0,19562035.
411,0.000,-81104349.940395,0.000,0.000,52.0,524.288,00211c00,52,7,2034612
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6131.981,0.000,-84563063.509442,0.000,0.000,50.0,524.288,00211c00,4,0,261
50070.425,0.000,-137419659.656870,0.000,0.000,39.0,524.288,00431c00,4,0,2
6150075.018,0.000,-105295656.424100,0.000,0.000,43.0,524.288,02231c00,4,0
,26150074.124,0.000,-102618585.989903,0.000,0.000,30.0,524.288,01831c00,1
9,0,22917354.746,0.000,-120431753.393638,0.000,0.000,43.0,524.288,00431c0
0,19,0,22917357.301,0.000,-92279022.910187,0.000,0.000,46.0,524.288,02231
c00,19,0,22917356.283,0.000,-89932725.464047,0.000,0.000,33.0,524.288,018
31c00,21,0,23876225.960,0.000,-125470692.823264,0.000,0.000,48.0,524.288,
00431c00,21,0,23876230.248,0.000,-96140058.167495,0.000,0.000,51.0,524.28
8,02231c00,21,0,23876229.712,0.000,-93695797.099575,0.000,0.000,38.0,524.
288,01831c00,23,0,24564825.324,0.000,-129089336.266803,0.000,0.000,46.0,5
24.288,00431c00,23,0,24564832.132,0.000,-98912803.777122,0.000,0.000,49.0
,524.288,02231c00,23,0,24564831.471,0.000,-96397985.996935,0.000,0.000,36
.0,524.288,01831c00,27,0,27253738.996,0.000,-143219661.017053,0.000,0.000
,34.0,524.288,00431c00,27,0,27253746.769,0.000,-109739807.871417,0.000,0.
000,41.0,524.288,02231c00,27,0,27253746.376,0.000,-106949897.682409,0.000
,0.000,28.0,524.288,01831c00,29,0,25403737.927,0.000,-133497591.467676,0.
000,0.000,43.0,524.288,00431c00,29,0,25403744.431,0.000,-102290378.900639
,0.000,0.000,46.0,524.288,02231c00,29,0,25403743.199,0.000,-99689762.9738
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088.406491,0.000,0.000,47.0,524.288,02a51c00,194,0,37538743.882,0.000,-14

7310387.061180,0.000,0.000,37.0,524.288,01c51c00,199,0,36875885.363,0.000
, -193784030.856118,0.000,0.000,40.0,524.288,00051c00,199,0,36875889.776,0
.000, -151000535.813890,0.000,0.000,48.0,524.288,02a51c00,199,0,36875889.0
79,0.000, -144708843.308976,0.000,0.000,38.0,524.288,01c51c00,10,0,2197758
5.750,0.000, -115493053.304735,0.000,0.000,45.0,524.288,00001c00,10,0,2197
7593.273,0.000, -89994603.209437,0.000,0.000,47.0,524.288,01201c00,10,0,21
977589.234,0.000, -86244822.093438,0.000,0.000,35.0,524.288,01c01c00,12,0,
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1c00,25,0,21225631.514,0.000, -83294217.664311,0.000,0.000,36.0,524.288,01
c01c00,26,0,22462444.427,0.000, -118041218.870345,0.000,0.000,41.0,524.288
,00001c00,26,0,22462454.666,0.000, -91980235.338655,0.000,0.000,43.0,524.2
88,01201c00,26,0,22462451.413,0.000, -88147759.187000,0.000,0.000,33.0,524
.288,01c01c00,28,0,21090153.335,0.000, -110829819.364446,0.000,0.000,46.0,
524.288,00001c00,28,0,21090155.712,0.000, -86360975.794402,0.000,0.000,44.
0,524.288,01201c00,28,0,21090157.266,0.000, -82762682.000492,0.000,0.000,3
9.0,524.288,01c01c00,29,0,23974235.799,0.000, -125985699.308182,0.000,0.00
0,36.0,524.288,00001c00,29,0,23974244.466,0.000, -98170696.853410,0.000,0.
000,39.0,524.288,01201c00,31,0,21800331.082,0.000, -114561865.708716,0.000
,0.000,44.0,524.288,00001c00,31,0,21800334.388,0.000, -89269090.859420,0.0
00,0.000,46.0,524.288,01201c00,32,0,20584237.677,0.000, -108171171.501800,
0.000,0.000,51.0,524.288,00001c00,32,0,20584241.805,0.000, -84289278.74116
5,0.000,0.000,51.0,524.288,01201c00,32,0,20584237.677,0.000, -80777170.546
083,0.000,0.000,38.0,524.288,01c01c00*f865768d

表3-24 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BASEOBS	报文头。 Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.		H	0
2	Base ID#	对应的基站 ID号。 ID of base station	Ushort	2	H
3	Vendor ID#	Vendor ID,see Vendor ID ID of vendor	Ushort	2	H+2
4	#obs	对应的观测信息个数。 Number of observations with information to follow	Ulong	4	H+4
5	PRN/slot	卫星 PRN 号 Satellite PRN number of range measurement	Ushort	2	H+8
6	glofreq	(GLONASS 频率+7) (GLONASS Frequency + 7) (see GLONASS Slot and Frequency Numbers section of this manual)	Ushort	2	H+10
7	psr	码伪距观测量，米。 Pseudorange measurement (m)	Double	8	H+12
8	psr σ	码伪距标准差，米。 Pseudorange measurement standard deviation (m)	Float	4	H+20
9	adr	载波相位，周。 Carrier phase, in cycles (accumulated Doppler range)	Double	8	H+24
10	adr σ	载波相位标准差，周。 Estimated carrier phase standard deviation (cycles)	Float	4	H+32

Field	Field type	Description	Format	Binary Bytes	Binary Offset
11	dopp	瞬时多普勒, Hz Instantaneous carrier Doppler frequency (Hz)	Float	4	H+36
12	C/No	载噪比。 Carrier to noise density ratio C/No = $10[\log_{10}(S/N_0)]$ (dB-Hz)	Float	4	H+40
13	locktime	连续跟踪时间, 无周跳。 Number of seconds of continuous tracking (no cycle slipping)	Float	4	H+44
14	ch-tr-status(HEX)	通道跟踪状态 Tracking status	Ulong	4	H+48
15...	Next PRN offset = $H + 4 + 4 + (\#obs \times 44)$				
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+4+4+(\#obs x 44)
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-25 Vendor ID

Binary	ASCII	Description
0	UNKNOWN	未知厂家
1	CHC_VRS	华测 SWAS
2	SWAS_VRS	华测 SWAS 一键固定
3	CHC_SAX	华测 M7xx 模组
4	CHC	华测
5	NOVATEL	诺瓦泰
6	UNICORE	和芯
7	TRIMBLE	天宝

Binary	ASCII	Description
8	MAGELLAN	Magellan
9	QXWZ_VRS	千寻位置差分服务
10	SIN	司南
11	HEMI	合众思壮
12	LEICA	莱卡
13	SWIFT	移动
14	SEPT	Septentrio

3.2.4. BASESTATION 基站位置和健康状况

适用产品：M720、M722

BASESTATION 输出基站位置和健康状况。该报文包含通过 RTCMV3 消息接收到的基站的 ECEF 笛卡尔位置。它还具有时间标签、基站的健康状态和基站 ID。此信息是在基站使用 SETFIXPOS 命令设置的。

- Message ID: 3032
- Log Type: Asynch

相对于 BASESTATION 报文而言，推荐优先使用 BASESTATIONINFO

主要由于



- BASESTATIONINFO 使用 LLH 来标识基站坐标，而 BASESTATION 使用 ECEF 坐标，更容易比对基站位置
- BASESTATIONINFO 还增加了天线相关信息，内容更丰富

BASESTATIONINFO Msg Header 的时间，取当前 BaseObs 的时间。不取当前接收机的时间。

输出示例：

```
#BASESTATIONA,29.5,0,0,COM2,FINE,2376,97541000,d323b507,00000000;00000001,RTCMV3,654,1,-2826742.374,4673641.240,3282482.203*969e277c
```

表3-26 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BASESTATION header	报文头。 Log header. For information about the log headers, see ASCII, or Bi-nary.		H	0
2	status	基站状态。 Status of the base station information, see Base Station Status	HEX	1	H+0
3	stn type	基站类型，见表 基站类型 Station type, see Base Station Type	Enum	1	H+1
4	stn ID	基站 ID Base station ID	Ushort	2	H+2
5	health	基站运行状况，见表 基站运行状况 Station health, see Base Station Health	Ulong	4	H+4
6	x	ECEF X value (m)	Double	8	H+8
7	y	ECEF X value (m)	Double	8	H+16
8	z	ECEF X value (m)	Double	8	H+24
9	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)		4	H+32
10	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-27 Base Station Status

Bit #	Mask	Description	Bit = 0	Bit = 1
0	0x00000001	Validity of the base station	Valid	Invalid

表3-28 Base Station Type

Binary	ASCII	Description
0	NONE	Base station is not used
2	RTCMV3	Base station is RTCMV3

Base Station Health:

基站健康状况字段是 8 个值（0 到 7）中的一个。值 0 到 5 表示与卫星 UDRE 一西格玛微分误差值相乘的比例因子。以下是 0 到 5 的值及其相应的 UDRE 比例因子：

0:1（健康正常） 1:0.75 2:0.5 3:0.3 4:0.2 5:0.1

基站健康字段仅适用于 RTCM 基站。值 6 表示不监视基站传输，值 7 表示基站不工作。

3.2.5. BASESTATIONINFO 基站位置信息

适用产品：M720、M722

BASESTATIONINFO 输出基站位置信息

- Message ID: 3033
- Log Type: Asynch

相对于 BASESTATION 报文而言，推荐优先使用 BASESTATIONINFO

主要由于



- BASESTATIONINFO 使用 LLH 来标识基站坐标，而 REFSTATION 使用 ECEF 坐标，更容易比对基站位置
- BASESTATIONINFO 还增加了天线相关信息，内容更丰富

BASESTATIONINFO Msg Header 的时间，取当前 BaseObs 的时间。不取当前接收机的时间。

输出示例：

```
#BASESTATIONINFOA,29.5,0,0,COM2,FINE,2376,97579000,d323b507,00000000;WGS84,RTCMV3,654,31.174684470,121.166663152,5.380408566,0.000,1,"ADVNULLANTENNA","*c70450be
```

表3-29 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BASESTATIONINFO header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
2	datum	基准 ID Datum ID number 0 = WGS84 1 = USER (default = WGS84)	Enum	1	H+0
3	Ref Stn Type	基站类型。 Base Station Type, see Base Station Type	Enum	1	H+1
4	stn ID	基站 ID Base Station ID	Ushort	2	H+2
5	latitude	纬度，单位：度。 Latitude (degrees)	Double	8	H+4
6	longitude	经度，单位：度。 Longitude (degrees)	Double	8	H+12
7	height	海拔高，单位：米。Ellipsoidal Height (m)	Double	8	H+20
8	ARP height	基站天线高，单位：米。Base Antenna ARP (m)	Float	4	H+28
9	health	基站运行状况。 Base Station Health, see Base Station Status	Ulong	4	H+32
10	Ant Model	基站天线型号名称 Base Antenna Model Name	Char[32]	32	H+36
11	Ant Serial	基站天线序列号 Base Antenna Serial Number	Char[32]	32	H+68

Field	Field type	Description	Format	Binary Bytes	Binary Offset
12	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+100
13	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.6. BD2EPH 北斗星历

适用产品：M720、M722

- Message ID: 3001
- Log Type: Asynch

输出示例：

```
#BD2EPHA,41.0,0,0,COM2,SATTIME,2376,96232000,d323b507,00000000;60,1020,2.00,7.878899616E-10,1,93600,0.00000000E+00,6.39488462E-14,-8.87666829E-07,1,691.1250000,9.5686128568E-09,-1.399904078,2.234661952E-05,1.5239918139E-04,7.803551853E-06,6493.407719,93600,1.206062734E-07,3.081616721E+00,-1.201406121E-07,8.6582461273E-02,-240.2812500,0.255222267,-8.44892336E-09,4.80E-08,4.80E-08,0*a10be1ab
```

表3-30 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	log header	BD2EPH 报文头 Log header.For information about the log headers, see ASCII, or Binary.		H	0
2	satellite ID	卫星 ID ID/ranging code	Ulong	4	H

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
3	Week	周计数, BDT Week number in BeiDou time (BDT) since the BeiDou system time start epoch (January 1, 2006)	Ulong	4	H+4
4	URA	用户距离精度 User range accuracy (m). This is the evaluated URAI/URA lookup-table value.	Double	8	H+8
5	IDOT	轨道倾角变化率, rad/s Rate of inclination angle (radians/second)	Double	8	H+16
6	AODE	星历数据龄期 Age of data, ephemeris	Ulong	4	H+24
7	toc	卫星钟差参数, s Reference time of clock parameters (s)	Ulong	4	H+28
8	a ₂	卫星钟差参数, s/s/s Quadratic term of clock correction polynomial (s/s/2)	Double	8	H+32
9	a ₁	卫星钟差参数, s/s Linear term of clock correction polynomial (s/s)	Double	8	H+40
10	a ₀	卫星钟差参数, s Constant term of clock correction polynomial (s)	Double	8	H+48
11	AODC	时钟数据龄期 Age of data, clock	Ulong	4	H+56
12	C _{rs}	轨道半径 (正弦振幅, m) Amplitude of sine harmonic correction term to the orbit radius (m)	Double	8	H+60

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
13	ΔN	卫星平均运动速率与计算值之差, rad/s Mean motion difference from computed value (radians/s)	Double	8	H+68
14	M_0	参考时间的平近角点, rad Mean anomaly at reference time (radians)	Double	8	H+76
15	C_{uc}	纬度幅角 (余弦振幅, rad) Amplitude of cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+84
16	ecc	偏心率 Eccentricity (dimensionless)	Double	8	H+92
17	C_{us}	纬度幅角 (正弦振幅, rad) Amplitude of sine harmonic correction term to the argument of latitude (radians)	Double	8	H+100
18	RootA	长半轴的平方根 Square root of semi-major axis (sqrt(m))	Double	8	H+108
19	toe	星历参考时间, s Reference time of ephemeris parameters (s)	Ulong	4	H+116
20	C_{ic}	倾角 (余弦振幅, rad) Amplitude of cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+120
21	Ω_0	按参考时间计算的升交点经度, rad Longitude of ascending node of orbital plane computed according to reference time (radians)	Double	8	H+128

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
22	C _{is}	倾角（正弦振幅， rad） Amplitude of sine harmonic correction term to the angle of inclination (radians)	Double	8	H+136
23	i ₀	参考时间的轨道倾角， rad Inclination angle at reference time (radians)	Double	8	H+144
24	C _{rc}	轨道半径（余弦振幅， m） Amplitude of cosine harmonic correction term to the orbit radius (m)	Double	8	H+152
25	ω	近地点幅角， rad Argument of perigee (radians)	Double	8	H+160
26	OmegaDot	升交点赤经变化率， rad/s Rate of right ascension (radians/s)	Double	8	H+168
27	tgdl	B1 信号星上设备时延差 Equipment group delay differential for the B1 signal (s)	Double	8	H+176
28	tgdl	B2 信号星上设备时延差 Equipment group delay differential for the B2 signal (s)	Double	8	H+184
29	health 1	卫星自主健康标示： 0-健康； 1-不健康 Autonomous satellite health flag. 0 means broadcasting satellite is good and 1 means not.	Ulong	4	H+192
30	xxxx	32 位CRC 校验 32-bit CRC (ASCII and Binary only)	Ulong	4	H+196
31	[CR][LF]	语句结束符（仅ASCII） Sentence terminator (ASCII only)	-	-	-

3.2.7. BD2IONO 北斗 Klobuchar 电离层延迟模型

适用产品：M720、M722

该日志包含北斗 B1I 信号传输的 Klobuchar 电离层模型参数。

- Message ID: 3015
- Log Type: Async

输出示例：

```
#BD2IONOA,30.0,0,0,COM2,FINESTEERING,2376,97704000,d323b507,00000000;2.32
8306436538696E-08,5.215406417846680E-08,-7.152557373046875E-07,1.13248825
0732422E-06,135168.0,-327680.0,2424832.0,-1835008.0*96a0edfe
```

表3-31 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BD2IONO	报文头。 Log header.For information about the log headers, see ASCII, or Binary.		H	0
2	α_0	Alpha 参数的常数项 Alpha parameter constant term	Double	8	H
3	α_1	Alpha 参数的 1 阶项 Alpha parameter 1st order term	Double	8	H+8
4	α_2	Alpha 参数的 2 阶项 Alpha parameter 2nd order term	Double	8	H+16
5	α_3	Alpha 参数的 3 阶项 Alpha parameter 3rd order term	Double	8	H+24
6	β_0	Beta 参数的常数项 Beta parameter constant term	Double	8	H+32
7	β_1	Beta 参数的 1 阶项 Beta parameter 1st order term	Double	8	H+40
8	β_2	Beta 参数的 2 阶项 Beta parameter 2nd order term	Double	8	H+48
9	β_3	Beta 参数的 3 阶项 Beta parameter 3rd order term	Double	8	H+56

10	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	H + 64	
11	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.8. BD3EPHCNAV1 北斗 CNAV1 星历

适用产品：M720、M722

- Message ID: 3002
- Log Type: Asynch

输出示例：

```
#BD3EPHCNAV1A,40.0,0,0,COM2,SATTIME,2376,96486000,d323b507,00000000;50,10
20,3.364425856E-10,26,93600,0.00000000,8.71214212E-12,-5.65361697E-04,26,
-119.1250000,3.3456750751E-09,2.176444581E-01,-5.929730833E-06,2.96427751
89E-04,1.026783139E-05,304.837891,-0.004540,93600,-8.009374142E-08,2.3650
450583E+00,-7.450580597E-09,9.6800866229E-01,156.1523438,7.948977236E-01,
-6.47026951E-09,3.422610462E-08,8.597271517E-08,00000061,3,2.8469726615E-
14,8.597271517E-08,0*5dadf501
```

表3-32 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	log header	BD3EPHCNAV1 报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	satellite ID	卫星 ID ID/ranging code	Ulong	4	H
3	WN	周计数, BDT Week number in BeiDou time (BDT). This will be offset from GPS time by 1356 weeks.	Ulong	4	H+4
4	i ₀ Dot	轨道倾角变化率, rad/s Rate of inclination angle (radians/s)	Double	8	H+8

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
5	IODE	星历数据期号 Issue of data, ephemeris	Ulong	4	H+16
6	toc	卫星钟差参数 (BDT) , s Reference time of clock parameters in BeiDou time (BDT). This will be offset from GPS time by 14 seconds. (s)	Ulong	4	H+20
7	a ₂	卫星钟差参数, s/s/s Quadratic term of clock correction polynomial (s/s/s)	Double	8	H+24
8	a ₁	卫星钟差参数, s/s Linear term of clock correction polynomial (s/s)	Double	8	H+32
9	a ₀	卫星钟差参数, s Constant term of clock correction polynomial (s)	Double	8	H+40
10	IODC	时钟数据期号 Issue of data, clock	Ulong	4	H+48
11	C _{rs}	轨道半径 (正弦振幅, m) Amplitude of sine harmonic correction term to the orbit radius (m)	Double	8	H+52
12	DeltaN	卫星平均角速度的改正值, rad/s Mean motion difference from computed value (radians/s)	Double	8	H+60
13	M ₀	参考时间的平近点角, rad Mean anomaly at reference time (radians)	Double	8	H+68
14	C _{uc}	纬度幅角 (余弦振幅, rad) Amplitude of cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+76
15	e	卫星轨道偏心率 Eccentricity (unitless)	Double	8	H+84

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
16	C _{us}	纬度幅角（正弦振幅， rad） Amplitude of sine harmonic correction term to the argument of latitude (radians)	Double	8	H+92
17	DeltaA	参考时刻的卫星长半轴偏差, m Semi-major axis difference at reference time (m)	Double	8	H+100
18	ADot	卫星长半轴变化率, m Change rate in Semi-major axis (m/s)	Double	8	H+108
19	toe	星历的参考时间, BDT Reference time of ephemeris parameters in BeiDou time (BDT). (s) This will be offset from GPS time.	Ulong	4	H+116
20	C _{ic}	倾角(余弦振幅,rad) Amplitude of cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+120
21	Omega ₀	升交点赤经, rad Longitude of ascending node of orbital of plane at weekly epoch (radians)	Double	8	H+128
22	C _{is}	倾角（正弦振幅， rad） Amplitude of sine harmonic correction term to the angle of inclination (radians)	Double	8	H+136
23	i ₀	参考时间的轨道倾角， rad Inclination angle at reference time (radians)	Double	8	H+144
24	C _{rc}	轨道半径（余弦振幅， m） Amplitude of cosine harmonic correction term to the orbit radius (m)	Double	8	H+152
25	Omega	近地点幅角， rad Argument of perigee (radians)	Double	8	H+160
26	OmegaDot	升交点赤经变化率， rad/s Rate of right ascension (radians/s)	Double	8	H+168

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
27	TGDB1Cp	B1C 组件和 B3I 信号之间的群延迟差, s Group delay differential between the B1C pilot component and the B3I signal (s)	Double	8	H+176
28	TGDB2Ap	B2a 组件和 B3I 信号之间的群延迟差, s Group delay differential between the B2a pilot component and the B3I signal (s)	Double	8	H+184
29	Satellite Status	卫星状态信息 Satellite Status Information: Bits 0 to 1: Health Status (HS) Bit 2: Data Integrity Flag (DIF) Bit 3: Signal Integrity Flag (SIF) Bit 4: Accuracy Integrity Flag (AIF) Bits 5 to 8: Signal In Space Monitoring Accuracy Index (SISMAI)	Hex Ulong	4	H+192
30	Satellite Type	2 bit 轨道类型 2 bit orbit type: 01 indicates GEO, 10 indicates IGSO, 11 indicates MEO	Ulong	4	H+196
31	NDot	平均角速度改正值变化率, rad/s/s Rate of Mean motion difference from computed value (radians/s/s)	Double	8	H+200
32	ISCB1Cd	B1C 数据和组件之间的群延迟差, s Group delay differential between the B1C data and pilot components (s)	Double	8	H+208
33	Reserved		Ulong	4	H+216
34	xxxx	32 位CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H+220
35	[CR][LF]	语句结束符 (仅ASCII) Sentence Terminator (ASCII only)	—	—	—

3.2.9. BD3EPHCNAV2 北斗 CNAV2 星历

适用产品：M720、M722

- Message ID: 3003
- Log Type: Asynch

输出示例：

```
#BD3EPHCNAV2A,55.5,10,0,COM1,SATTIME,2383,356790000,2b0d7715,00008000;22,1027,4.939491464E-10,15,356400,0.00000000,5.33262323E-12,3.84670217E-04,15,34.9101563,3.5072889498E-09,7.549586337E-01,1.681968570E-06,7.4567348929E-04,7.141381502E-06,21.259766,0.006664,356400,-4.097819328E-08,1.5018002321E+00,-8.847564459E-08,9.8401679040E-01,229.3359375,8.872952828E-01,-6.67742100E-09,1.740409061E-08,-3.899913281E-09,00000020,3,-5.0901080893E-14,-3.899913281E-09,0*36f0036f
```

表3-33 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	log header	BD3EPHCNAV2 报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	satellite ID	卫星 ID ID/ranging code	Ulong	4	H
3	WN	周计数，BDT Week number in BeiDou time (BDT). This will be offset from GPS time by 1356 weeks.	Ulong	4	H+4
4	i ₀ Dot	轨道倾角变化率，rad/s Rate of inclination angle (radians/s)	Double	8	H+8
5	IODE	星历数据期号 Issue of data, ephemeris	Ulong	4	H+16

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
6	toc	卫星钟差参数 (BDT) , s Reference time of clock parameters in BeiDou time (BDT). This will be offset from GPS time by 14 seconds. (s)	Ulong	4	H+20
7	a ₂	卫星钟差参数, s/s/s Quadratic term of clock correction polynomial (s/s/s)	Double	8	H+24
8	a ₁	卫星钟差参数, s/s Linear term of clock correction polynomial (s/s)	Double	8	H+32
9	a ₀	卫星钟差参数, s Constant term of clock correction polynomial (s)	Double	8	H+40
10	IODC	时钟数据期号 Issue of data, clock	Ulong	4	H+48
11	C _r s	轨道半径 (正弦振幅, m) Amplitude of sine harmonic correction term to the orbit radius (m)	Double	8	H+52
12	DeltaN	卫星平均角速度的改正值, rad/s Mean motion difference from computed value (radians/s)	Double	8	H+60
13	M ₀	参考时间的平近点角, rad Mean anomaly at reference time (radians)	Double	8	H+68
14	C _u c	纬度幅角 (余弦振幅, rad) Amplitude of cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+76
15	e	卫星轨道偏心率 Eccentricity (unitless)	Double	8	H+84

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
16	C_{uS}	纬度幅角（正弦振幅， rad） Amplitude of sine harmonic correction term to the argument of latitude (radians)	Double	8	H+92
17	DeltaA	参考时刻的卫星长半轴偏差, m Semi-major axis difference at reference time (m)	Double	8	H+100
18	ADot	卫星长半轴变化率, m Change rate in Semi-major axis (m/s)	Double	8	H+108
19	toe	星历的参考时间, BDT Reference time of ephemeris parameters in BeiDou time (BDT). (s) This will be offset from GPS time.	Ulong	4	H+116
20	C_{ic}	倾角（余弦振幅， rad） Amplitude of cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+120
21	Ω_{a0}	升交点赤经， rad Longitude of ascending node of orbital of plane at weekly epoch (radians)	Double	8	H+128
22	C_{is}	倾角（正弦振幅， rad） Amplitude of sine harmonic correction term to the angle of inclination (radians)	Double	8	H+136
23	i_0	参考时间的轨道倾角， rad Inclination angle at reference time (radians)	Double	8	H+144
24	C_{rc}	轨道半径（余弦振幅， m） Amplitude of cosine harmonic correction term to the orbit radius (m)	Double	8	H+152
25	Omega	近地点幅角， rad Argument of perigee (radians)	Double	8	H+160

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
26	OmegaDot	升交点赤经变化率, rad/s Rate of right ascension (radians/s)	Double	8	H+168
27	TGDB1Cp	B1C 组件和 B3I 信号之间的群延迟差, s Group delay differential between the B1C pilot component and the B3I signal (s)	Double	8	H+176
28	TGDB2Ap	B2a 组件和 B3I 信号之间的群延迟差, s Group delay differential between the B2a pilot component and the B3I signal (s)	Double	8	H+184
29	Satellite Status	卫星状态信息 Satellite Status Information: Bits 0 to 1: Health Status (HS) Bit 2: Data Integrity Flag (DIF) Bit 3: Signal Integrity Flag (SIF) Bit 4: Accuracy Integrity Flag (AIF) Bits 5 to 8: Signal In Space Monitoring Accuracy Index (SISMAI)	Hex Ulong	4	H+192
30	Satellite Type	2 bit 轨道类型 2 bit orbit type: 01 indicates GEO, 10 indicates IGSO, 11 indicates MEO	Ulong	4	H+196
31	NDot	平均角速度改正值变化率, rad/s/s Rate of Mean motion difference from computed value (radians/s/s)	Double	8	H+200
32	ISCB2ad	B2a 数据和组件之间的群延迟差, s Group delay differential between the B2a data and pilot components (s)	Double	8	H+208

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
33	Reserved		Ulong	4	H+216
34	xxxx	32 位CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H+220
35	[CR][LF]	语句结束符（仅ASCII） Sentence Terminator (ASCII only)	—	—	—

3.2.10. BD3EPHCNAV3 北斗 CNAV3 星历

适用产品：M720、M722

- Message ID: 3004
- Log Type: Asynch

输出示例：

```
#BD3EPHCNAV3A,60.5,0,0,COM1,SATTIME,2383,285241000,013cc608,00008000;40,1
027,-8.928943355E-13,284400,0.00000000,-2.81552559E-12,-1.67897262E-04,38
.0273438,1.9209728734E-09,1.742239175E+00,1.327134669E-06,4.5866309083E-0
3,-7.434748113E-06,2333.812500,-0.033169,284400,8.847564459E-08,-1.088223
5444E+00,3.343448043E-07,9.6413874731E-01,459.4765625,-2.968645365E+00,-3
.00333939E-09,-6.111804396E-09,000001e0,2,9.7573121232E-14,0*f46439b2
```

表3-34 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	BD3EPHCNAV3 header	BD3EPHCNAV3 报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	satellite ID	卫星 ID ID/ranging code	Ulong	4	H

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
3	WN	周计数, BDT Week number in BeiDou time (BDT). This will be offset from GPS time by 1356 weeks.	Ulong	4	H+4
4	i_0 Dot	轨道倾角变化率, rad/s Rate of inclination angle (radians/s)	Double	8	H+8
5	toc	卫星钟差参数 (BDT), s Reference time of clock parameters in BeiDou time (BDT). This will be offset from GPS time by 14 seconds. (s)	Ulong	4	H+16
6	a_2	卫星钟差参数, s/s/s Quadratic term of clock correction polynomial (s/s/s)	Double	8	H+20
7	a_1	卫星钟差参数, s/s Linear term of clock correction polynomial (s/s)	Double	8	H+28
8	a_0	卫星钟差参数, s Constant term of clock correction polynomial (s)	Double	8	H+36
9	C_{rs}	轨道半径 (正弦振幅, m) Amplitude of sine harmonic correction term to the orbit radius (m)	Double	8	H+44
10	DeltaN	卫星平均角速度的改正值, rad/s Mean motion difference from computed value (radians/s)	Double	8	H+52
11	M_0	参考时间的平近点角, rad Mean anomaly at reference time (radians)	Double	8	H+60

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
12	C_{uc}	纬度幅角（余弦振幅， rad） Amplitude of cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+68
13	e	卫星轨道偏心率 Eccentricity (unitless)	Double	8	H+76
14	C_{us}	纬度幅角（正弦振幅， rad） Amplitude of sine harmonic correction term to the argument of latitude (radians)	Double	8	H+84
15	DeltaA	参考时刻的卫星长半轴偏差, m Semi-major axis difference at reference time (m)	Double	8	H+92
16	ADot	卫星长半轴变化率, m Change rate in Semi-major axis (m/s)	Double	8	H+100
17	toe	星历的参考时间, BDT Reference time of ephemeris parameters in BeiDou time (BDT). (s) This will be offset from GPS time.	Ulong	4	H+108
18	C_{ic}	倾角（余弦振幅， rad） Amplitude of cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+112
19	Ω_{a0}	升交点赤经， rad Longitude of ascending node of orbital of plane at weekly epoch (radians)	Double	8	H+120
20	C_{is}	倾角（正弦振幅， rad） Amplitude of sine harmonic correction term to the angle of inclination (radians)	Double	8	H+128
21	i_0	参考时间的轨道倾角， rad Inclination angle at reference time (radians)	Double	8	H+136

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
22	C _{rc}	轨道半径（余弦振幅， m） Amplitude of cosine harmonic correction term to the orbit radius (m)	Double	8	H+144
23	Omega	近地点幅角， rad Argument of perigee (radians)	Double	8	H+152
24	OmegaDot	升交点赤经变化率， rad/s Rate of right ascension (radians/s)	Double	8	H+160
25	TGDB2 Bi	B2bl 和 B3I 信号之间的群延迟差, s Group delay differential between the B2bl and the B3I signal (s)	Double	8	H+168
26	Satellite Status	卫星状态信息 Satellite Status Information: Bits 0 to 1: Health Status (HS) Bit 2: Data Integrity Flag (DIF) Bit 3: Signal Integrity Flag (SIF) Bit 4: Accuracy Integrity Flag (AIF) Bits 5 to 8: Signal In Space Monitoring Accuracy Index (SISMAI)	Hex Ulong	4	H+176
27	Satellite Type	2 bit 轨道类型 2 bit orbit type: 01 indicates GEO, 10 indicates IGSO, 11 indicates MEO	Ulong	4	H+180
28	NDot	平均角速度改正值变化率， rad/s/s Rate of Mean motion difference from computed value (radians/s/s)	Double	8	H+184
29	Reserved		Ulong	4	H+192

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
30	XXXX	32 位CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H+196
31	[CR][LF]	语句结束符（仅ASCII） Sentence Terminator (ASCII only)	—	—	—

3.2.11. BD3IONO 北斗Klobuchar 电离层延迟模型

适用产品：M720、M722

该日志包含北斗 B1I 信号传输的 Klobuchar 电离层模型参数。

- Message ID: 3016
- Log Type: Async

输出示例：

```
#BD3IONOA,60.5,0,0,COM1,FINESTEERING,2383,285434000,013cc608,00008000;2.6
7500000000000000E+01,1.3750000000000000E+00,1.0125000000000000E+01,8.50000000
00000000E+00,-1.1000000000000000E+01,-1.2500000000000000E-01,7.500000000000
00E-01,2.2500000000000000E+00,8.7500000000000000E-01,0.0000000000000000E+00*
6aa638cf
```

表3-35 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BD3IONO	报文头。 Log header. For information about the log headers, see ASCII, or Binary.	H		0
2	A1	电离层延迟改正模型参数 1	Double	8	H
3	A2	电离层延迟改正模型参数 2	Double	8	H+8
4	A3	电离层延迟改正模型参数 3	Double	8	H+16
5	A4	电离层延迟改正模型参数 4	Double	8	H+24
6	A5	电离层延迟改正模型参数 5	Double	8	H+32

Field	Field type	Description	Format	Binary Bytes	Binary Offset
7	A6	电离层延迟改正模型参数 6	Double	8	H+40
8	A7	电离层延迟改正模型参数 7	Double	8	H+48
9	A8	电离层延迟改正模型参数 8	Double	8	H+56
10	A9	电离层延迟改正模型参数 9	Double	8	H+64
11	reserved	保留	Double	8	H+72
12	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	H + 80	
13	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.12. BESTDOPS 最佳 DOP

适用产品：M720、M722

BESTDOPS 输出最佳 DOP。

精度衰减因子由接收机当前跟踪并在位置解中使用的卫星的几何分布算出。本指令每 60 秒，或者当卫星星座改变时更新一次。因此本指令输出的数据字段总数可变，并取决于被跟踪的卫星数。

- **Message ID: 3029**
- **Log Type: Synch**

输出示例：

```
#BESTDOPSA,39.0,0,0,COM2,FINESTEERING,2376,96887000,d323b507,00000000;0.6
300,0.7189,0.3746,0.3462,0.7208,0.0,61,J196,C1,C59,C3,C60,C5,J194,J195,J1
99,C2,C4,C12,G19,C43,E11,G6,C33,G11,G21,G20,C56,G5,G13,C41,C7,C40,C10,G15
,E25,R24,G29,C21,C42,R1,E33,C38,C34,R23,C8,C13,E12,C11,E24,G9,E10,C50,E19
,G12,E16,C14,C49,R2,G25,R17,E4,E18,C28,E6,R3,G30,G7*dccea1c8
```

表3-36 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
-------	------------	-------------	--------	--------------	---------------

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTDOPS header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	pdop	位置精度因子 - 假设3-D 位置未知, 接收机钟差已知 Position dilution of precision - assumes 3D position is unknown and receiver clock offset is known	Float	4	H
3	gdop	几何精度因子 - 假设 3-D 位置和接收机钟差 (全部的4 个参数) 未知 Geometric dilution of precision - assumes 3D position and receiver clock offset (all 4 parameters) are unknown	Float	4	H+4
4	hdop	水平精度因子 Horizontal dilution of precision.	Float	4	H+8
5	tdop	时间精度因子 - 假设3-D 位置已知, 接收机钟差未知 GPS time dilution of precision - assumes 3D position is known and only the GPS receiver clock offset is unknown	Float	4	H+12
6	htdop	水平位置和时间精度因子 Horizontal position and time dilution of precision.	Float	4	H+16
7	elev mask	截止高度角, 单位: 度。 GPS elevation mask angle	Float	4	H+20

Field	Field type	Description	Format	Binary Bytes	Binary Offset
8	#sats	DOP 值计算用到的卫星总数 Number of satellites sats to follow	Ulong	4	H+24
9	sats	计算使用的卫星 ID Satellites in use at time of calculation	Ulong	4	H+28
10	Next satellite offset = H+28+(#sats x 4)				
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+28+(#sats x 4)
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.13. BESTECEF ECEF 坐标下的最佳位置和速度信息

适用产品：M720、M722

BESTECEF 输出 ECEF 坐标下的最佳位置和速度信息。

本指令包含接收机计算出的地心空间直角坐标系下最佳可用位置和速度信息。位置和速度的“status”字段表明了对应数据是否有效。

- **Message ID: 3028**
- **Log Type: Synch**

输出示例：

```
#BESTECEFA,40.0,0,0,COM2,FINESTEERING,2376,97166000,d323b507,00000000;SOL
_COMPUTED,PPP,0,-2828137.8635,4673843.3271,3281080.8292,0.0160,0.0239,0.0
148,SOL_COMPUTED,PPP,0,-0.0045,-0.0020,0.0039,0.0131,0.0153,0.0140,1.000,
11.000,1.000,63,55,50,51,3f20,37,17,07,00,400*858ec430
```

表3-37 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTECEF header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset
2	pos sol status	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位状态, Position Type Position type, see Position Type	Enum	1	H+1
4	Reserved		Ushort	2	H+2
5	P-X	X 轴坐标, 单位: 米。 Position X-coordinate (m)	Double	8	H+4
6	P-Y	Y 轴坐标, 单位: 米。 Position Y-coordinate (m)	Double	8	H+12
7	P-Z	Z 轴坐标, 单位: 米。 Position Z-coordinate (m)	Double	8	H+20
8	P-X σ	X 轴坐标标准差, 单位: 米。 Standard deviation of P-X (m)	Float	4	H+28
9	P-Y σ	Y 轴坐标标准差, 单位: 米。 Standard deviation of P-Y (m)	Float	4	H+32
10	P-Z σ	Z 轴坐标标准差, 单位: 米。 Standard deviation of P-Z (m)	Float	4	H+36
11	vel sol status	解算状态, Solution Status Solution status, see Solution Status	Enum	1	H+40
12	vel type	速度类型, 见 Position or Velocity Type Velocity type, see Position or Velocity Type	Enum	1	H+41
13	Reserved		Ushort	2	H+42

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset
14	V-X	X 轴速度，单位：米/秒。 Velocity vector along X-axis (m/s)	Double	8	H+44
15	V-Y	Y 轴速度，单位：米/秒。 Velocity vector along Y-axis (m/s)	Double	8	H+52
16	V-Z	Z 轴速度，单位：米/秒。 Velocity vector along Z-axis (m/s)	Double	8	H+60
17	V-X σ	X 轴速度标准差，单位：米/秒。 Standard deviation of V-X (m/s)	Float	4	H+68
18	V-Y σ	Y 轴速度标准差，单位：米/秒。 Standard deviation of V-Y (m/s)	Float	4	H+72
19	V-Z σ	Z 轴速度标准差，单位：米/秒。 Standard deviation of V-Z (m/s)	Float	4	H+76
20	V-latency	根据速度时标计算的延迟值，以秒为单位。用历元时间减去延迟可得到更准确的速度结果。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results	Float	4	H+80
21	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+84
22	sol_age	解算时间，单位：秒。 Solution age in seconds	Float	4	H+88
23	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+92

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset
24	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+93
25	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+94
26	#ggL1	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+95
27	ext sol stat	扩展解算状态 Extended Solution Status, see Extended Solution Status	Hex Ushort	2	H+96
28	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+98
29	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+99
30	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+100
31	Reserved	保留 Reserved	Hex	1	H+101
32	stn id	基准站 ID Base station ID	Ushort	2	H+102
33	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+104
34	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.14. BESTP 最佳位置信息

适用产品：M720、M722

BESTP 输出最佳位置信息。此外, BESTP 报告了一些状态指标, 包括差分龄期等。

该条报文包含了解算状态、定位类型、纬度、经度、海拔高、高程异常值、参考坐标系、纬度标准差、经度标准差、高程标准差、差分龄期、解算时间、跟踪到的卫星数、参与解算的卫星数、参与解算的 L1/E1/B1 卫星数、参与解算的多频信号卫星数、参与解算的信号、基站 ID 等信息。

- **Message ID: 3020**
- **Log Type: Synch**

输出示例:

```
#BESTPA,38.0,0,0,COM2,FINESTEERING,2376,96962000,d323b507,00000000;SOL_COMPUTED,PPP,WGS84,0,31.15969059779,121.17808997152,36.0953,10.0212,0.0139,0.0143,0.0311,12.000,1.000,61,55,50,51,3f20,37,17,07,00,400*04ad884c
```

表3-38 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	pos sol stat	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型, 见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度, 单位: 度。 Latitude (degrees)	Double	8	H+4

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset
7	lon	经度，单位：度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高，单位：米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值，单位：米。 Undulation - the relationship between the geoid and the ellipsoid (m) of the chosen datum When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差，单位：米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差，单位：米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差，单位：米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间，单位：秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

补充:

差分龄期为 0 表示未使用差分校正。

不同的定位模式具有不同的最大记录速率，这些速率也由模型选项控制，最大速率: RTK 为 20Hz, PPP 为 20Hz，SPP 定位为 100Hz。

表3-39 Solution Status

Binary	ASCII	Description
0	SOL_COMPUTED	已解出 Solution computed
1	INSUFFICIENT_OBS	观测数据不足 Insufficient observations
2	NO_CONVERGENCE	无法收敛 No convergence
3	SINGULARITY	参数矩阵奇异 Singularity at parameters matrix
4	COV_TRACE	协方差矩阵的迹超过最大值（迹>1000 米） Covariance trace exceeds maximum (trace > 1000 m)
5	VARIANCE	方差超过限制 Variance exceeds limits
6	PENDING	当输入 SETFIXPOS 指令时，接收机会计算自己的位置并确定固定位置是否有效 When a SETFIXPOS position command is entered, the receiver computes its own position and determines if the fixed position is valid
7	INVALID_FIX	使用 SETFIXPOS 指令输入的固定位置无效 The fixed position, entered using the SETFIXPOS position command, is not valid
8	RESIDUALS	残差过大 Residuals are too large
9	INTEGRITY_WARNING	残差过大导致位置不可靠 Large residuals make position unreliable
10~20	Reserved	

表3-40 Position or Velocity Type

Binary	ASCII	Description
0	NONE	无解 No solution
1	SINGLE	单点定位 Solution calculated using only data supplied by the GNSS satellites
2	SPPDIFF	伪距差分 Solution calculated using pseudorange differential(DGPS, DGNSS) corrections
3	Reserved	
4	NARROW_INT	窄巷固定解 Multi-frequency RTK solution with carrier phase ambiguities resolved to narrow-lane integers
5	NARROW_FLOAT	窄巷浮点解 Multi-frequency RTK solution with unresolved, float carrier phase ambiguities
6	Reserved	
7	FIXEDPOS	位置由SETFIXPOS 指令指定 Position has been fixed by the SETFIXPOS position command or by position averaging.
8	Reserved	
9	SBAS	SBAS 定位 Solution calculated using corrections from an SBAS satellite
10	PPP	收敛的 B2b 、 HAS 、 MADOCA PPP 定位解 Converged B2b or HAS or MADOCA PPP solution
11	PPP_FIXED	收敛的 PointSky 、 CLAS PPP 固定解 Converged PointSky or CLAS PPP solution
12	PPP_CONVERGING	收敛中的 PPP 定位解 Converging PPP

Binary	ASCII	Description
		solution
13	WIDE_INT	宽巷固定解 Multi-frequency RTK solution with carrier phase ambiguities resolved to wide-lane integers
14	L1_INT	L1 固定解 Single-frequency RTK solution with carrier phase ambiguities resolved to integers
15	L1_FLOAT	L1 浮点解 Single-frequency RTK solution with unresolved, float carrier phase ambiguities
16	DOPPLER_VELOCITY	速度由即时多普勒信息导出 Velocity computed using instantaneous Doppler
17-19	Reserved	
20	INS	纯惯导定位解 INS positioning solution
21	INS_SBAS	惯导与SBAS 定位组合解 INS position, where the last applied position update used a GNSS solution computed using corrections from an SBAS solution
22	INS_SPP	惯导与单点定位组合解 INS position, where the last applied position update used a single point GNSS(SINGLE) solution

Binary	ASCII	Description
23	INS_SPPDIFF	惯导与伪距差分定位组合解 INS position, where the last applied position update used a pseudorange differential GNSS (SPPDIFF) solution
24	INS_RTKFLOAT	惯导与载波相位差分浮点解的组合解 INS position, where the last applied position update used a floating ambiguity RTK (L1_FLOAT or NARROW_FLOAT) solution
25	INS_RTKFIXED	惯导与载波相位差分固定解的组合解 INS position, where the last applied position update used a fixed integer ambiguity RTK (L1_INT, WIDE_INT or NARROW_INT) solution
26	INS_PPP	惯导与 B2b 、 HAS 、 MADOCA PPP 定位解的组合解 INS position, where the last applied position update used B2b or HAS or MADOCA ppp converged solution
27	INS_PPPFIXED	惯导与 PointSky 、 CLAS PPP 固定解的组合解 INS position, where the last applied position update used PointSky or CLAS ppp converged solution

表3-41 Extended Solution Status

Nibble	Bit	Mask	Description	Value
NO	0	0x0001	伪距电离层改正	0 = 未知 1 = Klobuchar 广播星历改正 2 = SBAS 电离层格网改正

Nibble	Bit	Mask	Description	Value
	1	0x0002		3 = 多频改正 4 = 伪距差分改正 Pseudorange Ionospheric Correction 0 = Unknown or default Klobuchar model 1 = Klobuchar Broadcast 2 = SBAS Broadcast 3 = Multi-frequency Computed 4 = SPPDiff Correction
	2	0x0004		
	3	0x0008	云端同源使用标志	1: 使用, 0: 未使用
N1	4	0x0010	云端同源使用版本	0 = 2.0 版本
	5	0x0020		1 = 3.0 版本
	6	0x0040		2 = 4.0 版本
	7	0x0080		
N2	8	0x0100	云端同源龄期	0-61s
	9	0x0200		
	10	0x0400		
	11	0x0800		
N3	12	0x1000	双天线基线长异常	0: 正常 1: 异常
	13	0x2000		
	14	0x4000		
	15	0x8000	Reserved	

表3-42 BeiDou Signal - Used Mask

Bit	Mask	Description
0	0x01	BeiDou B1 used in Solution (B1I)
1	0x02	BeiDou B2 used in Solution (B2I)
2	0x04	BeiDou B3 used in Solution (B3I)

3	0x08	BeiDou B1C used in Solution
4	0x10	Beidou B2a used in Solution
5	0x20	Beidou B2b used in Solution
6-7	0xC0	Reserved

表3-43 Galileo Signal - Used Mask

Bit	Mask	Description
0	0x01	Galileo E1 used in Solution
1	0x02	Galileo E5a used in Solution
2	0x04	Galileo E5b used in Solution
3	0x08	Galileo ALTBOC used in Solution
4	0x10	Galileo E6 used in Solution (E6B and E6C)
5-7	0xE0	Reserved

表3-44 GPS and GLONASS Signal - Used Mask

Bit	Mask	Description
0	0x01	使用GPS L1 解算 GPS L1 used in Solution
1	0x02	使用GPS L2 解算 GPS L2 used in Solution
2	0x04	使用GPS L5 解算 GPS L5 used in Solution
3	0x08	Reserved
4	0x10	使用GLONASS L1 解算 GLONASS L1 used in Solution
5	0x20	使用GLONASS L2 解算 GLONASS L2 used in Solution
6	0x40	使用GLONASS L3 解算 GLONASS L3 used in Solution
7	0x80	保留 Reserved

3.2.15. BESTSATINFO 参与定位的卫星信息

适用产品：M720、M722

BESTSATINFO 输出参与 BESTP 解算的已使用和未使用卫星信息，它还描述了所用卫星被使用或未被使用的原因。

- Message ID: 3031
- Log Type: Synch

输出示例：

```
#BESTSATINFOA,39.0,0,0,COM2,FINESTEERING,2376,97023000,d323b507,00000000;
63,J,196,00000007,GOOD,C,1,00000007,GOOD,C,59,00000005,GOOD,C,3,00000007,
GOOD,C,60,00000005,GOOD,C,5,00000007,GOOD,J,194,00000007,GOOD,J,195,00000
007,GOOD,J,199,00000007,GOOD,C,2,00000007,GOOD,C,4,00000007,GOOD,C,12,000
0007,GOOD,G,19,00000003,GOOD,C,43,00000015,GOOD,E,11,00000007,GOOD,G,6,0
0000007,GOOD,C,33,00000015,GOOD,G,11,00000007,GOOD,G,21,00000007,GOOD,G,2
0,00000003,GOOD,C,56,00000000,NOEPHEMERIS,G,5,00000003,GOOD,G,13,00000003
,GOOD,C,41,00000000,NOEPHEMERIS,C,7,00000007,GOOD,C,40,00000015,GOOD,C,10
,00000007,GOOD,G,15,00000003,GOOD,E,25,00000007,GOOD,R,24+3,00000003,GOOD
,G,29,00000003,GOOD,C,21,00000015,GOOD,C,42,00000015,GOOD,R,1-7,00000001,
GOOD,E,33,00000007,GOOD,C,38,00000015,GOOD,C,34,00000015,GOOD,R,23-3,0000
0000,ELEVATIONERROR,C,8,00000007,GOOD,C,13,00000007,GOOD,E,12,00000007,GO
OD,C,11,00000007,GOOD,E,24,00000007,GOOD,G,9,00000007,GOOD,E,10,00000007,
GOOD,C,50,00000000,NOEPHEMERIS,E,19,00000007,GOOD,G,12,00000003,GOOD,E,16
,00000007,GOOD,C,14,00000007,GOOD,C,49,00000000,NOEPHEMERIS,R,2+1,0000000
3,GOOD,G,25,00000007,GOOD,R,17-1,00000003,GOOD,E,4,00000007,GOOD,E,18,000
00000,NOEPHEMERIS,C,28,00000015,GOOD,E,6,00000007,GOOD,R,3-4,00000003,GOO
D,G,30,00000007,GOOD,G,7,00000003,GOOD,C,39,00000000,ELEVATIONERROR,R,18+
4,00000000,ELEVATIONERROR*b6038918
```

表3-45 报文格式说明

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTSATINF header	报文头。 Log header. For information about the log headers, see ASCII, or Binary		H	0
2	#entries	Number of records to follow	Ulong	4	H
3	System	GNSS 卫星系统列表。 see Satellite System  卫星系统使用与 GNSSOBS 报文一致的标识符定义	Enum	4	H+4
		卫星 PRN 号 • GPS: 1 到32 , • GLONASS: 1 到 24, • Galileo 1 到 36, • BDS 1 到63, • SBAS 120 到 158 及 183 到 192, • QZSS 193 到202 在二进制消息中, 卫星 ID 字段由 1 个 Short 和 1 个 Ushort 组成。 • 最低的两个字节是系统标识符: (如 GPS 的 PRN , GLONASS 的通道号), 为 USHORT 类型; • 最高的两个字节是 GLONASS 的频率通道, 其他系统这两个字节值为零。 在 ASCII 消息中, 卫星 ID 字段是系统标识符。 如果系统是 GLONASS, 而频率通道不是零, 则 在系统标识符后加上频率通道数。例如, 系统 标识符是 13, 频率通道-2, 输出			

Field	Field type	Description	Format	Binary Bytes	Binary Offset
4	Satellite ID	为 13-2, 数据 类型为 Short 类型。 In binary logs, the satellite ID field is 4 bytes. The 2 lowest-order bytes, interpreted as a USHORT, are the system identifier: for instance, the PRN for GPS, Galileo, BeiDou, NavIC and QZSS, or the slot for GLONASS. The 2 highest-order bytes are the frequency channel for GLONASS, interpreted as a SHORT and zero for all other systems. In ASCII and abbreviated ASCII logs, the satellite ID field is the system identifier. If the system is GLONASS and the frequency channel is not zero, then the signed channel is appended to the sys- tem identifier. For example, slot 13, frequency channel -2 is output as 13-2.	Ulong	4	H+8
5	Signal mask	See GPS Signal Mask , GLONASS Signal Mask , Galileo Signal Mask , BeiDou Signal Mask , QZSS Signal Mask and NavIC Signal Mask	Hex	4	H+12
6	Status	在二进制消息中, 数值是 “0” ; 在 ASCII 消息 中为: “GOOD” Satellite status. See Observation Statuses	Ulong	4	H+16
7	Next satellite offset = H + 4 + (#entries x 16)				
8	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+4 (#en-tries x 16)
9	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-46 Observation Statuses

Value	Name	Description
0	GOOD	观测数据正常 Observation is good
1	BADHEALTH	卫星在星历表或历书中被标记为健康状况不佳 Satellite is flagged as bad health in ephemeris or almanac
2	OLDEPHEMERIS	星历时间大于 3 小时 Ephemeris >3 hours old
6	ELEVATIONERROR	卫星位于高度截止角以下 Satellite was below the elevation cutoff
7	MISCLOSURE	观察值与预测值相差大 Observation was too far from predicted value
8	NODIFFCORR	无差分校正可用 No differential correction available
9	NOEPHEMERIS	无有效星历 No ephemeris available
10	INVALIDIODE	使用的 IODE 无效 IODE used is invalid
11	LOCKEDOUT	卫星已被锁定 Satellite has been locked out
12	LOWPOWER	卫星信号功率低 Satellite has low signal power
13	OBSL2	解算中未直接使用的 L2 观测值 An L2 observation not directly used in the solution
15	UNKNOWN	未使用观察结果，因为其类型未知 Observation was not used because it was of an unknown type
16	NOIONOCORR	电离层延迟校正不可用 No ionosphere delay correction was available

Value	Name	Description
17	NOTUSED	解算中未使用的观测数据 Observation was not used in the solution
18	OBSL1	解算中未使用 L1 观测值 An L1 observation not directly used in the solution
19	OBSE1	解算中未使用 E1 观测值 An E1 observation not directly used in the solution
20	OBSL5	解算中未使用 L5 观测值 An L5 observation not directly used in the solution
21	OBSE5	解算中未使用 E5 观测值 An E5 observation not directly used in the solution
22	OBSB2	解算中未使用 B2 观测值 A B2 observation not directly used in the solution
23	OBSB1	解算中未使用 B1 观测值 A B1 observation not directly used in the solution
24	OBSB3	解算中未使用 B3 观测值 A B3 observation not directly used in the solution
25	NOSIGNALMATCH	信号类型不匹配 Signal type does not match
26	SUPPLEMENTARY	观测数据在解算中提供补偿 Observation contributes supplemental information to the solution
99	NA	无有效观测数据 No observation available
100	BAD_INTEGRITY	观察数据异常，在解算中排除 Observation was an outlier and was eliminated from the solution
101	LOSSOFLOCK	此信号失锁 Lock was broken on this signal

Value	Name	Description
102	NOAMBIGUITY	未匹配 RTK 模糊类型 No RTK ambiguity type resolved

3.2.16. BESTV 最佳可用速度

适用产品：M720、M722

本指令包含接收机计算出的最佳可用速度信息。此外，接收机还报告了速度状态指示符，对指示相应的数据是否有效非常有用。该速度测量有时会带来相关的延迟。

- **Message ID: 3021**
- **Log Type: Synch**

输出示例：

```
#BESTVA,40.5,0,0,COM2,FINESTEERING,2376,97125000,d323b507,00000000;SOL_COMPUTED,DOPPLER_VELOCITY,0,0.0101,184.850358,-0.0009,0.000,10.000,0.000*e372eb64
```

表3-47 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTV header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	vel sol stat	解算状态，见 Vel Solution Status Solution status, see Vel Solution Status	Enum	1	H+0
3	vel type	速度类型，见 Velocity Type Velocity type, see Velocity Type	Enum	1	H+1
4	reserved		Ushort	2	H+2
5	hor spd	对地水平速度，单位：米/秒。 Horizontal speed over ground, in metres per second	Double	8	H+4

Field	Field type	Description	Format	Binary Bytes	Binary Offset
6	trk gnd	相对于真北的实际对地运动方向（相对地面轨迹），单位：度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Double	8	H+12
7	vert spd	垂直速度，正值表示高度增加（向上），负值表示高度下降（向下），单位：米/秒。 Vertical speed, in metres per second, where positive values indicate increasing altitude (up) and negative values indicate decreasing altitude (down)	Double	8	H+20
8	latency	根据速度时标计算的延迟值，用历元时间减去延迟可得到更准确的速度结果，单位：秒。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results (s)	Float	4	H+28
9	age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+32
10	Reserved		Float	4	H+36
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.17. CPUUSAGE CPU 使用信息

适用产品：M720、M722

- Message ID: 23901
- Log Type: ASync

输出示例：

```
#CPUUSAGEA,65.5,0,0,COM2,FINESTEERING,2381,20977000,cba257c1,00008000;1,0,3432,5830,13544532,2937,3823,78453024,174388283*abdc7388
```

表3-48 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	CPUUSAGE	报文头。 Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.		H	0
2	# cpu	CPU数量	Ulong	4	H
3	CPU index	CPU编号	Ulong	4	H+4
4	CPU total usage	CPU总使用率*10000 (0~10000)	unsigned short	2	H+6
5	CPU max usage	CPU最大总使用率*10000 (0~10000)	unsigned short	2	H+8
	CPU smooth time	CPU线程平滑使用时间（单位为CPU时钟周期）	Ulong	4	H+12
6	lstr total usage	中断总使用率*10000 (0~10000)	unsigned short	2	H+14
7	lstr max usage	中断最大使用率*10000 (0~10000)	unsigned short	2	H+16
8	lstr smooth time	中断平滑使用时间（单位为CPU时钟周期）	Ulong	4	H+20
9	ldel smooth time	空闲线程平滑使用时间（单位为CPU时钟周期）	Ulong	4	H+24
Next CPU offset = H + 4 + (#cpu x 24)					
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H + 4 + (#cpu x 24)
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.18. DANTHEADING 双天线航向信息

适用产品：M720、M722

DANTHEADING 双天线 Heading 信息，Heading 是主天线的真北到顺时针方向从天线矢量的角度。

- Message ID: 3051
- Log Type: Synch

输出示例：

```
#DANTHEADINGA,38.5,0,0,COM2,FINESTEERING,2376,98061000,d323b507,00000000;  
SOL_COMPUTED,NARROW_INT,0,0.001143,126.363876,39.962364,0.0,0.000000,0.00  
0000,63,56,56,51,3f20,08,37,17,07,0*a57e6af2
```

表3-49 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	DANTHEADING header	报文头。 Log header. For information about the log headers, see ASCII , or Binary .		H	0
2	sol stat	解算状态，见 Solution Status Solution status, see Solution Status	Enum	1	H+0
3	pos type	定位状态，见 Position or Velocity Type Position type, see Position or Velocity Type	Enum	1	H+1
4	Reserved		Ushort	2	H+2
5	length	Baseline 长度，单位：米。 Baseline length in metres For Heading models, this field is -1. For Relative Positioning models with a fixed position, this field is -1. \\For Relative Positioning models, this field	Float	4	H+4

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		is the baseline length in metres, unless the position is fixed.			
6	heading	航向, 单位: 度。 Heading in degrees (0° to 359.999°)	Float	4	H+8
7	pitch	pitch, 单位: 度。 Pitch (±90 degrees)	Float	4	H+12
8	Reserved		Float	4	H+16
9	hdg std dev	航向标准差, 单位: 度。 Heading standard deviation in degrees	Float	4	H+20
10	ptch std dev	俯仰标准差, 单位: 度。 Pitch standard deviation in degrees	Float	4	H+24
11	#SVs	跟踪卫星数 Number of satellites tracked	Uchar	1	H+28
12	#solnSVs	参与解算卫星数 Number of satellites used in solution	Uchar	1	H+29
13	#obs	高度截止角以上的卫星数 Number of satellites above the elevation mask angle	Uchar	1	H+30
14	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+31
15	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+32
16	sol source	解算源 Solution source	Hex	1	H+34
17	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志	Hex	1	H+35

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		GPS and GLONASS signals used mask			
18	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+36
19	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+37
20	stn id	基准站 ID Base station ID	Ushort	2	H+38
21	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
22	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表 3-50 Solution Source

Bit	Mask	Description
0-1	0x03	保留 Reserved
2-3	0x0C	天线源 0 = 主天线 1 = 副天线 Source antenna 0=Primary antenna 1=Secondary antenna
4-7	0xF0	保留 Reserved

3.2.19. ENVSTATUS 环境感知

适用产品：M720、M722

ENVSTATUS 输出设备所处环境感知信息，包括环境质量、卫星可视比例、基站位移状态、跟踪卫星数、电离层活跃状态等信息。

- **Message ID: 3074**
- **Log Type:Synch**

输出示例：

```
#ENVSTATUSA,66.0,0,0,COM2,FINESTEERING,2381,21104000,cba257c1,00008000;38
,SINGLE,84,87,20.00,0,0,63,30,0,0,0,0.000,0.7,0.000,0.000,0.000,0.000*0
2dcb158
```

表3-51 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	ENVSTATUS header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	env score	环境质量分值： • 0~59：环境质量不合格 • 60~84：一般可用 • 85~100：优秀 Environmental quality score: • 0~59: The quality of the base station is not qualified • 60-84: Generally available • 85~100: Excellent	Uchar	1	H+0
3	pos type	定位类型，见 Position Type Position type, see Position Type	Enum	1	H+1
4	sat vis rate	卫星可视比率,0~100（适用于移动站模式），计算方式：跟踪卫星数/理论可视卫星数（理论可视卫星数排除低于仰角的可视卫星数）	Uchar	1	H+2

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Satellite visibility ratio, 0~100 (applicable to rover mode)			
5	sat sol rate	参与解算的卫星比率,0~100（适用于移动站模式），计算方式：参与解算卫星数/跟踪卫星数。 Solution satellite ratio, 0~100 (applicable to rover mode)	Uchar	1	H+3
6	sol confidence	解的置信度（适用于移动站模式） Confidence level of solution (applicable to rover mode)	Float	4	H+4
7	moved	基站配置后是否发生过位移 0：没有被移动过 1：被移动过 2：无法判断是否发生位移(当前定位结果与 fix pos 偏差过大) Has there been any displacement after configuring the base station 0: It has not been moved. 1: It has been moved. 2: It cannot be determined whether displacement has occurred	Uchar	1	H+8
8	reserved		Uchar	1	H+9
9	#SVs	跟踪卫星数 Number of track satellite	Uchar	1	H+10
10	cv sats	基站和 Rover 的共视卫星数（适用于移动站模式）	Uchar	1	H+11

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Number of co view satellites between the base station and rover (applicable to rover mode)			
11	IONO level	电离层活跃状态见 IONO level Ionospheric activity status, see IONO level	Enum	1	H+12
12	avg SNR	平均 SNR ≥38dBHz: 开阔地 ≥32dBHz && ≤37dBHz: 半遮挡 ≤31dBHz: 严重遮挡 Average SNR ≥38dBHz: Open ground ≥32dBHz & ≤37dBHz: Semi occluded ≤31dBHz: Severely occluded	Uchar	1	H+13
13	reserved		Uchar	1	H+14
14	reserved		Uchar	1	H+15
15	diff pos	当前位置与 fixed pos 的差值 (适用于基站模式) The difference between the current location and the fixed POS (applicable to base station mode)	Float	4	H+16
16	pdop	位置精度因子 ≤2.5: 卫星分布较好 >2.5&& ≤5.0: 卫星分布一般 >5.0: 卫星分布较差	Float	4	H+20

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Position dilution of precision ≤ 2.5: good satellite distribution >2.5&≤ 5.0: average satellite distribution >5.0: poor satellite distribution			
17	reserved		Float	4	H+24
18	reserved		Float	4	H+28
19	reserved		Float	4	H+32
20	reserved		Float	4	H+36
21	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
22	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-52 IONO level

Binary	ASCII	Description
0	QUIET	平静 Quiet
1	WEAK	轻微 Weak
2	MODERATE	中等 Moderate
3	STRONG	强烈 Strong
4	EXTREME	超强 Extreme
255	UNKNOWN	未知 Unknown

3.2.20. EVENT1P/EVENT2P/EVENT3P/EVENT4P

适用产品：M720、M722

该报文包含检测到的标记输入脉冲的前沿时间。

- EVENT1P 当 MK1I (EVENT_IN1) 输入端出现脉冲时生成
- EVENT2P 当 MK2I (EVENT_IN2) 输入端出现脉冲时生成
- EVENT3P 当 MK3I (EVENT_IN3) 输入端出现脉冲时生成
- EVENT4P 当 MK4I (EVENT_IN4) 输入端出现脉冲时生成
- Message ID:
 - 3035 (EVENT1P)
 - 3036 (EVENT2P)
 - 3037 (EVENT3P)
 - 3039 (EVENT4P)
- Log Type: Asynch

- EVENTxT 或 EVENTxP 报文应使用 ONNEW
1. 只有 EVENTxP, EVENTxT 和“polled” 报文会在标记的准确时间“on the fly” 生成

2. ONMARK 仅适用于 MK1I, 使用 ONMARK 时, MK2I (如有) 上的事件不会触 发 报文。EVENT1T 、 EVENT2T 、 EVENT1P 或 EVENT2P 报文应使用 ONNEW。



3. 对于 EVENTxT 和 EVENTxP 报文, 一旦 1PPS 信号达到上升沿, 测量值的分辨 率均为 10 ns。对于其他使用 ONMARK 的测量延迟的报文 (如BESTP 等), 报 文 从 1PPS 信号输出信息一般需要20-30 ms (最长 50 ms)。延迟是指从接收 到 1PPS 脉冲到相关报文的第一个字节之间的时间。

 通过这些报文, 可以测量其他设备发生事件的位置

输出示例:

```
#EVENT1PA,31.0,0,0,COM2,FINE,2376,98440285,d323b507,00000000;SOL_COMPUTED,NARROW_INT,WGS84,0,31.15969320452,121.17808121391,36.0993,10.0212,0.0039,0.0034,0.0099,1.000,0.000,60,53,48,49,0928,37,17,07,00,654*054a66d7
```

表3-53 报文格式说明

Field	Field type	Description	Format	Binary	Binary
				Bytes	Offset

1	Log header	报文头。 EVENT1P /EVENT2P / EVENT3P/ EVENT4P header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态, 见 Solution Status Solution status, see Solution Status	Enum	1	H+0
3	pos type	定位状态, 见 Position or Velocity Type Position type, see Position or Velocity Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度, 单位: 度。 Latitude (degrees)	Double	8	H+4
7	lon	经度, 单位: 度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the ellipsoid (m) of the chosen datum When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32

11	lon σ	经度标准差，单位：米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差，单位：米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间，单位：秒 。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61

24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.21. EVENT1T/EVENT2T/EVENT3T/EVENT4T

适用产品：M720、M722

该报文包含检测到的标记输入脉冲的前沿时间。

- EVENT1T 当 MK1I (EVENT_IN1) 输入端出现脉冲时生成
- EVENT2T 当 MK2I (EVENT_IN2) 输入端出现脉冲时生成
- EVENT3T 当 MK3I (EVENT_IN3) 输入端出现脉冲时生成
- EVENT4T 当 MK4I (EVENT_IN4) 输入端出现脉冲时生成
- Message ID:
 - 3040 (EVENT1T)
 - 3041 (EVENT2T)
 - 3044 (EVENT3T)
 - 3045 (EVENT4T)

- Log Type: Asynch

输出示例：

```
#EVENT1TA,31.0,0,0,COM2,FINE,2376,98440285,d323b507,00000000;VALID,0,2376,98440.284608465,2.329200160E-04,0.000000000E+00,18.000000001*3bae4f51
```

表3-54 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	Log header	EVENT1T / EVENT2T / EVENT3T /EVENT4T 报文头。 EVENT1T / EVENT2T / EVENT3T / EVENT4T header Log header. For information about the log headers, see ASCII , or Binary .		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
2	status	时钟模型状态。 Clock model status, see Clock Model Status	Enum	1	H+0
3	Reserved		Uchar	1	H+1
4	week	GPS 参考周计数。 GPS reference week number	Ushort	2	H+2
5	seconds	接收机时钟测量的一周内的秒数。 Seconds into the week as measured from the receiver clock, coincident with the time of electrical closure on the Mark Input port	Double	8	H+4
6	offset	接收机钟差，单位：秒。 Receiver clock offset, in seconds. A positive offset implies that the receiver clock is ahead of GPS system time. To derive GPS system time, use the following formula: GPS system time = GPS reference time - (offset) Where GPS reference time can be obtained from the log header	Double	8	H+12
7	offset std	接收机钟差标准差。 Standard deviation of receiver clock offset (s)	Double	8	H+20
8	utc offset	该字段表示 GPS 系统时间和 UTC 时间的偏差，使用历书参数计算。UTC 时间是 GPS 基准时间加上当前 UTC 偏移量再减去接收机时钟偏移量。 This field represents the offset of GPS system time from UTC time (s), computed using almanac parameters. UTC time is GPS reference time plus the current UTC offset minus the receiver clock offset. UTC time = GPS reference time - offset + UTC offset 0 indicates that UTC time is unknown because there is	Double	8	H+28

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		no almanac available in order to acquire the UTC offset.			
9	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Ulong	4	H+36
10	[CR][LF]	语句结束符 (仅 ASCII) Sentence terminator (ASCII only)	-	-	-

3.2.22. GALEPHNAVF Galileo FNAV 星历

适用产品: M720、M722

- Message ID: 3011
- Log Type: Asynch

输出示例:

```
#GALEPHNAVF,40.5,14,0,COM2,SATTIME,2376,96193000,d323b507,00000000;4,27,107,0,7.4110E-10,92400,0.0000E+00,-1.010392E-11,-2.910305629E-04,-6.328E+01,3.0566E-09,-2.885784461E-01,-2.7567E-06,2.650833922E-04,1.0401E-05,5.44060909E+03,92400,1.3039E-08,3.00749214E+00,5.0291E-08,9.655307307E-01,1.197E+02,-4.049565753E-01,-5.26736226E-09,-3.027E-09,0,0,0,0*e61346d5
```

表3-55 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	GALEPHNAVF header	报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	SatId	卫星 ID 编号 Satellite identifier	Ulong	4	H

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
3	IODnav	星历数据期号 Issue of data ephemeris	Ushort	2	H+4
4	SISA Index	空间信号精度 Signal in space accuracy (unitless)	Uchar	1	H+6
5	Reserved	保留字段	Uchar	1	H+7
6	IDot	轨道倾角变化率, rad/s Rate of inclination angle (radians/s)	Double	8	H+8
7	T0c	卫星钟差参数, s Clock correction data reference time of week from the F/NAV message (s)	Ulong	4	H+16
8	Af2	卫星钟差参数, s/s/s SV clock drift rate correction coefficient from the F/NAV message (s/s/s)	Double	8	H+20
9	Af1	卫星钟差参数, s/s SV clock drift correction coefficient from the F/NAV message (s/s)	Double	8	H+28
10	Af0	卫星钟差参数, s SV clock bias correction coefficient from the F/NAV message (s)	Double	8	H+36
11	Crs	轨道半径 (正弦振幅, m) Amplitude of the sine harmonic correction term to the orbit radius (m)	Double	8	H+44
12	DeltaN	卫星平均角速度的改正值, rad/s Mean motion difference (radians/s)	Double	8	H+52
13	M0	参考时间的平近点角, rad	Double	8	H+60

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Mean anomaly at ref time (radians)			
14	Cuc	纬度幅角（余弦振幅，rad） Amplitude of the cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+68
15	Ecc	卫星轨道偏心率 Eccentricity (unitless)	Double	8	H+76
16	Cus	纬度幅角（正弦振幅，rad） Amplitude of the sine harmonic correction term to the argument of latitude (radians)	Double	8	H+84
17	RootA	卫星轨道长半轴的平方根 Square root of semi-major axis	Double	8	H+92
18	T0e	星历的参考时间，单位：s Ephemeris reference time (s)	Ulong	4	H+100
19	Cic	倾角（余弦振幅，rad） Amplitude of the cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+104
20	Omega0	升交点赤经，rad Longitude of ascending node of orbital plane at weekly epoch (radians)	Double	8	H+112
21	Cis	倾角（正弦振幅，rad） Amplitude of the sine harmonic correction term to the angle of inclination (radians)	Double	8	H+120

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
22	I0	参考时间轨道倾角, rad Inclination angle at ref time (radians)	Double	8	H+128
23	Crc	轨道半径 (余弦振幅, m) Amplitude of the cosine harmonic correction term to the orbit radius (m)	Double	8	H+136
24	Omega	近地点幅角, rad Argument of perigee (radians)	Double	8	H+144
25	OmegaDot	升交点赤经变化率, rad/s Rate of right ascension (radians/s)	Double	8	H+152
26	E1E5aBGD	E1,E5a 广播群延迟 E1, E5a broadcast group delay	Double	8	H+160
27	E5aHealth	E5a 健康状态 E5a health status bits	Uchar	1	H+168
28	E5aDVS	E5a 数据有效状态 E5a data validity status	Uchar	1	H+169
29	Reserved	保留字段	Uchar	1	H+170
30	Reserved	保留字段	Uchar	1	H+171
31	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H+172
32	[CR][LF]	语句结束符 (仅 ASCII) Sentence terminator (ASCII only)	-	-	-

3.2.23. GALEPHNAVI Galileo INAV 星历

适用产品：M720、M722

- Message ID: 3009
- Log Type: Asynch

输出示例：

```
#GALEPHNAVIA,40.5,14,0,COM2,SATTIME,2376,96151000,d323b507,00000000;4,27,
107,0,7.4110E-10,92400,0.0000E+00,-1.011813E-11,-2.910303883E-04,-6.328E+
01,3.0566E-09,-2.885784461E-01,-2.7567E-06,2.650833922E-04,1.0401E-05,5.4
4060909E+03,92400,1.3039E-08,3.00749214E+00,5.0291E-08,9.655307307E-01,1.
197E+02,-4.049565753E-01,-5.2673623E-09,-3.027E-09,-3.725E-09,0,0,0,0,0,0
,3,0*65ca576c
```

表3-56 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	GALEPHNAVI header	报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	SatId	卫星 ID 编号 Satellite identifier	Ulong	4	H
3	IODnav	星历数据期号 Issue of data ephemeris	Ushort	2	H+4
4	SISA Index	空间信号精度 Signal in space accuracy (unitless)	Uchar	1	H+6
5	Reserved	保留字段	Uchar	1	H+7
6	IDot	轨道倾角变化率, rad/s Rate of inclination angle (radians/s)	Double	8	H+8
7	Toc	卫星钟差参数, s Clock correction data reference time of	Ulong	4	H+16

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		week from the I/NAV message (s)			
8	Af2	卫星钟差参数, s/s/s SV clock drift rate correction coefficient from the I/NAV message (s/s ²)	Double	8	H+20
9	Af1	卫星钟差参数, s/s SV clock drift correction coefficient from the I/NAV message (s/s)	Double	8	H+28
10	Af0	卫星钟差参数, s SV clock bias correction coefficient from the I/NAV message (s)	Double	8	H+36
11	Crs	轨道半径 (正弦振幅, m) Amplitude of the sine harmonic correction term to the orbit radius (m)	Double	8	H+44
12	DeltaN	卫星平均角速度的改正值, rad/s Mean motion difference (radians/s)	Double	8	H+52
13	M0	参考时间的平近点角, rad Mean anomaly at ref time (radians)	Double	8	H+60
14	Cuc	纬度幅角 (余弦振幅, rad) Amplitude of the cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+68
15	Ecc	卫星轨道偏心率 Eccentricity (unitless)	Double	8	H+76
16	Cus	纬度幅角 (正弦振幅, rad) Amplitude of the sine harmonic correction	Double	8	H+84

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		term to the argument of latitude (radians)			
17	RootA	卫星轨道长半轴的平方根 Square root of semi-major axis	Double	8	H+92
18	Toe	星历的参考时间, 单位: s Ephemeris reference time (s)	Ulong	4	H+100
19	Cic	倾角 (余弦振幅, rad) Amplitude of the cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+104
20	Omega0	升交点赤经, rad Longitude of ascending node of orbital plane at weekly epoch (radians)	Double	8	H+112
21	Cis	倾角 (正弦振幅, rad) Amplitude of the sine harmonic correction term to the angle of inclination (radians)	Double	8	H+120
22	IO	参考时间轨道倾角, rad Inclination angle at ref time (radians)	Double	8	H+128
23	Crc	轨道半径 (余弦振幅, m) Amplitude of the cosine harmonic correction term to the orbit radius (m)	Double	8	H+136
24	Omega	近地点幅角, rad Argument of perigee (radians)	Double	8	H+144
25	OmegaDot	升交点赤经变化率, rad/s Rate of right ascension(radians/s)	Double	8	H+152

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
26	E1E5aBGD	E1,E5a 广播群延迟 E1, E5a broadcast group delay	Double	8	H+160
27	E1E5bBGD	E1,E5b 广播群延迟 E1, E5b broadcast group delay	Double	8	H+168
28	E5bHealth	E5b 健康状态 E5b health status bits	Uchar	1	H+176
29	E5bDVS	E5b 数据有效状态 E5b data validity status	Uchar	1	H+177
30	Reserved	保留字段	Uchar	1	H+178
31	Reserved	保留字段	Uchar	1	H+179
32	E1bHealth	E1b 健康状态 E1b health status bits	Uchar	1	H+180
33	E1bDVS	E1b 数据有效状态 E1b data validity status	Uchar	1	H+181
34	INAV Source	来源信号标识 Identifies the source signal: 0 = Unknown 1 = E1b 2 = E5b 3 = E1b and E5b	Uchar	1	H+182
35	Reserved	保留字段	Uchar	1	H+183
36	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H+184
37	[CR][LF]	语句结束符 (仅 ASCII)	-	-	-

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Sentence terminator (ASCII only)			

3.2.24. GLOEPH GLONASS 星历

适用产品：M720、M722

- Message ID: 3006
- Log Type: Asynch

输出示例：

#GLOEPHA,40.5,5,0,COM2,SATTIME,2376,96100000,d323b507,00000000;38,8,29,19800,0,23,6.6945552825927734E+02,-1.1823899902343751E+07,0.000000000000000000,-2.8393735885620117E+03,7.5853779296874997E+06,4.65661287307739258E-06,1.3849868774414063E+03,2.1283201171875001E+07,-0.000000000000000000E+00,9.09494701772928238E-13,3,-1.32960267364978790E-04,9.313225746E-09,0,2,568,0,0,1,0,2376,96318000,10782*6c323e15

表3-57 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	Log header	GLOEPH 报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sloto	Slot 信息偏移量 - PRN 标识(Slot + 37) Slot information offset - PRN identification (Slot + 37). This is also called SLOTO in Connect	Ushort	2	H
3	frequ	卫星的频率信道偏移范围为 0 至 20 Frequency channel offset for satellite in the range 0 to 20	Ushort	2	H+2

Field	Field type	Description	Format	Binary Bytes	Binary Offset
4	Flags	信号标志, 见表 GLONASS Ephemeris Flags Coding Information flags	Ulong	4	H+4
5	Tk	帧开始时间 (自 GLONASS 日开始以来) Time of frame start (since start of GLONASS day) (s)	Ulong	4	H+8
6	health	星历健康状态 Ephemeris health where 0-3 = GOOD 4-15 = BAD	Ulong	4	H+12
7	issue	对应于星历参考时间的 15 分钟间隔数 15 minute interval number corresponding to ephemeris reference time	Ulong	4	H+16
8	vel x	参考时间卫星 x 轴方向速度(PZ-90.02) X coordinate for satellite velocity at reference time (PZ-90.02) (m/s)	Double	8	H+20
9	pos x	参考时间卫星的 x 坐标 (PZ-90.02) X coordinate for satellite at reference time (PZ-90.02) (m)	Double	8	H+28
10	LS acc x	参考时间卫星 x 轴方向引力加速度 X coordinate for lunisolar acceleration at reference time (PZ-90.02), (m/s/s)	Double	8	H+36
11	vel y	参考时间卫星 y 轴方向速度(PZ-90.02) Y coordinate for satellite velocity at reference time (PZ-90.02) (m/s)	Double	8	H+44

Field	Field type	Description	Format	Binary Bytes	Binary Offset
12	pos y	参考时间卫星的 Y 坐标 (PZ-90.02) Y coordinate for satellite at reference time (PZ-90.02) (m)	Double	8	H+52
13	LS acc y	参考时间卫星 Y 轴方向引力加速度 Y coordinate for lunisolar acceleration at reference time (PZ-90.02) (m/s/s)	Double	8	H+60
14	vel z	参考时间卫星 Z 轴方向速度(PZ-90.02) Z coordinate for satellite velocity at reference time (PZ-90.02), (m/s)	Double	8	H+68
15	pos z	参考时间卫星的 Z 坐标 (PZ-90.02) Z coordinate for satellite at reference time (PZ-90.02) (m)	Double	8	H+76
16	LS acc z	参考时间卫星 Z 轴方向引力加速度 Z coordinate for lunisolar acceleration at reference time (PZ-90.02) (m/s/s)	Double	8	H+84
17	gamma	频率校正 Frequency correction (s/s)	Double	8	H+92
18	P	技术参数 Technological parameter	Ulong	4	H+100
19	tau_n	第 n 个卫星时间 t_n 相对于 GLONASS 时间 t_c 的校正 Correction to the nth satellite time t_n relative to GLONASS time t_c (s)	Double	8	H+104
20	delta_tau_n	第 n 颗卫星在 L2 子频段传输的导航 RF 信号与在 L1 子频段传输的导航 RF 信号之间	Double	8	H+112

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		的时间差 Time difference between navigation RF signal transmitted in L2 sub-band and navigation RF signal transmitted in L1 sub-band by nth satellite (s)			
21	age	数据年龄（天） Age of data (days)	Ulong	4	H+120
22	Ft	用户范围 User range	Ulong	4	H+124
23	Nt	从闰年的 1 月 1 日开始的 4 年内的日历天数 Calendar number of day within 4 year interval starting at Jan 1 of a leap year	Ushort	2	H+128
24	Reserved	保留字段	Uchar	1	H+130
25	Reserved	保留字段	Uchar	1	H+131
26	sat type	卫星类型 Satellite type where 0 = GLO_SAT 1 = GLO_SAT_M (M type) 2 = GLO_SAT_K (K type)	Uchar	1	H+132
27	Reserved	保留字段	Uchar	1	H+133
28	e week	星历参考周（GPS 参考时间） Reference week of ephemeris (GPS reference time)	Ushort	2	H+134
29	e time	星历参考时间（GPS 参考时间） Reference time of ephemeris (GPS reference time) (ms)	Ulong	4	H+136

Field	Field type	Description	Format	Binary Bytes	Binary Offset
30	t offset	GPS 和 GLONASS 时间之间的整数秒。正值表示 GLONASS 早于 GPS 参考时间。 Integer seconds between GPS and GLONASS time. A positive value implies GLONASS is ahead of GPS reference time.	Ulong	4	H+140
31	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Ulong	4	H+144
32	[CR][LF]	语句结束符（仅 ASCII） Sentence terminator (ASCII only)	-	-	-

表3-58 GLONASS Ephemeris Flags Coding

Nibble Number	Bit	Description	Range Values	Hex Value
N0	0	P1 Flag - Time interval between adjacent values of (tb) parameter in minutes	See P1 Flag Range Values below	00000001
	1			00000002
	2	P2 Flag - Flag of oddness or evenness of the value of (tb) (for intervals of 30 or 60 minutes)	0 = even 1 = odd	00000004
	3	P3 Flag - Number of satellites with almanac information within current subframe	0 = four 1 = five	00000008
N1	4	P4 Flag - Flag to show that ephemeris parameters are present. "1" indicates that updated ephemeris or frequency/time parameters have been uploaded by the control	0 = no updated ephemeris 1 = updated ephemeris present	00000010
	5 ... 7	Reserved		

Nibble Number	Bit	Description	Range Values	Hex Value
N2 through N7	8 ... 31	Reserved		

表3-59 P1 Flag Range Values

State	Description
00	0 minutes
01	30 minutes
10	45 minutes
11	60 minutes

3.2.25. GNSSOBS 卫星观测值信息

适用产品：M720、M722

GNSSOBS 报文传输原始观测数据信息。

在精确相位处理中使用载波相位测量值时，如果在 ch-tr-status 字段中的 parity known 标志未被设置，测量值可能存在半个周期(1/2 cycle)的不确定性，此时的载波相位不应该使用。

伪距和载波相位的跟踪误差估计仅包含接收机跟踪环路的热噪声，没有考虑到可能的多路径误差或大气延迟。如果对一个给定的 PRN 正在跟踪多个信号，则在 GNSSOBS 报文中出现具有相同 PRN 的每个信号的信息，用 signal ID 观测信号的类型 来区分。

- **Message ID: 3000**
- **Log Type: Sync**

输出示例：

```
#GNSSOBSA,40.5,0,0,COM2,FINESTEERING,2376,95756000,d323b507,00000000;214,6,G1C,0,5297,22209238.425,-116710325.243511,-1374.951,9,138,21621.002,00017633,6,G2L,0,5249,22209248.537,-90943099.372810,-1071.427,9,139,21708.002,00017633,6,G5Q,0,5734,22209250.895,-87153856.963999,-1026.772,8,124,21829.002,00017633,6,G2W,0,5541,22209248.421,-90943134.127941,-1071.348,8,130,21491.002,00017633,21,G1C,0,5692,20543721.293,-107958019.439836,-372.566,8,125,12684.001,00027633,21,G2L,0,6113,20543726.880,-84123181.093918,-2
```

90.343,7,112,12681.001,00027633,21,G5Q,0,6463,20543733.010,-80618080.9030
 09,-278.226,6,100,12681.001,00027633,21,G2W,0,5284,20543727.124,-84123184
 .860151,-290.319,9,138,11584.001,00027623,11,G1C,0,5807,21010334.995,-110
 410062.427816,288.541,8,121,16151.001,00037633,11,G2L,0,5821,21010339.652
 ,-86033864.142667,224.793,8,121,16170.001,00037633,11,G5Q,0,6495,21010346
 .582,-82449124.372779,215.457,6,99,16344.001,00037633,11,G2W,0,5201,21010
 339.904,-86033874.895768,224.854,9,141,14941.001,00037633,9,G1C,0,5016,23
 543153.975,-123720108.899084,-2907.286,9,147,15009.001,00057623,9,G2L,0,5
 108,23543162.573,-96405351.402925,-2265.602,9,144,15336.001,00057633,9,G5
 Q,0,5368,23543166.130,-92388479.725758,-2171.125,8,135,15338.001,00057633
 ,9,G2W,0,5284,23543162.521,-96405342.159475,-2265.444,9,138,15000.001,000
 57623,20,G1C,0,5763,20810017.042,-109357411.480892,98.603,8,123,12170.001
 ,00067633,20,G2W,0,5283,20810021.015,-85213605.889855,76.753,9,138,9925.0
 00,00067633,12,G1C,0,5336,24071177.962,-126494879.331792,-2136.584,9,136,
 11407.001,00077633,12,G2L,0,4682,24071185.677,-98567463.023736,-1664.928,
 10,157,11400.001,00077623,12,G2W,0,5119,24071185.650,-98567463.766793,-16
 64.837,9,143,11190.001,00077623,13,G1C,0,4941,23053566.499,-121147318.767
 814,3758.354,9,149,2965.000,00087623,13,G2W,0,3848,23053574.054,-94400555
 .815200,2928.735,13,1062,723.000,00087553,5,G1C,0,5486,21746873.046,-1142
 80642.375151,1542.767,8,132,7802.000,00097633,5,G2L,0,5298,21746879.040,-
 89049906.573802,1202.177,9,138,7858.000,00097633,5,G2W,0,5586,21746878.66
 9,-89049895.318171,1202.189,8,128,7795.000,00097633,25,G1C,0,4865,2453121
 0.319,-128912347.537008,-1169.268,10,152,6379.000,000a7623,25,G2L,0,4759,
 24531222.219,-100451259.286897,-911.152,10,155,6459.000,000a7623,25,G5Q,0
 ,4943,24531224.456,-96265767.280773,-873.210,9,149,6575.000,000a7623,25,G
 2W,0,4831,24531222.065,-100451228.048705,-911.049,10,153,5742.000,000a761
 3,19,G1C,0,5152,22959445.788,-120652663.223105,-3687.286,9,142,23912.002,
 000b7623,19,G2W,0,4339,22959452.211,-94015045.502760,-2873.183,11,190,218
 79.002,000b75e3,7,G1C,0,5082,25446067.360,-133720014.514844,1124.355,9,14
 5,2372.000,000c7633,7,G2L,0,4750,25446078.223,-104197453.306076,875.906,1

0,155,2138.000,000c7623,7,G2W,0,4914,25446077.999,-104197458.031385,876.1
36,9,150,2023.000,000c7633,30,G1C,0,4977,25044801.878,-131611357.205703,1
933.739,9,148,2186.000,000d7633,30,G2L,0,4564,25044818.944,-102554384.713
922,1506.643,10,161,2399.000,000d7613,30,G5Q,0,5457,25044823.527,-9828130
7.917169,1443.998,8,133,2446.000,000d7633,30,G2W,0,4874,25044818.913,-102
554385.474842,1506.830,9,151,1653.000,000d7613,29,G1C,0,5078,24877998.253
, -130734774.894326,3480.245,9,145,1195.000,000e7633,29,G2L,0,4695,2487800
6.367,-101871279.238406,2711.702,10,157,1191.000,000e7623,29,G2W,0,4942,2
4878006.332,-101871279.995513,2711.896,9,149,1181.000,000e7623,60,R1C3,0,
4823,23430893.034,-125339376.877480,-2955.772,10,153,25159.002,00017623,3
8,R1C1,0,5812,20086148.239,-107372090.764190,-2482.326,8,121,12217.001,00
027633,40,R1C5,0,5440,23265442.044,-124541782.775267,4101.922,8,133,1758.
000,00037633,40,R2C5,0,5329,23265452.506,-96865890.530506,3190.361,9,137,
1714.000,00037633,39,R1C-4,0,6271,19547623.887,-104309958.287333,1467.741
,7,107,7555.000,00047633,39,R2C-4,0,6099,19547628.733,-81130049.373316,11
41.647,7,112,7698.000,00047633,54,R1C4,0,6056,21181314.941,-113345527.943
052,2990.085,7,113,6183.000,00087633,54,R2C4,0,5957,21181318.705,-8815768
8.794568,2325.637,7,117,6127.000,00087633,61,R1C2,0,6153,19780699.564,-10
5776330.474917,-56.496,7,110,15389.001,000a7633,61,R2C2,0,6102,19780702.5
72,-82270502.569580,-43.900,7,112,15465.001,000a7633,12,E1C,0,5302,239904
05.595,-126070432.744645,465.693,9,138,19488.000,00017633,12,E5Q,0,5475,2
3990412.880,-94143530.746716,347.649,8,132,19496.000,00017633,12,E7Q,0,57
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表3-60 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	GNSSOBS	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	# obs	对应的观测信息个数。 Number of observations with information to follow	Ulong	4	H
3	PRN/slot	卫星 PRN 号 Satellite PRN number of range measurement	Ushort	2	H+4
4	satellite system	卫星系统 (为了减少 ASCII 报文时的编码长度，减少通讯带宽占用，将“satellite	Uchar	1	H+6

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		system"/"signal ID"/"glo freq"三个字段在映射成 ASCII 时合并成一个字符串，卫星系统名占用一个字符，信号名占用两个字符，GLONASS 频点如果有则加在后面，比如 GPS L1C/A，对应字符串"G1C"，GLONSS L1C/A slot -3，对应字符串"R1C-3")			
5	signal ID	Signal ID 频点，Follow RTCM 的对每个卫星系统的 ID 定义，对应 ASCII follow RINEX 的定义，固定两个字符表示	Uchar	1	H+7
6	glo freq	GLONASS 频率(-7~+6)	Char	1	H+8
7	reserved	保留	Uchar	1	H+9
8	C/No	载噪比。 Carrier to noise density ratio C/No = $10[\log_{10}(S/N_0)]$ (dB-Hz) C/No *128	Ushort	2	H+10
9	psr	码伪距观测量，米。 Pseudorange measurement (m)	Double	8	H+12
10	adr	载波相位，周。 Carrier phase, in cycles (accumulated Doppler range)	Double	8	H+20
11	dopp	瞬时多普勒， Hz Instantaneous carrier Doppler frequency	Float	4	H+28

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		(Hz)			
12	psr σ	码伪距标准差，米。 Pseudorange measurement standard deviation (m) psr std * 128	Ushort	2	H+32
13	adr σ	载波相位标准差，周。 Estimated carrier phase standard deviation (cycles) adt std * 8192	Ushort	2	H+34
14	locktime	连续跟踪时间，无周跳。 Number of seconds of continuous tracking (no cycle slipping)	Float	4	H+36
15	ch-tr-status	通道跟踪状态 Tracking status	Ulong	4	H+40
16...	下一个 OBS offset = H + 4 + (#obs x 40) Next PRN offset = H + 4 + (#obs x 40)				
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+4+(#obs x 40)
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-61 PRN Numbers

Command/Log	GPS PRN	SBAS PRN	SBAS QZSS L1S PRN	GLONASS Slot	Galileo SVID	QZSS PRN	BDS PRN	NavIC PRN
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Command/Log	GPS PRN	SBAS PRN	SBAS QZSS L1S PRN	GLONASS Slot	Galileo SVID	QZSS PRN	BDS PRN	NavIC PRN
GNSOBS	1-32	120-158	183-192	38-61	1-36	193-202	1-63	1-14
GNSOBSOMP	1-32	120-158	183-192	38-61	1-36	193-202	1-63	1-14
SATINVIEW	1-32	120-158	183-191	1-24	1-36	193-202	1-63	1-14
BESTDOPS	G1-G32	S120-S158	S183-S191	R1-R24	E1-E36	J193-J202	C1-C63	I1-I14
UVNAV	G1-G32	S120-S158	S183-S191	R1-R24	E1-E36	J193-J202	C1-C63	I1-I14

表3-62 Satellite system

Satellite system	System ID	ASCII name	Comments
GPS	0	'G'	
GLONASS	1	'R'	
GALLIEO	2	'E'	
SBAS	3	'S'	
QZSS	4	'J'	
BDS	5	'C'	
NavIC	6	'I'	
Other	7	'L'	Lband, not defined in RINEX

表3-63 Channel Tracking Status

Nibble	Bit	Mask	Description	Range Value
N0	0	0x00000001	基带跟踪状态 BB Tracking state	0-23, see Tracking State
	1	0x00000002		
	2	0x00000004		
	3	0x00000008	保留 Reserved	
N1	4	0x00000010		
	5	0x00000020		
	6	0x00000040		

Nibble	Bit	Mask	Description	Range Value
N2	7	0x00000080	PLD	
	8	0x00000100		
	9	0x00000200		
	10	0x00000400		
	11	0x00000800	保留 Reserved	
	12	0x00001000	伪距平滑标志 PSR Smooth Flag	0 = 未平滑, 1 = 平滑 0 = Not smoothed, 1 = smoothed
	13	0x00002000	载波有效标志 ADR Valid Flag	0 = 无效, 1 = 有效 0 = invalid, 1 = valid
	14	0x00004000	载波相位偏差版本号 PhaseShiftVer	0 = 版本0, 1 = 版本1 0 = version0, 1 = version1 always set to 1 since 202411
	15	0x00008000	保留 Reserved	
N4	16	0x00010000	频道号 SV channel number	1 ~ n
	17	0x00020000		
	18	0x00040000		
	19	0x00080000		
N5	20	0x00100000		
	21	0x00200000		
	22	0x00400000		
	23	0x00800000		
N6	24	0x01000000	保留	
	25	0x02000000		
	26	0x04000000		
	27	0x08000000		

Nibble	Bit	Mask	Description	Range Value
N7	28	0x10000000	Reserved	
	29	0x20000000		
	30	0x40000000		
	31	0x80000000		

表3-64 Tracking State

State	Description
0	0x20
1	0x21, 环路快速收敛
2	0x22, 高动态跟踪
3	0x23, FINE TRACK
4	0x24, 高灵敏度 PLL 跟踪
5	0x25, 高灵敏度 FLL 跟踪
6	0x26
7	0x27

下表为 CHC 扩展的 RTCM SignalID 与 扩展的 Range SignalType 之间的转换关系:

表3-65 SignalID Mapping of RTCM and Range

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
GPS	1					Reserved
	2	L1	C/A	1C	0 - L1C/A	
	3	L1	P	1P	4 - L1P	
	4	L1	Z-trackin g or	1W	8 - L1W	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
			similar			
	5-7					Reserved
	8	L2	C/A	2C	1 - L2C/A	
	9	L2	P	2P	5 - L2P	
	10	L2	Z-trackin g or similar	2W	9 - L2P(Y),semi-codeles s	
	11-14					Reserved
	15	L2	L2C(M)	2S	17 - L2C(M)	
	16	L2	L2C(L)	2L	13 - L2C(L)	
	17	L2	L2C(M+L)	2X	21 - L2CX	
	18-21					Reserved
	22	L5	I	5I	6 - L5I	
	23	L5	Q	5Q	14 - L5Q	
	24	L5	I+Q	5X	10 - L5X	
	25-29					Reserved
	30	L1	L1C-D	1S	12 - L1CD	
	31	L1	L1C-P	1L	16 - L1CP	
	32	L1	L1C-(D+P)	1X	20 - L1CX	
GLONASS	1					Reserved
	2	G1	C/A	1C	0 - L1C/A	
	3	G1	P	1P	4 - L1P	
	4-7					Reserved

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
	8	G2	C/A	2C	1 - L2C/A	
	9	G2	P	2P	5 - L2P	
	10-14					Reserved
	15	G2a	D	6A	13 - L2SCI	
	16	G2a	P	6B	17 - L2OCP	
	17	G2a	D+P	6X	21 - L2OCX	
	18-21					Reserved
	22	G3	I	3I	2 - L3I	
	23	G3	Q	3Q	6 - L3Q	
	24	G3	I+Q	3X	10 - L3X	
	25-29					Reserved
	30	G1a	D	4A	12 - L1OCD	
	31	G1a	P	4B	16 - L1OCP	
	32	G1a	D+P	4X	20 - L1OCX	
Galileo	1					Reserved
	2	E1	C no data	1C	2 - E1C	
	3	E1	A	1A	0 - E1A	
	4	E1	B I/NAV OS /CS/SoL	1B	1 - E1B	
	5	E1	B+C	1X	3 - E1X	
	6	E1	A+B+C	1Z	4 - E1Z	
	7					Reserved

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
	8	E6	C	6C	7 - E6C	
	9	E6	A	6A	5 - E6A	
	10	E6	B	6B	6 - E6B	
	11	E6	B+C	6X	8 - E6X	
	12	E6	A+B+C	6Z	9 - E6Z	
	13					Reserved
	14	E5B	I	7I	16 - E5bI	
	15	E5B	Q	7Q	17 - E5bQ	
	16	E5B	I+Q	7X	18 - E5bX	
	17					Reserved
	18	E5(A+B)	I	8I	19 - E5AltBOC-I	
	19	E5(A+B)	Q	8Q	20 - E5AltBOC-Q	
	20	E5(A+B)	I+Q	8X	21 - E5AltBOC-X	
	21					Reserved
	22	E5A	I	5I	11 - E5aI	
	23	E5A	Q	5Q	12 - E5aQ	
	24	E5A	I+Q	5X	13 - E5aX	
	25-32					Reserved
SBAS	1					Reserved
	2	L1	C/A	1C	0 - L1C/A	
	3-21					Reserved
	22	L5	I	5I	6 - L5I	
	23	L5	Q	5Q	14 - L5Q	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
QZSS	24	L5	I+Q	5X	10 - L5X	
	25-32					Reserved
	1					Reserved
	2	L1	C/A	1C	0 - L1C/A	
	3-8					Reserved
	9	L6	D	6S	26 - L6D	
	10	L6	P	6L	27 - L6P	
	11	L6	D+P	6X	28 - L6X	
	12	L6	E	6E	29 - L6E	
	13	L6	D+E	6Z	30 - L6Z	
	14					Reserved
	15	L2	L2C(M)	2S	17 - L2CM	
	16	L2	L2C(L)	2L	13 - L2CL	
	17	L2	L2C(M+L)	2X	21 - L2CX	
	18-21					Reserved
	22	L5	I	5I	6 - L5I	
	23	L5	Q	5Q	14 - L5Q	
	24	L5	I+Q	5X	10 - L5X	
	25-29					Reserved
	30	L1	L1C(D)	1S	12 - L1CD	
	31	L1	L1C(P)	1L	16 - L1CP	
	32	L1	L1C(D+P)	1X	20 - L1CX	
BDS	1					Reserved

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
	2	B1	I	2I	0 or 4 B1I	
	3	B1	Q	2Q	20 - B1Q	
	4	B1	I+Q	2X	B1X ¹⁾	
	5-7					Reserved
	8	B3	I	6I	2 or 6 B3I	
	9	B3	Q	6Q	22 - B3Q	
	10	B3	I+Q	6X	B3X ²⁾	
	11-13					Reserved
	14	B2	I	7I	1 or 5 B2I	
	15	B2	Q	7Q	21 - B2Q	
	16	B2	I+Q	7X	B2X ³⁾	
	17-21					Reserved
	22	B2a	D	5D	12 - B2aD	
	23	B2a	P	5P	9 - B2aP	
	24	B2a	D+P	5X	15 - B2aX	
	25	B2b	I	7D	11 - B2bI	
	26	B2b	Q	7P	8 - B2bP	
	27	B2b	I+Q	7Z	14 - B2bX	
	28-29					Reserved
	30	B1C	D	1D	10 - B1CD	
	31	B1C	P	1P	7 - B1CP	
	32	B1C	D+P	1X	13 - B1CX	
NIC	1-21					Reserved

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Range Signal Type	Comments/Notes
	22	L5	A	5A	0 - L5A	
	23-32					Reserved
Other		XW1L1	P	1P	0 - XW1L1L	
		XW1L1	D	1D	4 - XW1L1H	
		XW1L1	D+P	1X		
		XW1B2	P	2P	1 - XW1B2L	
		XW1B2	D	2D	5 - XW1B2H	
		XW1B2	D+P	2X		
		L-Band			19 - L-Band	

3.2.26. GNSSOBSCMP 压缩格式的卫星观测值信息

适用产品：M720、M722

压缩格式的 GNSSOBS 报文。

- Message ID: 3058
- Log Type: Sync

输出示例：

```
#GNSSOBSCMPA,37.5,0,0,COM2,FINESTEERING,2376,95972000,d323b507,00000000;2
14,33760100FF6DFA9F7D1E9E0A171B86866106A0A92A1F1000,33760100CFA8FB1FC31E9
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001,33760100DAA8FBAFC21E9E0A2733B890610660996A1F9000,337602002D45FE5F18D4
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```

1E9000,23760500FC59F4DF7BB3480B7EC278968109206F671E1000,3376050016ECF6EFC
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41335BC1610E201E251F1500, 33760B00EA0002F037AB5C0A56E5F584510EA023841F7500
, 33760B001CE8015035AB5C0AE744D9C8510EE048A51FD500, 23760C006D290070C1317D1
15D72C29D71034017D31E1500, 23760C00A2210090C7317D111CD2F0BF6103C0931C1F750
0, 23760C00162000B0AB317D11384C86B261030017131FD500, 33760D0002A2FD8F3B454A
114855E7BE710AE050DB1E1500, 33760D008513FEFF34454A112A4ADFDA710A20CBFB1E75

00, 33760D005E2BFEEAF3B454A1131F726CC610A00CB5B1FD500, 23760E00601B00701AB1D
A11F6BCE6E071019F17D31E1500, 23760E005216009033B1DA11ACFE7C8E7101C093FC1E7
500, 33760E002A1500101CB1DA116109778361018017331FD500, 33760F00B948FC1F81D6
5B11120A78B36128406E3B1F1500, 33760F00E6FAFCAF8ED65B1187AD94D15128E0E49B1F
7500, 30760F005620FD6F84D65B11FD0450C361282043791F8501, 33760F000540FC1F52D
65B11E0C6EE987128806EDB1EE501, 33760F002833FD6FA5D65B1182D32CFC6128C0E45B1
F6501, 33761000E8F5FE4F894CB0112EBE7EFC7108805CAB1E1500, 33761000C327FFAF77
4CB011BF3EE9A4610860BF0B1F7500, 337610004B32FF2F744CB011B570CD986108A0EB4B
1FD500, 33761200C44000E0B9E6BC1182D84AF4613B9F17331F1500, 33761200A13400F06
7E6BC119BF53E9E613BA0C3691F7500, 337612001F3200C062E6BC113AC37592513B60179
31F8501, 3376130092380B70D158FF0BD901D6B0711CE0A0E11E1500, 337613003F1E09E0
F458FF0B6ED683A7611CA062011F7500, 1076130039AD08A0EB58FF0B57A51EF6711CE01D
C01E8501, 33761300EC520B80A458FF0B55DA7E9E711CC09A811EE501, 23761300BD7408A
00059FF0B4EC56B9D811C409E611E6501, 33761400F6EEFB4FAD7C890A39252FA4612AA00
9661F1500, 337614001CB2FCFF067D890A613F41ED512A20BDA51F7500, 3376140006DBFC
4F0B7D890A7D6A4AB2502A8009C61F8501, 3376140069E5FB8F867C890A4E791394612A40
09461FE501, 3376140075EFFCDF297D890A51F9CED4512A8009861F6501, 33761500374EF
FEFB11CD911E9F7EDE1710D8020CB1E1500, 33761500766FFF8FD51CD911ECFA528F610DA
0650A1F7500, 337615006276FF3FD91CD911C78A4284610DA02E0B1FD500, 237616001144
008099644C12A376E496713C0017B31E1500, 337616003A3700F04B644C124ADA59D2613C
A0C3291F7500, 23761600813400504C644C1288C83CCA613CA016131F8501, 03761700025
100C0D2FE3D12147C43A08202A077201E1500, 23761700EC4100A0E1FE3D12B418F7D9710
2A093BC1E7500, 23761700AE3E00B0BCFE3D12FFB47BD17102E015D31ED500, A3751800E7
2D0070FB311B135A964890F2058091B01D1500, F3751800692500D01F321B130800F8E482
05A002001E7500, 037618008E2300F00D321B13560826E281058016531ED500, 33761A000
55AFC4FA1B45F11DD5DF3B071070026DC1E1500, 33761A00F108FDFF87B45F1154B388CF6
10760933C1F7500, 33761A00C02DFDEF81B45F1143125DC1610720BF3E1FD500, 33761D00
7277F43FCAFE170C33F7CAA071152005E61E1500, 33761D00E5A0F64FC1FE170C92B87A9A
7115E051E51E7500, 23761D00F714F74FD4FE170C44D1B6E961150005061F8501, 33761D0
0415CF4EF9EFE170C66324E8E7115C004861EE501, 33761D00BA4EF7EFE8FE170C9AD2549

```

171150005861E6501,33761E002B27FDFF323C710AA357F8B3610B606B281F1500,33761E
00CEAFFD1F343C710A7B5315FA500B2029C91F7500,33761E005ECCFD5F333C710A0FAB7F
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ECBF561C36016731F6401,237604004A2DFEDFB82AA01176110DEC81C26016531E1400,33
7604006F94FE6F0B2BA011D5D76DFA71C20016931EF400,33760400A7A3FE4F332BA011B4
2EA9DA61C26016131F6401,236601005C4100807EE6BC11A52D2DD971828017F31E1300,3
3660100D030004081E6BC11241591CC61824017331F5301,336602002C22008007DEA0110
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236603008F4400305F644C12396BEBFA719000B8921E1300,13660300573300006D644C12
F42A2E8671904017D31E5301,3366020026D3FFFF37AC9911B7CBD8DD610AE014331F5601
,336606008C71FE8F3FBFD8129F5E54C17102E079B71E5601*c4f306ab

```

表3-66 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	GNSSOBSCMP	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	#obs	对应的观测信息个数。 Number of satellite observations with information to follow	Ulong	4	H
3	1st gnssobs record	GNSSOBS 压缩信息。 Compressed gnssobs log in format of Table GNSSOBS Record Format (GNSSOBSCMP only)	Hex	24	H+4
4	Next gnssobscmp offset = H+4 (#obs x 24)				

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
5	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+4+(#obs x 24)
6	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-67 GNSSOBS Record Format (GNSSOBSCOMP only)

Data	Description	Bits first to last	Length(bits)	Scale Factor	Units
Channel Tracking Status	Channel tracking status word	0-31	32	see Table Channel Tracking Status	-
Doppler Frequency	Instantaneous carrier Doppler frequency	32-59	28	1/256	Hz
Pseudorange (PSR)	Pseudorange measurement	60-95	36	1/128	m
ADR	ADR (Accumulated Doppler Range) is calculated as follows: ADR_ROLLS = (GNSSOBSCOMP_PSR / WAVELENGTH + GNSSOBSCOMP_ADR) / MAX_VALUE Round to the closest integer IF (ADR_ROLLS ≤ 0) ADR_ROLLS = ADR_ROLLS - 0.5 ELSE ADR_ROLLS = ADR_ROLLS + 0.5 At this point integerise	96-127	32	1/256	cycles

Data	Description	Bits first to last	Length(bits)	Scale Factor	Units
	ADR_ROLLS CORRECTED_ADR = GNSSOBSCMP_ ADR - (MAX_VALUE*ADR_ROLLS) where ADR has units of cycles WAVELENGTH = 0.1902936727984 for GPS L1 WAVELENGTH = 0.2442102134246 for GPS L2 MAX_VALUE = 8388608 GLONASS satellites emit L1 and L2 carrier waves at a satellite-specific frequency				
StdDev-PSR	Pseudorange measurement standard deviation	128-131	4	See Table StdDev-PSR Values	m
StdDev-ADR	ADR measurement standard deviation	132-135	4	(n+1)/512	cycles
PRN/Slot	Refer to Table PRN Numbers	136-143	8	1	-
Lock Time	Number of seconds of continuous tracking (no cycle slipping) This field is constrained to a maximum value of 2,097,151 which represents a lock time of 65535.96875 s (2097151/32).	144-164	21	1/32	s
C/No	Carrier to noise density ratio The C/No is constrained to a	165-169	5	(20+n)	dB-Hz

Data	Description	Bits first to last	Length(bits)	Scale Factor	Units
	value between 20-51 dB-Hz. Thus, if it is reported that C/No = 20 dB-Hz, the actual value could be less. Likewise, if it is reported that C/No = 51, the true value could be greater.				
GLONASS Frequency number	GLONASS Frequency number	170-175	6	n+7	1
Satellite system	Satellite system, see Satellite system	176-179	4	1	1
Signal type	Signal type, see GNSSOBS Signal Type	180-185	6	n-1	1
Reserved		186-191	6		

表3-68 StdDev-PSR Values

Code	StdDev-PSR(m)
0	0.050
1	0.075
2	0.113
3	0.169
4	0.253
5	0.380
6	0.570
7	0.854
8	1.281
9	2.375
10	4.750

Code	StdDev-PSR(m)
11	9.500
12	19.000
13	38.000
14	76.000
15	152.000

3.2.27. GPSEPH GPS 星历

适用产品：M720、M722

- Message ID: 3005
- Log Type: Asynch

输出示例：

```
#GPSEPHA,40.0,20,0,COM2,SATTIME,2376,96035000,d323b507,00000000;1,2376,4.00000000E+00,5.714523747E-11,38,86400.0,0.00000E+00,7.50333E-12,3.19299E-04,806,-5.00000000E+00,4.587691096E-09,2.480301415E+00,-2.216547728E-07,6.0117477551E-04,3.371387720E-07,2.656078990E+07,86400.0,-4.284083843E-08,8.662112049E-01,-1.676380634E-08,9.5925149716E-01,3.69468750e+02,1.5204550423E-01,-8.47070998E-09,-8.847564459E-09,0,79218.0,38,2376,TRUE,1.458549256E-04*f045aed2
```

表3-69 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	GPSEPH header	报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	PRN	卫星 PRN 号 Satellite PRN number	Ulong	4	H
3	week	由 z 计数计算得到的参考周	Ulong	4	H+4

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		toe week number computed from Z count week (GPS reference week number)			
4	URA	用户测距精度 (m) ² , 原始 URA 索引与输出值之间的对应关系见表 URA Variance User Range Accuracy variance (m) ² The ICD specifies that the URA index transmitted in the ephemerides can be converted to a nominal standard deviation value using an algorithm listed there. We publish the square of the nominal value (variance).	Double	8	H+8
5	I^0	轨道倾角变化率, rad/s Rate of inclination angle (radians/s)	Double	8	H+16
6	IODE1	星历数据期号 1 Issue of ephemeris data 1	Ulong	4	H+24
7	toc	卫星钟差参数, s SV clock correction term (s)	Double	8	H+28
8	a_{f2}	卫星钟差参数, s/s/s Clock aging parameter (s/s/s)	Double	8	H+36
9	a_{f1}	卫星钟差参数, s/s Clock aging parameter (s/s)	Double	8	H+44
10	a_{f0}	卫星钟差参数, s Clock aging parameter (s)	Double	8	H+52
11	iodc	时钟数据期号 Issue of data clock	Ulong	4	H+60

Field	Field type	Description	Format	Binary Bytes	Binary Offset
12	crs	轨道半径（正弦振幅，m） Amplitude of sine harmonic correction term to the orbit radius (m)	Double	8	H+64
13	ΔN	卫星平均角速度的改正值，rad/s Mean motion difference (radians/s)	Double	8	H+72
14	M_0	参考时间的平近点角，rad Mean anomaly of reference time (radians)	Double	8	H+80
15	cuc	纬度幅角（余弦振幅，rad） Amplitude of cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+88
16	ecc	卫星轨道偏心率 Eccentricity, dimensionless	Double	8	H+96
17	cus	纬度幅角（正弦振幅，rad） Amplitude of sine harmonic correction term to the argument of latitude (radians)	Double	8	H+104
18	A	卫星轨道长半轴 Semi-major axis (m)	Double	8	H+112
19	toe	星历的参考时间，单位：s Reference time for ephemeris (s)	Double	8	H+120
20	cic	倾角（余弦振幅，rad） Amplitude of cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+128
21	Ω_0	升交点赤经，rad Longitude of ascending node of orbit plane (radians)	Double	8	H+136

Field	Field type	Description	Format	Binary Bytes	Binary Offset
22	cis	倾角（正弦振幅，rad） Amplitude of sine harmonic correction term to the angle of inclination (radians)	Double	8	H+144
23	lo	参考时间的轨道倾角，rad Inclination angle at reference time (radians)	Double	8	H+152
24	crc	轨道半径（余弦振幅，m） Amplitude of cosine harmonic correction term to the orbit radius (m)	Double	8	H+160
25	ω	近地点幅角，rad Argument of perigee (radians)	Double	8	H+168
26	$\dot{\omega}$	升交点赤经变化率，rad/s Rate of right ascension (radians/s)	Double	8	H+176
27	tgd	估计的群延迟差 Estimated group delay difference (s)	Double	8	H+184
28	health	健康状态 Health status - bits 0-5: 6-bit health code as defined in IS-GPS-200, - bit 6: Alert flag	Ulong	4	H+192
29	tow	子帧时间戳 Time stamp of subframe 1 (s)	Double	8	H+196
30	IODE2	星历数据期号 2 Issue of ephemeris data 2	Ulong	4	H+204
31	z week	Z 计数 Z count week number. This is the week number from subframe 1 of the ephemeris. The 'toe week'	Ulong	4	H+208

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		(field #7) is derived from this to account for rollover			
32	AS	反欺骗 Anti-spoofing on: 0 = FALSE 1 = TRUE	Bool	4	H+212
33	N	改正的卫星平均角速度 Corrected mean motion (radians/s) This field is computed by the receiver.	Double	8	H+216
34	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Ulong	4	H+224
35	[CR][LF]	语句结束符 (仅 ASCII) Sentence terminator (ASCII only)	-	-	-

表3-70 URA Variance

Index Value (m)	A: Standard Deviations (m)	Variance: A ² (m ²)
0	2.0	4
1	2.8	7.84
2	4.0	16
3	5.7	32.49
4	8	64
5	11.3	127.69
6	16.0	256
7	32.0	1024
8	64.0	4096
9	128.0	16384
10	256.0	65536
11	512.0	262144

Index Value (m)	A: Standard Deviations (m)	Variance: A ² (m ²)
12	1024.0	1048576
13	2048.0	4194304
14	4096.0	16777216
15	8192.0	67108864

3.2.28. GPSIONO 电离层模型

适用产品：M720、M722

GPSIONO 输出电离层模型参数（IONO）。

- Message ID: 3017
- Log Type: Async

输出示例：

```
#GPSIONOA,39.5,0,0,COM2,FINESTEERING,2376,96633000,d323b507,00000000;1.30
3851604461670e-08,2.235174179077148e-08,-5.960464477539063e-08,-1.1920928
95507813e-07,1.105920000000000e+05,1.474560000000000e+05,-6.5535999999999
99e+04,-3.932160000000000e+05,0*ec1b8eeb
```

表3-71 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	GPSIONO header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	a0	Alpha 参数常数项。 Alpha parameter constant term	Double	8	H
3	a1	Alpha 参数的 1 阶项。 Alpha parameter 1st order term	Double	8	H+8
4	a2	Alpha 参数的 2 阶项。 Alpha parameter 2nd order term	Double	8	H+16

Field	Field type	Description	Format	Binary Bytes	Binary Offset
5	a3	Alpha 参数的 3 阶项。 Alpha parameter 3rd order term	Double	8	H+24
6	b0	Beta 参数常数项。 Beta parameter constant term	Double	8	H+32
7	b1	Beta 参数的 1 阶项。 Beta parameter 1st order term	Double	8	H+40
8	b2	Beta 参数的 2 阶项。 Beta parameter 2nd order term	Double	8	H+48
9	b3	Beta 参数的 3 阶项。 Beta parameter 3rd order term	Double	8	H+56
10	Reserved			4	H+64
11	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+68
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.29. GPSUTC UTC 时间参数

适用产品：M720、M722

GPSUTC UTC 数据。

- Message ID: 3018
- Log Type: Asynch

输出示例：

```
#GPSUTCA,38.5,0,0,COM2,FINESTEERING,2376,96693000,d323b507,00000000;2376,319488,-9.3132257461547852e-10,-1.065814104e-14,2185,7,18,18,1*0702806f
```

表3-72 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	GPSUTC header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	utc wn	UTC 参考周数。 UTC reference week number	Ulong	4	H+0
3	tot	UTC 参数的参考时间。 Reference time of UTC parameters (s)	Ulong	4	H+4
4	A0	UTC 参数的常数项。 UTC constant term of polynomial (s)	Double	8	H+8
5	A1	UTC 参数的 1 阶项。 UTC 1st order term of polynomial (s)	Double	8	H+16
6	wn lsf	新的闰秒生效的周计数。 Future week number	Ulong	4	H+24
7	dn	新的闰秒生效的周内日计数。 Day number (the range is 1 to 7 where Sunday = 1 and Saturday = 7)	Ulong	4	H+28
8	deltat ls	新的闰秒生效前的累积闰秒改正数。 Delta time due to leap seconds	Long	4	H+32
9	deltat lsf	新的闰秒生效后的累积闰秒改正数。 Future delta time due to leap seconds	Long	4	H+36
10	Reserved			4	H+40
11	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+44
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.30. HWSTATUS 硬件状态

适用产品：B760

HWSTATUS： 监测硬件状态，当前支持温度(仅支持 B760)

- Message ID: 3065
- Log Type: Polled

输出示例：

```
#HWSTATUSA,31.0,0,0,COM2,FINESTEERING,2376,98296000,d323b507,00000000;0*88fd9c87
```

表3-73 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	Log header	报文头 HWSTATUS header For information about the log headers, see ASCII, or Binary.		H	0
2	# stats	硬件状态数量	Ulong	4	H+0
3	value	状态值，温度(摄氏度)	Float	4	H+4
4	status	硬件状态，包括类型和状态，（见表：硬件状态）。	Hex Ulong	4	H+8
5	下一个状态值偏移 = H + 4 + (# stats x 8)				
6	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H + 4 + (# stats x 8)
7	[CR][LF]	语句结束符 Sentence terminator (ASCII only)	-	-	-

表3-74 HW Status

Nibble	Bit	Mask	Description
--------	-----	------	-------------

Nibble	Bit	Mask	Description
N0	0	0x00000001	边界限制状态: 0x00 = 值在正常范围 0x01 = 值超过警告范围 0x02 = 值超过严重范围
	1	0x00000002	
	2	0x00000004	
	3	0x00000008	
N1	4	0x00000010	
	5	0x00000020	
	6	0x00000040	
	7	0x00000080	
N2	8	0x00000100	状态类型: 0x00 = Reserved 0x01 = 片上温度
	9	0x00000200	
	10	0x00000400	
	11	0x00000800	
N3	12	0x00001000	
	13	0x00002000	
	14	0x00004000	
	15	0x00008000	
N4	16	0x00010000	Reserved
	17	0x00020000	
	18	0x00040000	
	19	0x00080000	
N5	20	0x00100000	
	21	0x00200000	
	22	0x00400000	
	23	0x00800000	
N6	24	0x01000000	

Nibble	Bit	Mask	Description
	25	0x02000000	
	26	0x04000000	
	27	0x08000000	
	28	0x10000000	
N7	29	0x20000000	
	30	0x40000000	
	31	0x80000000	

3.2.31. IMUOBS IMU 原始数据

适用产品：M720、M722

IMUOBS 为后处理输出扩展的 IMU 原始数据，包括 IMU 信息。

- Message ID: 3048
- Log Type: Asynch

输出示例：

```
#IMUOBSA,21.5,0,0,COM2,FINESTEERING,2376,97801485,d323b507,00000000;00,81,2376,97801.485000,160000,210000,200000,40980000,-10000,-190000,0dfe0000*5649e9a7
```

表3-75 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	IMUOBS Header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.	-	H	0
2	IMU Info	IMU 信息位。位 0：如果设置，则 IMU 有误；位 1：如果设置，则 IMU 数据加密，不可使用；位 2-7：保留。 IMU Info Bits	Hex Uchar	1	H

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Bit 0: If set, an IMU error was detected. Check the IMU Status field for details. Bit 1: If set, the IMU data is encrypted and should not be used. Bits 2 to 7: Reserved			
3	IMU Type	IMU 类型标识符。 IMU Type identifier.	Uchar	1	H+1
4	Week	GNSS 周。 GNSS Week	Ushort	2	H+2
5	Seconds into Week	周秒。 Seconds from week start	Double	8	H+4
6	Z Gyro Output	z 轴角度计数的变化（右手系）。 Change in angle count along z axis Right-handed	Long	4	H+12
7	-(Y Gyro Output)	y 轴角度计数的变化（右手系）。 负值表示 IMU 标记 y 轴右手系旋转的正方向，正值表示 IMU 标记 y 轴右手系旋转的反方向。 - (Change in angle count around y axis) Right-handed A negative value implies a positive right-handed rotation about the y-axis marked on the IMU. A positive value implies a negative right-handed rotation about the y-axis marked on the IMU.	Long	4	H+16
8	X Gyro Output	x 轴角度计数的变化（右手系）。 Right-handed rotation from local level	Long	4	H+20

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		around x-axis in degrees			
9	Z Accel Output	z 轴速度计数的变化。 Change in velocity count along z axis	Long	4	H+24
10	-(Y Accel Output)	-(Y 轴速度计数的变化)。 负值表示输出沿 IMU 标记的 y 轴正方向， 正值表示变化与 IMU 标记的 y 方向相反。 - (Change in velocity count along y axis) A negative value implies the output is along the positive y-axis marked on the IMU. A positive value implies the change is in the direction opposite to that of the y-axis marked on the IMU.	Long	4	H+28
11	X Accel Output	x 轴速度计数的变化。 Change in velocity count along x axis	Long	4	H+32
12	IMU Status	IMU 状态。 The status of the IMU. This field is given in a fixed length (n) array of bytes in binary but in ASCII or Abbreviated ASCII is converted into 2 character hexadecimal pairs. For the raw IMU status, see one of the following tables: · IIM-42652 IMU Status · ADIS16505 IMU Status · SCH1633 IMU Status	Hex Ulong	4	H+36
13	xxxx	32 位 CRC 校验。 32-bit CRC	Hex	4	H+40
14	[CR][LF]	语句结束符。	-	-	-

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Sentence Terminator (ASCII only)			

表3-76 IMU Type(undetermined)

Binary	ASCII	IMU Name	Description
0	UNKNOWN		Unknown IMU type(default)
81	IIM42652	IIM-42652	InvenSense IIM-42652
91	ADIS16505	ADIS-16505	Analog Devices ADIS16505
92	SCH1633	SCH1633	muRata SCH1633
101	SCH1633_EX	SCH1633_EX	external muRata SCH1633

表3-77 IIM-42652 IMU Status(undetermined)

Nibble	Bit	Mask	Description	Range Value
N0	0	0x00000001	Reserved	
	1	0x00000002	Reserved	
	2	0x00000004	Reserved	
	3	0x00000008	Reserved	
N1	4	0x00000010	Reserved	
	5	0x00000020	Initial Self Test Failure	0 = Passed, 1 = Failed
	6	0x00000040	Reserved	
	7	0x00000080	Reserved	
N2	8	0x00000100	Reserved	
	9	0x00000200	Reserved	
	10	0x00000400	Reserved	
	11	0x00000800	Reserved	
N3	12	0x00001000	Reserved	
	13	0x00002000	Reserved	
	14	0x00004000	IMU Data Abnormal	0 = Normal, 1 = Abnormal

Nibble	Bit	Mask	Description	Range Value
	15	0x00008000	Reserved	
N4	16	0x00010000	IMU temperature reading as follows: Signed 2-byte value (SHORT) = Temperature * 100	
	17	0x00020000		
	18	0x00040000		
	19	0x00080000		
N5	20	0x00100000		
	21	0x00200000		
	22	0x00400000		
	23	0x00800000		
N6	24	0x01000000		
	25	0x02000000		
	26	0x04000000		
	27	0x08000000		
N7	28	0x10000000		
	29	0x20000000		
	30	0x40000000		
	31	0x80000000		

IMU 异常判断条件：

请求 RAWIMUX 报文返回 OK 但没有报文输出；

IMU ID 读取不正确；

IMU 一个或多个轴一段时间内一直满量程；

IMU 一个或多个轴一段时间内一直为零。

表3-78 ADIS16505 IMU Status(undetermined)

Nibble	Bit	Mask	Description	Range Value
N0	0	0x00000001	Alarm Status Flag	H
	1	0x00000002	Reserved	
	2	0x00000004		

Nibble	Bit	Mask	Description	Range Value
	3	0x00000008	SPI Communication Error	0 = Passed, 1 = Failed
N1	4	0x00000010	Sensor Over-Range	0 = Passed, 1 = One of more sensors overranged
	5	0x00000020	Initial Self Test Failure	0 = Passed, 1= Failed
	6	0x00000040	Flash Memory Failure	0 = Passed, 1= Failed
	7	0x00000080	Processing Overrun	0 = Passed, 1= Failed
N2	8	0x00000100	Self Test Failure – X-axis gyro	0 = Passed, 1= Failed
	9	0x00000200	Self Test Failure – Y-axis gyro	0 = Passed, 1= Failed
	10	0x00000400	Self Test Failure – Z-axis gyro	0 = Passed, 1= Failed
	11	0x00000800	Self Test Failure – X-axis accelerometer	0 = Passed, 1= Failed
N3	12	0x00001000	Self Test Failure – Y-axis accelerometer	0 = Passed, 1= Failed
	13	0x00002000	Self Test Failure – Z-axis accelerometer	0 = Passed, 1= Failed
	14	0x00004000	Reserved	
	15	0x00008000		
N4	16	0x00010000	IMU temperature reading as follows: Signed 2-byte value (SHORT) 25°C = 0x0000 1 LSB = 0.00565°C	
	17	0x00020000		
	18	0x00040000		
	19	0x00080000		
N5	20	0x00100000		
	21	0x00200000		
	22	0x00400000		
	23	0x00800000		
N6	24	0x01000000		

Nibble	Bit	Mask	Description	Range Value
N7	25	0x02000000		
	26	0x04000000		
	27	0x08000000		
	28	0x10000000		
	29	0x20000000		
	30	0x40000000		
	31	0x80000000		

表3-79 SCH1633 IMU Status(undetermined)

Nibble	Bit	Mask	Description	Range Value
N0	0	0x00000001	Reserved	
	1	0x00000002	Reserved	
	2	0x00000004	Reserved	
	3	0x00000008	Reserved	
N1	4	0x00000010	Reserved	
	5	0x00000020	Reserved	
	6	0x00000040	Reserved	
	7	0x00000080	Reserved	
N2	8	0x00000100	Reserved	
	9	0x00000200	Reserved	
	10	0x00000400	Reserved	
	11	0x00000800	Reserved	
N3	12	0x00001000	Reserved	
	13	0x00002000	Reserved	
	14	0x00004000	Reserved	
	15	0x00008000	Reserved	

Nibble	Bit	Mask	Description	Range Value
N4	16	0x00010000	Reserved	
	17	0x00020000	Reserved	
	18	0x00040000	Reserved	
	19	0x00080000	Reserved	
N4	16	0x00010000	IMU temperature reading as follows: Signed 2-byte value (SHORT) = Temperature * 100	
	17	0x00020000		
	18	0x00040000		
	19	0x00080000		
N5	20	0x00100000		
	21	0x00200000		
	22	0x00400000		
	23	0x00800000		
N6	24	0x01000000		
	25	0x02000000		
	26	0x04000000		
	27	0x08000000		
N7	28	0x10000000		
	29	0x20000000		
	30	0x40000000		
	31	0x80000000		

表3-80 Raw IMU Scale Factors

Gyroscope Scale Factor		Acceleration Scale Factor
IIM42652	1/65.5/1e4 deg/s/LSB	1/4096/1e4 g/LSB
ADIS16505	1/2621440 deg/s/LSB	1/26783550 m/s/s/LSB
SCH1633	1/1600/1e3 deg/s/LSB	1/31360/1e3 g/LSB

3.2.32. INSSOL 组合导航信息

适用产品：M720、M722

INSSOL 输出组合导航位置、速度及姿态信息。

- Message ID: 3049
- Log Type: Synch

输出示例：

```
#INSSOLA,31.0,0,0,COM2,FINESTEERING,2376,97855000,d323b507,00000000;INS_I
NACTIVE,71,78,71,2376,97855.000000000,0.00000000000,0.00000000000,0.0000,
0.0000,0.0000,0.0000,0.000000000,0.000000000,0.000000000*a41d9585
```

表3-81 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	IINSSOL Header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.	-	H	0
2	Status	INS 状态。 INS Status	Enum	1	H+0
3	Reserved		Uchar[3]	3	H+1
4	Week	GNSS 周。 GNSS Week	Ulong	4	H+4
5	Seconds	周秒。 Seconds from week start	Double	8	H+8
6	Latitude	纬度，度。 Latitude [degrees]	Double	8	H+16
7	Longitude	经度，度。 Longitude [degrees]	Double	8	H+24
8	Height	椭球高，米。 Ellipsoidal Height [m]	Double	8	H+32

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
9	North Velocity	北向速度（负值为南向），m/s。 Velocity in a northerly direction (a negative value implies a southerly direction) [m/s]	Double	8	H+40
10	East Velocity	东向速度（负值为西向），m/s。 Velocity in an easterly direction (a negative value implies a westerly direction) [m/s]	Double	8	H+48
11	Up Velocity	天向速度，m/s。 Velocity in an up direction [m/s]	Double	8	H+56
12	Roll	横滚角，沿 y-axis 右手螺旋，单位为度。 Right-handed rotation from local level around y-axis in degrees	Double	8	H+64
13	Pitch	俯仰角，沿 x-axis 右手螺旋，单位为度。 Right-handed rotation from local level around x-axis in degrees	Double	8	H+72
14	Azimuth	航向角，沿 z-axis 左手螺旋，单位为度。从北向顺时针为正。这是由 IMU 陀螺经组合滤波器计算出的惯性方位角。 Left-handed rotation around z-axis in degrees clockwise from North This is the inertial azimuth calculated from the IMU gyros and the SPAN filters.	Double	8	H+80
15	xxxx	32 位 CRC 校验。 32-bit CRC	Hex	4	H+88
16	[CR][LF]	语句结束符。 Sentence Terminator (ASCII only)	-	-	-

表3-82 Inertial Solution Status

Binary	ASCII	Description
0	INS_INACTIVE	IMU logs are present, but the alignment routine has not started;INS is inactive.
1	INS_ALIGNING	INS is in alignment mode.
2	INS_HIGH_VARIANCE	The INS solution uncertainty contains outliers and the solution may be outside specifications. The solution is still valid but you should monitor the solution uncertainty in the INSSTDEV log. It may be encountered during times when GNSS is absent or poor.
3	INS_SOLUTION_GOOD	The INS filter is in navigation mode and the INS solution is good.
6	INS_SOLUTION_FREE	The INS Filter is in navigation mode and the GNSS solution is suspected to be in error. The inertial filter will report this status when there are no available updates (GNSS or other) being accepted and used.
7	INS_ALIGNMENT_COMPLETE	The INS filter is in navigation mode, but not enough vehicle dynamics have been experienced for the system to be within specifications.
8	DETERMINING_ORIENTATION	INS is determining the IMU axis aligned with gravity.
9	WAITING_INITIALPOS	The INS filter has determined the IMU orientation and is awaiting an initial position estimate to begin the alignment process.
10	WAITING_AZIMUTH	The INS filter has orientation, initial biases, initial position and valid roll/pitch estimated. Will not proceed until initial azimuth is entered.
11	INITIALIZING_BIASES	The INS filter is estimating initial biases during the first 10 seconds of stationary data.
12	MOTION_DETECT	The INS filter has not completely aligned, but has detected motion.
14	WAITING_ALIGNMENTORIENTATION	The INS filter is waiting to start alignment until the

Binary	ASCII	Description
		current Vehicle Frame roll and pitch estimates are within the configured threshold of the expected orientation. Note: This requires an accurate RBV rotation to be configured.
255	IMU_ABNORMAL	IMU self-test failed. For detailed information, refer to IMU Status in RAWIMUX log.

3.2.33. INSSOLX 组合导航扩展信息

适用产品：M720、M722

INSSOLX 输出组合导航位置、速度、姿态信息以及位置标准偏差等信息。

INSSOLX 是 INSSOL 扩展报文版本。

- **Message ID: 3050**
- **Log Type: Synch**

输出示例：

```
#INSSOLXA,36.0,0,0,COM2,FINESTEERING,2376,97897000,d323b507,00000000;INS_
INACTIVE,NONE,0,0.000000000000,0.000000000000,0.0000,0.0000,0.0000,0.0000,0
.0000,0.0000000000,0.0000000000,0.0000000000,0.0000,0.0000,0.0000,0.0000,0.0
000,0.0000,0.0000,0.0000,0.0000,00000000*45b16194
```

表3-83 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	INSSOLX Header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.	-	H	0
2	INS Status	Solution status See Table: Inertial Solution Status	Enum	1	H+0
3	Pos Type	Position type See Table: Position or Velocity Type	Enum	1	H+1
4	Time Since	Elapsed time since the last ZUPT or	Ushort	2	H+2

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
	Update	position update (seconds)			
5	Latitude	纬度，度。 Latitude [degrees]	Double	8	H+4
6	Longitude	经度，度。 Longitude [degrees]	Double	8	H+12
7	Height	椭球高，米。 Ellipsoidal Height [m]	Double	8	H+20
8	Undulation	Undulation (m)	Float	4	H+28
9	North Velocity	北向速度（负值为南向），m/s。 Velocity in a northerly direction (a negative value implies a southerly direction) [m/s]	Double	8	H+32
10	East Velocity	东向速度（负值为西向），m/s。 Velocity in an easterly direction (a negative value implies a westerly direction) [m/s]	Double	8	H+40
11	Up Velocity	天向速度，m/s。 Velocity in an up direction [m/s]	Double	8	H+48
12	Roll	横滚角，沿 y-axis 右手螺旋，单位为度。 Right-handed rotation from local level around y-axis in degrees	Double	8	H+56
13	Pitch	俯仰角，沿 x-axis 右手螺旋，单位为度。 Right-handed rotation from local level around x-axis in degrees	Double	8	H+64
14	Azimuth	航向角，沿 z-axis 左手螺旋，单位为度。 从北向顺时针为正。这是由 IMU 陀螺经组合滤波器计算出的惯性方位角。	Double	8	H+72

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Left-handed rotation around z-axis in degrees clockwise from North This is the inertial azimuth calculated from the IMU gyros and the SPAN filters.			
15	Lat σ	Latitude standard deviation (m)	Float	4	H+80
16	Long σ	Longitude standard deviation (m)	Float	4	H+84
17	Height σ	Height standard deviation (m)	Float	4	H+88
18	North Vel σ	North velocity standard deviation (m/s)	Float	4	H+92
19	East Vel σ	East velocity standard deviation (m/s)	Float	4	H+96
20	Up Vel σ	Up velocity standard deviation (m/s)	Float	4	H+100
21	Roll σ	Roll standard deviation (degrees)	Float	4	H+104
22	Pitch σ	Pitch standard deviation (degrees)	Float	4	H+108
23	Azimuth σ	Azimuth standard deviation (degrees)	Float	4	H+112
24	Ext sol stat	Extended solution status see Table: Extended Solution Status	Hex	4	H+116
25	xxxx	32 位 CRC 校验。 32-bit CRC	Hex	4	H+120
26	[CR][LF]	语句结束符。 Sentence Terminator (ASCII only)	-	-	-

3.2.34. JAMPSDDETECT 功率频谱密度样本

适用产品：M720、M722

该协议用于输出干扰检测结果,包括干扰个数, 干扰所在的频率以及干扰强度。

- Message ID: 1968
- Log Type: Async

输出示例：

```
#JAMPSDDETECTA,COM2,0,81.0,APPROXIMATE,2331,221991.000,00008000,5103,50425;L1,NBI,1575419375,46*78aca40c
```

表3-84 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	JAMPSDDETECT	JAMPSDDETECT 报文头。 Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.		H	0
2	RFChannel	射频通道 L1/L2/L5	Ulong	4	H
3	JamNum	干扰个数	Ulong	4	H+4
4	JamFrequency	干扰所在的频率	Ulong	4	H+8
5	JamPower	干扰强度(单位: dB)	Ulong	4	H+12
Next jam offset = H + 4 + (#jam x 8)					
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+ 4 + (#jam x 8)
variable	CR][LF] 语句结束符。				

干扰类型检测报文从 S33.5 开始引入。

表3-85 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	JAMPSDDETECT	JAMPSDDETECT 报文头。		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.			
2	RFChannel	射频通道: L1/L2/L5	Ulong	4	H
3	JamType	干扰类型: UNKNOWN/NBI/CHIRP/PULSE/WBI	Ushort	2	H+4
4	JamFrequency	干扰所在的频率	Ulong	4	H+6
5	JamPower	干扰强度(单位: dB)	Ulong	4	H+10
variable	xxxx	32 位 CRC 校验。32-bit CRC (ASCII and Binary only)	Ulong	4	H+14
variable	CR][LF] 语句结束符。				

表3-86 RF Channel

Binary	ASCII
0	NONE
1	L5
2	L1
4	L2

表3-87 Jam Type

Binary	ASCII
0	UNKNOWN
1	NBI
2	CHIRP
3	PULSE

Binary	ASCII
4	WBI

3.2.35. JAMPSDFINAL 光谱分析样本

适用产品：M720、M722

JAMPSDFINAL 用于输出频谱分析结果，输出不同点数的 FFT 频谱结果。包括抗干扰前的和抗干扰后的频谱数据，抗干扰之后的频谱只支持 1024 点，抗干扰前的频谱支持最大 65536 点。

- **Message ID: 2063**
- **Log Type: Async**

输出示例：

```
#JAMPSDFINALA,52.0,0,0,COM2,FINESTEERING,2381,352258000,26ffe8ae,00008000
;FILTERBEFORE-L1,1534.000000,101562.000000,1024,7331,2165,1623,1650,499,4
95,513,1334,1500,241,399,723,1048,725,710,1004,1848,1718,935,977,2042,194
9,1651,1655,1842,1714,1770,2819,2663,1758,1563,731,808,963,1094,1701,3400
,3102,1268,2966,1942,1768,1514,1884,2747,1803,1280,664,1024,3155,3232,186
8,2247,1468,320,118,533,1100,1773,2707,2427,1995,1804,1260,1944,2795,2775
,3548,2836,1940,1478,1325,2813,2916,1881,770,616,1216,1359,2040,2849,2090
,832,774,1791,1236,1056,1650,1706,2071,2619,1538,966,1377,1130,1773,1523,
1146,1577,1327,964,297,1269,370,664,819,600,850,1603,1107,375,375,1776,14
17,844,1373,1166,1906,1502,1016,469,471,1425,2912,3856,2965,1008,828,1411
,1434,1119,580,1000,1046,1203,2261,3064,1862,1964,1213,966,917,1323,2565,
3066,1856,309,1719,1745,1452,1049,832,1147,590,236,261,376,892,1102,1302,
493,302,1246,1096,568,284,1347,2522,2507,2074,1163,472,940,1034,1075,1251
,992,638,1680,1942,1126,1849,1563,256,739,1893,1521,874,1388,723,847,1103
,882,851,790,695,1319,1842,1091,211,797,884,413,1183,869,497,323,1028,122
1,652,994,587,637,403,449,938,1366,1009,656,454,1240,2410,1663,501,1354,1
104,235,341,472,733,1040,701,719,1106,659,951,1322,972,1323,927,814,1864,
526,1065,942,1643,545,1538,1432,1428,2111,1238,683,235,1623,2525,1836,177
```

8,238,608,605,1166,1125,1508,2155,1546,658,527,372,1357,1480,979,805,822,
598,618,900,1134,1101,942,1054,1164,955,2161,1784,600,922,1969,1088,973,5
30,29,449,974,1130,1123,611,977,1565,993,773,102,371,369,465,619,320,601,
360,333,193,289,706,894,927,894,372,468,406,408,530,434,155,873,899,167,2
82,152,720,854,710,377,626,465,432,645,598,561,580,744,497,34,373,263,54,
175,105,124,172,130,187,316,327,205,110,102,157,91,57,141,74,81,107,287,3
82,214,74,206,212,187,303,244,19,85,163,144,61,169,101,89,164,103,30,91,6
0,25,121,212,125,63,65,40,63,57,61,97,34,45,100,117,100,69,6,50,77,50,56,
78,93,35,6,79,47,52,58,72,58,28,11,33,34,61,89,48,54,47,16,64,99,129,86,2
3,8,56,71,85,11,62,50,84,70,59,53,61,59,49,70,69,38,89,83,73,50,41,50,66,
85,76,38,61,77,94,73,69,46,17,54,81,51,25,39,50,82,36,24,26,45,34,28,16,3
2,44,20,105,65,11,59,26,56,76,72,57,35,68,70,69,68,15,35,87,76,59,37,38,2
2,34,21,50,70,108,69,98,96,73,90,136,91,51,42,25,59,60,45,6,26,23,52,87,6
6,83,41,34,56,37,11,63,41,14,34,37,43,59,69,45,11,54,46,90,97,109,93,107,
79,64,41,78,97,57,60,34,39,42,39,56,28,29,21,19,81,79,69,74,54,49,58,61,6
1,103,92,49,43,17,49,38,37,60,11,43,46,102,50,9,86,102,108,34,45,11,53,15
,18,75,143,147,171,80,37,26,103,104,112,203,185,57,47,112,98,10,36,132,14
0,118,50,92,109,108,13,76,176,41,106,69,20,248,350,225,159,242,347,392,43
0,690,879,785,576,526,605,467,438,465,333,99,117,209,244,369,245,53,294,4
83,658,1028,836,574,670,856,670,344,643,787,807,420,792,776,571,1575,1378
,646,391,340,831,356,1407,2500,1840,806,857,882,1541,2268,2273,1167,1037,
751,321,451,517,497,1066,1007,1164,903,862,295,1054,1977,2207,2703,1887,3
08,714,299,784,1230,2142,2295,1754,1045,339,1173,1707,911,756,1032,1089,1
258,1841,2354,1491,517,2242,1547,1133,2368,1876,637,1221,2220,2914,1973,1
620,2622,2412,2187,2269,1225,548,1474,1505,1141,821,1491,1228,1165,753,13
70,1630,889,2513,2221,1096,1122,420,1382,3857,4543,1830,4778,5856,4101,36
0,3215,3965,2786,308,1779,1175,1346,1821,1848,1893,1472,747,946,293,998,1
971,2119,1691,1477,1667,2436,3281,2087,2010,1716,1358,1217,1148,1490,1444
,1576,1024,480,606,655,897,1014,1399,566,402,388,585,378,870,685,33,1094,
625,838,253,1161,737,793,1082,1463,812,792,393,480,741,833,1209,909,242,1

016,780,564,595,687,1373,1092,1019,1810,1565,2091,1401,974,636,1426,3075,
 1996,1040,603,723,1006,1417,1052,459,362,370,996,563,995,894,1470,2143,16
 63,870,1598,1120,527,665,998,1186,1622,1390,383,543,1142,837,784,727,908,
 2427,2284,1132,555,579,585,1305,2023,1925,1212,938,318,1325,2141,1429,877
 ,86,678,1033,462,246,531,2135,3373,2989,2168,975,125,245,1206,1269,973,80
 2,287,918,1584,1356,1842,2805,2490,2976,2895,1733,735,645,1488,1034,2093,
 2199,2258,845,786,1513,2477,1835,328,1246,984,632,747,1297,928,1196,1736,
 1093,338,1251,1034,1506,2345,1789,1685,1505,1604,961,181,430,668,890,842,
 677,654,602,947,239,451,1094,17,1412,641,352,631,1201,2220,2471,715,458,2
 63,706,1307,1344,1002,732,526,946,1147,748,263,478,503,538,728,808,1421,1
 565,1050,546,1154,1819,1711,875,185,850,1058,1145,2503,1789,1178,1087,128
 8,824,345,306,276,960,2312,2273,1844,1844,1844*d1561d16

表3-88 报文格式说明

Field	Field type	Description	Forma t	Binary Bytes	Binary Offset
1	JAMPSDFINAL	JAMPSDFINAL 报文头。 Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.	-	H	0
2	StatusWord	status word containing information about the configuration of the spectral analysis datasource: FILTERBEFORE or FILTERAFTER RFPATH :	Ulong	4	H

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		L1/L2/L5/LBAND			
3	Frequency start	Frequency represented by first data sample(MHz)	Float	4	H+4
4	Resolution bandwidth	the resolution bandwidth(HZ)	Float	4	H+8
5	#samples	number of spectral density samples	Ulong	4	H+12
6	Sample	density samples	Ulong	4*NumOfSamples	H+16
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+16+4*NumOfSamples
variable	[CR][LF] 语句结束符。 Sentence terminator (ASCII only) - - -				

3.2.36. LBDBEAMSTAT LBand 波束列表

适用产品：M720、M722

LBDBEAMSTAT 已知的 LBand 波束列表。

- Message ID: 3057
- Log Type:

输出示例：

```
#LBDBEAMSTAT,30.5,0,0,COM2,FINESTEERING,2376,98030000,d323b507,00000000;
1,"SIO",1,1,1,1,83.50,1545790000,2400,C685,5C08,5000*8719cafe
```

表3-89 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
-------	------------	-------------	--------	--------------	---------------

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	LDBEAMSTAT header	报文头。 Log header,see ASCII, or Binary.		H	0
2	#BeamNumber	波束数量 Beam number	Ulong	4	H
3	BeamName	波束名称见表波束名称说明 Beam/transmitting satellite name, see BeamName	Char[]	8	H+4
4	Access	波束可用标志 Beam service availability flag	Uchar	1	H+12
5	UseServiceID	解码时是否判断服务码 Enable/disable service ID check to confirm the correct service is received.	Uchar	1	H+13
6	UseDescramble	解码时是否使用干扰向量 Enables/disables the descrambler.	Uchar	1	H+14
7	UsePrescramble	解码时是否使用预扰向量 Enables/disables the prescrambling	Uchar	1	H+15
8	Longitude	GEO 卫星经度 Transmitting satellite longitude (degrees)	Float	4	H+16
9	Frequency	频率 Frequency (Hz)	Ulong	4	H+20
10	DataRate	数据率 Baud rate (bps)	Ulong	4	H+24
11	ServiceID	报文服务码 Service id	Hex Ushort	2	H+28
12	DescramblerInit	报文干扰向量	Hex	2	H+30

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		The intialisation value for the descrambler.	Ushort		
13	DopplerRange	频率搜索范围 Search window from the center frequency	Ulong	4	H+
18...	H + 4 + 32 * #TrackNumber				
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)			
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)			

3.2.37. LBDTRACKINFO LBand 各通道跟踪状态

适用产品：M720、M722

LBDTRACKINFO 输出 LBand 各通道跟踪状态。

该条报文包含了跟踪通道数、各通道跟踪信息。

- **Message ID: 3056**
- **Log Type:**

输出示例：

```
#LBDTRACKINFOA,52.0,0,0,COM2,COARSESTEERING,2381,390227000,3d7cea15,00008000;0*65961a69
```

表3-90 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	LBDTRACKINFO header	报文头。 Log header, see ASCII, or Binary.		H	0
2	#TrackNumber	跟踪通道数 Number of records to follow	Ulong	4	H
3	BeamName	波束名称说明	Char[]	8	H+4

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Beam/transmitting satellite name, see BeamName			
4	Status	通道状态 Tracking status word. See LBand Signal Tracking Status	Hex Ushort	2	H+12
5	ServiceID	服务 ID Service ID of the assigned beam	Hex Ushort	2	H+14
6	Frequency	频率 Frequency assigned to this L-Band beam (Hz)	Ulong	4	H+16
7	BautRate	数据符号率 Baud rate of assigned beam	Ushort	2	H+20
8	reserved	保留 Reserved	Ushort	2	H+22
9	LockTime	锁定时长 Lock time (s)	Float	4	H+24
10	CNO	载噪比 Carrier to noise density ratio (dB-Hz)	Float	4	H+28
11	Doppler	多普勒 Signal Doppler (Hz)	Float	4	H+32
12	PhaseStd	环路误差 Phase error standard deviation (cycles)	Float	4	H+36
13	BitErrorRate	误码率 Estimated pre-Viterbi Bit Error Rate (BER)	Float	4	H+40
14	ViterbiSymbols	viterbi 解码数量	Ulong	4	H+44

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Total Viterbi symbols			
15	CorrectedViterbi	viterbi 纠正数量 Corrected Viterbi symbols	Ulong	4	H+48
16	UniqueBitNum	特定位数量 Total unique word bits	Ulong	4	H+52
17	BadUniqueWordsBit	特定位错误数 Bad unique word bits	Ulong	4	H+56
18...	H + 4 + 56 * #TrackNumber				
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)			
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)			

表3-91 通道状态

Bit	Description	Range Value
0	tracking state	0 = close
1		1 = idle
2		2 = searching
3		3 = track
4		4 = hold
5		
6		
7		
8	frame synchronize flag	0 = not found frame head 1 = found frame head

Bit	Description	Range Value
9	reserved	

3.2.38. NICEPH NAVIC 星历

适用产品：M720、M722

- Message ID: 3014
- Log Type: Asynch

输出示例：

```
#NICEPHA,38.5,2,0,COM2,SATTIME,2376,96592000,d323b507,00000000;2,1352,-7.49779399E-04,-8.48103809E-11,0.00000000e+00,0,95760,-1.86E-09,2.9901245507E-09,171,0,0,0,4.682689905E-06,2.332776785E-05,-1.452863216E-07,-2.197921276E-07,-625.1250000,145.6250000,-1.188620939E-09,-7.507562004E-01,95760,2.1423683502E-03,6493.435717,-9.046985227E-01,-2.967571005,-2.51867634E-09,5.0731753702E-01,0,0,0,0*75a224b7
```

表3-92 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	Log header	NICEPH 报文头 Log header. For information about the log headers, see ASCII, or Binary.	-	H	0
2	PRN	卫星标识 Satellite Identifier	Ulong	4	H
3	WN	自 NAVIC 系统时间开始以来的周数 Week number since the NavIC system time start epoch (August 22nd 1999)	Ulong	4	H+4
4	Af0	卫星钟差参数, s Clock bias (s)	Double	8	H+8
5	Af1	卫星钟差参数, s/s	Double	8	H+16

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Clock drift (s/s)			
6	Af2	卫星钟差参数, s/s/s Clock drift rate (s/s ²)	Double	8	H+24
7	URA	卫星精度 SV Accuracy	Ulong	4	H+32
8	toc	卫星钟差参数, s Reference time for the satellite clock corrections(s)	Ulong	4	H+36
9	TGD	群延迟 Total group delay (s)	Double	8	H+40
10	DeltaN	卫星平均角速度的改正值, rad/s Mean motion difference (radian/s)	Double	8	H+48
11	IODEC	星历和时钟数据期号 Issue of data ephemeris and clock	Ulong	4	H+56
12	Reserved	保留字段	Ulong	4	H+60
13	L5 Health	L5 SPS 信号导航数据健康状态 Health status of navigation data on L5 SPS signal 0=OK; 1=bad	Ulong	4	H+64
14	S Health	S SPS 信号导航数据健康状态 Health status of navigation data on S SPS signal 0=OK; 1=bad	Ulong	4	H+68
15	Cuc	纬度幅角 (余弦振幅, rad) Amplitude of the cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+72
16	Cus	纬度幅角 (正弦振幅, rad)	Double	8	H+80

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Amplitude of the sine harmonic correction term to the argument of latitude (radians)			
17	Cic	轨道半径（余弦振幅，m） Amplitude of the cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+88
18	Cis	轨道半径（正弦振幅，m） Amplitude of the sine harmonic correction term to the angle of inclination (radians)	Double	8	H+96
19	Crc	倾角（余弦振幅，rad） Amplitude of the cosine harmonic correction term to the orbit radius (m)	Double	8	H+104
20	Crs	倾角（正弦振幅，rad） Amplitude of the sine harmonic correction term to the orbit radius (m)	Double	8	H+112
21	IDOT	轨道倾角变化率，rad/s Rate of inclination angle (radians/s)	Double	8	H+120
22	M0	参考时间的平近点角，rad Mean anomaly (radians)	Double	8	H+128
23	toe	星历时间 Time of ephemeris (s)	Ulong	4	H+136
24	Ecc	卫星轨道偏心率 Eccentricity (dimensionless)	Double	8	H+140
25	RootA	卫星轨道长半轴的平方根 Square root of semi-major axis (sqrt(m))	Double	8	H+148
26	Omega0	升交点赤经，rad	Double	8	H+156

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Longitude of ascending node (radians)			
27	Omega	近地点幅角, rad Argument of perigee (radians)	Double	8	H+164
28	OmegaDot	升交点赤经变化率, rad/s Rate of RAAN (radians/s)	Double	8	H+172
29	IO	轨道倾角, rad Inclination angle (radians)	Double	8	H+180
30	Spare	备用字段 spare bits after IDOT	Ulong	4	H+188
31	Spare	备用字段 spare bits after IO	Ulong	4	H+192
32	Alert flag	使用导航数据的风险由用户自行承担 The utilization of navigation data shall be at the users' own risk. 1=Alert; 0=OK	Ulong	4	H+196
33	AutoNav flag	设置为 1 时, 卫星处于 AutoNav 模式。卫星广播来自 AutoNav 数据集的主要导航参数, 最长 7 天没有地面上行链路 When set to 1, satellite is in AutoNav mode. Satellite broadcasts primary navigation parameters from AutoNav data sets with no uplink from ground for maximum of 7 days	Ulong	4	H+200
34	xxxx	32 位 CRC 校验 32-bit CRC (ASCII and Binary only)	Hex	4	H+204
35	[CR][LF]	语句结束符 (仅 ASCII) Sentence terminator (ASCII only)	-	-	-

3.2.39. PPPP PPP 定位的位置信息

适用产品：M720、M722

PPPP 输出 PPP position 定位的位置信息。

本指令包含接收机 PPP 定位的位置及定位精度、状态等信息。

- **Message ID: 3026**

- **Log Type: Synch**

输出示例：

```
#PPPPA,49.0,0,0,COM2,FINESTEERING,2381,25770000,cba257c1,00008000;SOL_COM  
PUTED,PPP,WGS84,0,31.15969067616,121.17808968070,36.1041,10.0212,0.0105,0  
.0111,0.0205,10.000,0.000,60,53,14,17,3f20,37,00,00,00,400*9fa6b876
```

表3-93 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	PPPP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态，见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型，见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度，单位：度。 Latitude (degrees)	Double	8	H+4
7	lon	经度，单位：度。	Double	8	H+12

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Longitude (degrees)			
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the WGS84 ellipsoid (m) When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差, 单位: 米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差, 单位: 米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间, 单位: 秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。	Uchar	1	H+55

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Number of satellites with L1/E1/B1 signals used in solution			
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.40. QZSEPH QZSS 星历

适用产品：M720、M722

- Message ID: 3013
- Log Type: Asynch

输出示例：

```
#QZSEPHA,40.0,3,0,COM2,SATTIME,2376,96540000,d323b507,00000000;194,97200.
```

0,0.00000E+00,-3.41061E-13,-1.60933E-06,105,3.50687500E+02,1.691499029E-09,-5.452766756E-01,1.179054379E-05,7.5452667777E-02,1.606717706E-05,4.216372363E+07,97200.0,-2.073124051E-06,-8.710803186E-01,7.133930922E-07,6.9682963723E-01,-2.98750000E+02,-1.5809571253E+00,-1.71221418E-09,-1.667569461E-09,2376,7.84000000E+00,0,1.396983862E-09,873,0,0,0,0,93618.0,105,2376,FALSE,7.292401242E-05*f4aa5f33

表3-94 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	Log header	QZSEPH 报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	PRN	卫星 PRN 号 Satellite PRN number	Ulong	4	H
3	toc	卫星钟差参数, s SV clock correction term (s)	Double	8	H+4
4	a _{f2}	卫星钟差参数, s/s/s Clock aging parameter (s/s/s)	Double	8	H+12
5	a _{f1}	卫星钟差参数, s/s Clock aging parameter (s/s)	Double	8	H+20
6	a _{f0}	卫星钟差参数, s Clock aging parameter (s)	Double	8	H+28
7	IODE1	星历数据期号 1 Issue of ephemeris data 1	Ulong	4	H+36
8	crs	轨道半径 (正弦振幅, m) Amplitude of sine harmonic correction term to the	Double	8	H+40

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		orbit radius (m)			
9	ΔN	卫星平均角速度的改正值, rad/s Mean motion difference (radians/s)	Double	8	H+48
10	M_0	参考时间的平近点角, rad Mean anomaly of reference time (radius)	Double	8	H+56
11	cuc	纬度幅角 (余弦振幅, rad) Amplitude of cosine harmonic correction term to the argument of latitude (radians)	Double	8	H+64
12	ecc	卫星轨道偏心率 Eccentricity (dimensionless)	Double	8	H+72
13	cus	纬度幅角 (正弦振幅, rad) Amplitude of sine harmonic correction term to the argument of latitude (radians)	Double	8	H+80
14	A	卫星轨道长半轴 Semi-major axis (m)	Double	8	H+88
15	toe	星历的参考时间, 单位: s Reference time for ephemeris (s)	Double	8	H+96
16	cic	倾角 (余弦振幅, rad) Amplitude of cosine harmonic correction term to the angle of inclination (radians)	Double	8	H+104
17	ω_0	赤经, rad Right ascension (radians)	Double	8	H+112
18	cis	倾角 (正弦振幅, rad) Amplitude of sine harmonic correction term to the angle of inclination (radians)	Double	8	H+120

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
19	l_0	参考时间的轨道倾角, rad Inclination angle at reference time (radians)	Double	8	H+128
20	crc	轨道半径 (余弦振幅, m) Amplitude of cosine harmonic correction term to the orbit radius (m)	Double	8	H+136
21	ω	近地点幅角, rad Argument of perigee (radians)	Double	8	H+144
22	$\dot{\omega}$	赤经变化率, rad/s Rate of right ascension (radians/s)	Double	8	H+152
23	$\dot{i}$	轨道倾角变化率, rad/s Rate of inclination angle (radians/s)	Double	8	H+160
24	week	GPS 参考周 GPS reference week number	Ulong	4	H+168
25	URA	用户测距精度 User Range Accuracy variance, m^2 . The ICD specifies that the URA index transmitted in the ephemerides can be converted to a nominal standard deviation value using an algorithm listed there. We publish the square of the nominal value (variance)	Double	8	H+172
26	health	健康状态 Health status - bits 0-5: 6-bit health code as defined in QZSS Interface Specification - bit 6: Alert flag	Ulong	4	H+180
27	tgd	估计的群延迟差	Double	8	H+184

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		Estimated group delay difference (s)			
28	iodc	时钟数据期号 Issue of data clock	Ulong	4	H+192
29	Fit Interval	星历数据有效期 Curve fit interval: 0 = Ephemeris data are effective for 2 hours 1 = Ephemeris data are effective for more than 2 hours	Uchar	1	H+196
30	Reserved	保留字段	Uchar	1	H+197
31	Reserved	保留字段	Uchar	1	H+198
32	Reserved	保留字段	Uchar	1	H+199
33	tow	子帧时间戳 Time stamp of subframe 0 (s)	Double	8	H+200
34	IODE2	星历数据期号 2 Issue of ephemeris data 2	Ulong	4	H+208
35	z week	z 计数 Z count week number. This is the week number from subframe 1 of the ephemeris. The 'toe week' (field #7) is derived from this to account for rollover	Ulong	4	H+212
36	AS	反欺骗 Anti-spoofing on: 0= FALSE 1=TRUE	Enum	4	H+216
37	N	改正的卫星平均角速度 Corrected mean motion (radians/s)	Double	8	H+220
38	xxxx	32 位 CRC 校验	Ulong	4	H+228

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
		32-bit CRC (ASCII and Binary only)			
39	[CR][LF]	语句结束符（仅 ASCII） Sentence terminator (ASCII only)	-	-	-

3.2.41. RTCMINSTATUS 物理端口 RTCM 输入统计

适用产品：M720、M722

RTCMINSTATUS 所有接收到 RTCM 数据的 RTCM 端口所接收到的 RTCM message 数据包的统计。

- **Message ID: 23921**
- **Log Type: ASync**
注：仅支持 3 个物理端口输入的 RTCM 数据包统计，超过 3 个的不支持

输出示例：

```
#RTCMINSTATUSA,60.5,0,0,COM2,FINESTEERING,2381,26140000,cb257c1,00008000;1,COM2,8,1005,135,4063,459,1124,270,1084,135,1094,135,1114,135,1074,135,1033,14*9383c95d
```

表3-95 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	RTCMINSTATUS	报文头。 Log header. For information about the log headers, see ASCII, Abbreviated ASCII or Binary.		H	0
2	#RTCMPortNum	RTCM 端口个数	Ulong	4	H
3	RTCMPortName	RTCM 端口名称 (ASCII 字符) binary	Ulong	4	H+4

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		为端口 ID 号			
4	#InputMessageTypeNum	输入报文类型个数	Ulong	4	H+8
5	InputMessageID	输入报文 ID	Ulong	4	H+12
6	InputMessageCount	输入报文计数	Ulong	4	H+16
Next RTCMPortName offset = H + 4 + (4 + 4 + InputMessageNumber x 8)					
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+ 4 + (4 + 4 + InputMessageTypeNum x 8) + (4 + 4 + InputMessageCount x 8).....
variable	[CR][LF] 语句结束符。 Sentence terminator (ASCII only) - - -				

3.2.42. RTKP RTK 位置信息

适用产品：M720、M722

RTKP 输出 RTK 位置信息。

本指令包含接收机 RTK 定位的位置及定位精度、状态等信息。

- **Message ID: 3024**
- **Log Type: Synch**

输出示例：

```
#RTKPA,59.5,0,0,COM2,FINESTEERING,2381,26222000,cba257c1,00008000;SOL_COM
PUTED,NARROW_INT,WGS84,0,31.15969322594,121.17808142758,36.1054,10.0212,0
.0020,0.0023,0.0051,2.000,0.000,60,53,45,47,0528,37,1f,07,00,654*2553efcd
```

表3-96 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	RTKP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型, 见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度, 单位: 度。 Latitude (degrees)	Double	8	H+4
7	lon	经度, 单位: 度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the WGS84 ellipsoid (m) When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差, 单位: 米。	Float	4	H+36

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Longitude standard deviation (m)			
12	hgt σ	高度标准差，单位：米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间，单位：秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留	Hex	1	H+61

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Reserved			
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.43. RTKV RTK 速度信息

适用产品：M720、M722

RTKV 输出 RTK 速度。

本指令包含接收机计算出的 RTK 速度信息。此外，接收机还报告了速度状态指示符，对指示相应的数据是否有效非常有用。该速度测量有时会带来相关的延迟。

- **Message ID: 3025**
- **Log Type: Synch**

输出示例：

```
#RTKVA,60.5,0,0,COM2,FINESTEERING,2381,26258000,cba257c1,00008000;SOL_COM  
PUTED,NARROW_INT,0,0.0068,337.573303,-0.0017,0.000,1.000,0.000*54c8416b
```

表3-97 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	RTKV header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态，见 Vel Solution Status Solution status, see Vel Solution Status	Enum	1	H+0
3	vel type	速度类型，见 Velocity Type	Enum	1	H+1

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Velocity type, see Velocity Type			
4	reserved		Ushort	2	H+2
5	hor spd	对地水平速度，单位：米/秒。 Horizontal speed over ground, in metres per second	Double	8	H+4
6	trk gnd	相对于真北的实际对地运动方向（相对地面轨迹），单位：度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Double	8	H+12
7	vert spd	垂直速度，正值表示高度增加（向上），负值表示高度下降（向下），单位：米/秒。 Vertical speed, in metres per second, where positive values indicate increasing altitude (up) and negative values indicate decreasing altitude (down)	Double	8	H+20
8	latency	根据速度时标计算的延迟值，用历元时间减去延迟可得到更准确的速度结果，单位：秒。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results (s)	Float	4	H+28
9	age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+32
10	Reserved		Float	4	H+36
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
11	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.44. SATINVIEW 可见卫星

适用产品：M720、M722

SATINVIEW 输出可见卫星。

该消息包含可见卫星列表及卫星信息。

- **Message ID: 3054**
- **Log Type: Asynch**

输出示例：

```
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;G,
TRUE,FALSE,14,5,0,50.6,289.3,1028.796,1027.914,6,0,34.5,97.1,-1806.257,-1
806.469,7,0,10.7,75.8,605.218,605.006,9,0,9.1,38.6,-3378.872,-3379.896,11
,0,63.2,54.5,-186.310,-187.193,12,0,11.9,239.1,-2419.830,-2420.443,13,0,3
8.3,182.2,3357.870,3356.987,15,0,14.3,212.4,3821.185,3822.174,19,0,11.9,1
55.5,-3956.713,-3957.737,20,0,61.2,347.7,-266.378,-267.260,21,0,63.1,0.7,
-867.489,-868.372,25,0,7.1,269.1,-1745.827,-1746.686,29,0,20.5,317.8,3066
.892,3065.869,30,0,14.9,108.1,1738.150,1737.267*546edc1e
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;R,
TRUE,FALSE,7,1+1,0,39.4,27.1,-3292.574,-3291.567,2-4,0,83.3,265.2,111.067
,110.171,3+5,0,26.7,216.3,3526.137,3527.146,17+4,0,50.1,345.3,2575.890,25
74.991,18-3,0,7.3,324.0,3734.675,3735.680,23+3,0,3.4,131.8,-3457.309,-345
6.301,24+2,0,58.5,101.8,-1063.629,-1064.527*ab3e85a6
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;E,
TRUE,TRUE,14,4,0,47.4,176.8,2306.788,2305.906,6,0,25.1,193.8,2949.915,295
0.905,9,0,4.2,202.6,3074.840,3075.830,10,0,53.6,339.1,1150.350,1149.467,1
1,0,31.4,313.9,1790.245,1791.235,12,0,62.6,27.1,305.487,304.604,16,0,16.5
,263.9,-1640.427,-1639.437,18,5,35.9,300.6,5058.817,5058.524,19,0,54.2,71
.3,-524.210,-524.699,24,0,16.3,237.7,-2206.520,-2205.531,25,0,12.5,287.3,
-845.068,-844.078,29,0,14.3,40.1,-2406.453,-2407.311,31,0,2.1,191.0,-2172
.055,-2171.065,33,0,26.8,94.9,-1405.779,-1406.661*44ceafe5
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;C,
```

```

TRUE,TRUE,29,1,0,47.2,138.7,23.363,22.577,2,0,35.7,235.3,79.551,78.765,3,
0,53.0,200.5,36.777,35.991,4,0,36.9,120.0,24.443,23.657,5,0,14.3,253.8,47
.340,46.554,7,0,54.1,205.1,-1117.544,-1118.030,8,0,54.7,298.6,-289.906,-2
90.391,10,0,57.8,229.9,-820.202,-820.687,11,0,60.1,127.6,-1349.065,-1349.
939,12,0,8.8,135.3,-2978.874,-2979.749,13,0,48.6,285.1,-210.931,-211.416,
14,0,82.3,294.0,-52.092,-52.967,21,0,12.2,42.4,-3012.362,-3013.236,28,0,2
7.3,317.5,2748.321,2747.446,33,0,57.2,237.3,1426.275,1425.400,34,0,35.1,1
37.4,-2535.878,-2535.942,38,0,62.6,325.1,-314.391,-315.266,39,0,5.7,169.6
,1358.357,1357.483,40,0,55.0,185.1,-1183.196,-1183.681,41,0,8.9,227.5,289
6.933,2896.058,42,0,61.3,31.5,-1589.481,-1590.355,43,0,89.5,192.6,-37.150
,-38.025,49,1,58.7,319.7,1632.824,1632.535,50,1,34.8,40.6,-2557.209,-2557
.499,56,0,0.0,0.0,0.000,0.000,59,0,49.3,145.7,60.808,59.933,60,0,33.8,240
.7,64.200,63.414,61,0,53.0,200.7,30.622,29.608,62,0,34.4,123.2,-28.271,-2
9.145*edfa2c31
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;S,
TRUE,FALSE,3,130,0,49.3,145.7,0.062,-0.230,143,0,53.0,200.7,0.059,-0.233,
144,0,33.8,240.7,0.034,-0.258*849cb00a
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;J,
TRUE,FALSE,4,194,0,18.1,171.6,-507.822,-508.680,195,0,71.9,88.2,795.131,7
94.248,196,0,74.3,126.9,-704.556,-705.439,199,0,53.3,168.8,1.616,2.069*cf
dc5793
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;I,
TRUE,TRUE,7,2,0,20.9,267.2,-470.087,-470.305,3,0,33.1,234.4,113.456,113.2
38,4,0,37.5,199.8,-592.270,-592.911,5,0,67.5,224.5,236.790,236.572,7,0,53
.5,163.7,80.029,79.811,9,0,4.2,239.9,384.136,383.918,10,0,50.8,165.1,-41.
743,-41.961*2c551da5
#SATINVIEWA,31.0,0,0,COM2,FINESTEERING,2376,97453000,d323b507,00000000;L,
TRUE,FALSE,1,1,1,35.0,236.0,0.000,0.000*e024e9e5

```

表3-98 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	SATINVIEW header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	Satellite System	GNSS 卫星系统列表 GNSS satellite system identifier. See Satellite System 卫星系统使用与 GNSSOBS 报文一致的标识符定义	Enum	4	H
3	sat vis	卫星是否可视 Is satellite visibility valid? 0 = FALSE 1 = TRUE	Enum	4	H+4
4	almanac flag	是否使用了完整历书 Was complete almanac used? 0 = FALSE 1 = TRUE	Enum	4	H+8
5	#sat	所有数据的卫星数 Number of satellites with data to follow	Ulong	4	H+12
6	Satellite ID	卫星 PRN 号 (GPS: 1 到 32 , GLONASS: 1 到 24, Galileo 1 到 36, BDS 1 到 63, SBAS 120 到 158 及 183 到 192, QZSS 193 到 202)	Ulong	4	H+16

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		在二进制消息中,卫星 ID 字段由 1 个 Short 和 1 个 Ushort 组成。 最低的两个字节是系统标识符: (如 GPS 的 PRN, GLONASS 的通道号), 为 USHORT 类型; 最高的两个字节是 GLONASS 的频率通道, 其他系统这两个字节值为零。 在 ASCII 消息中, 卫星 ID 字段是系统标识符。如果系统是 GLONASS, 而频率通道不是零, 则在系统标识符后加上频率通道数。例如, 系统标识符是 13, 频率通道-2, 输出为 13-2,数据类型为 Short 类型。 In binary logs, the satellite ID field is 4 bytes. The 2 lowest order bytes, interpreted as a USHORT, are the system identifier: for instance, the PRN for GPS or the slot for GLONASS. The 2 highest-order bytes are the frequency channel for GLONASS, interpreted as a SHORT and zero for all other systems. In ASCII and abbreviated ASCII logs, the satellite ID field is the system identifier. If the system is GLONASS and the frequency channel is not zero, then the signed channel is appended to the system identifier. For example, slot 13, frequency channel -2 is output as 13-2. For more information, refer to PRN Numbers			

Field	Field type	Description	Format	Binary Bytes	Binary Offset
7	health	卫星健康 Satellite health as defined in System ICD	Ulong	4	H+20
8	elev	仰角，单位：度。 Elevation (degrees)	Double	8	H+24
9	az	方位角，单位：度。 Azimuth (degrees)	Double	8	H+32
10	true Doppler ¹⁾	卫星多普勒理论值-基于卫星相对于接收机的运动得出的预期多普勒频率。它是通过卫星的坐标和速度，及接收机的坐标和速度计算的， Hz Theoretical Doppler of satellite - the expected Doppler frequency based on a satellite's motion relative to the receiver. It is computed using the satellite's coordinates and velocity along with the receiver's coordinates and velocity (Hz).	Double	8	H+40
11	apparent Doppler ²⁾	该接收机的真实多普勒-与上面的理论多普勒相似，但增加了钟差改正， Hz Apparent Doppler for this receiver - the same as Theoretical Doppler above but with clock drift correction added (Hz).	Double	8	H+48
12	Next satellite offset = H + 16 + (#sat x 40)				
13	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+16+(#sat x 40)
14	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.45. SPPP 伪距位置信息

适用产品：M720、M722

SPPP 输出伪距位置信息。

本指令包含接收机伪距定位的位置及定位精度、状态等信息。

- Message ID: 3022
- Log Type: Synch

输出示例：

```
#SPPPA,40.0,0,0,COM2,FINESTEERING,2376,97207000,d323b507,00000000;SOL_COM
PUTED,SINGLE,WGS84,0,31.15968442808,121.17809062690,36.4498,10.0212,1.084
3,1.2235,2.6260,0.000,0.000,63,55,0,0,0000,37,17,07,00,0*6a622e2d
```

表3-99 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	SPPP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态，见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型，见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度，单位：度。 Latitude (degrees)	Double	8	H+4
7	lon	经度，单位：度。 Longitude (degrees)	Double	8	H+12

Field	Field type	Description	Format	Binary Bytes	Binary Offset
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the WGS84 ellipsoid (m) When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差, 单位: 米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差, 单位: 米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间, 单位: 秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	Reserved		Uchar	1	H+54
18	Reserved		Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig	GPS 和 GLONASS 信号标志	Hex	1	H+58

Field	Field type	Description	Format	Binary Bytes	Binary Offset
	mask	GPS and GLONASS signals used mask			
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.46. SPPV 伪距速度信息

适用产品：M720、M722

SPPV 输出伪距速度。

本指令包含接收机计算出的伪距速度信息。此外，接收机还报告了速度状态指示符，对指示相应的数据是否有效非常有用。该速度测量有时会带来相关的延迟。

- **Message ID: 3023**
- **Log Type: Synch**

输出示例：

```
#SPPVA,40.0,0,0,COM2,FINESTEERING,2376,97246000,d323b507,00000000;SOL_COMPUTED,DOPPLER_VELOCITY,0,0.0023,43.139939,-0.0081,0.000,0.000,0.000*3dee0783
```

表3-100 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	SPPV header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态, 见 Vel Solution Status Solution status, see Vel Solution Status	Enum	1	H+0
3	vel type	速度类型, 见 Velocity Type Velocity type, see Velocity Type	Enum	1	H+1
4	reserved		Ushort	2	H+2
5	hor spd	对地水平速度, 单位: 米/秒。 Horizontal speed over ground, in metres per second	Double	8	H+4
6	trk gnd	相对于真北的实际对地运动方向 (相对地面轨迹), 单位: 度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Double	8	H+12
7	vert spd	垂直速度, 正值表示高度增加 (向上), 负值表示高度下降 (向下), 单位: 米/秒。 Vertical speed, in metres per second, where positive values indicate increasing altitude (up) and negative values indicate decreasing altitude (down)	Double	8	H+20
8	latency	根据速度时标计算的延迟值, 用历元时间减去延迟可得到更准确的速度结果, 单位: 秒。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results (s)	Float	4	H+28

Field	Field type	Description	Format	Binary Bytes	Binary Offset
9	age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+32
10	Reserved		Float	4	H+36
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.47. TIMES 时间信息

适用产品: M720、M722

TIMES 输出各系统的时间信息, 包含接收机时钟偏差以及 UTC 时间和偏差等。TIME 也可用于计算与 GPS 参考时间相关的 PPS 信号偏差。

- Message ID: 3052
- Log Type: Synch

输出示例:

```
#TIMESA,40.0,0,0,COM2,FINESTEERING,2376,96830000,d323b507,00000000;VALID,
2025,7,21,2,53,32000,VALID,4,0,2.330890303e-04,-5.926223899e-10,0.0002330
8903,1,-2.871172855e-08,-5.926223899e-10,-0.00000002871,2,-9.128038566e-0
9,-5.926223899e-10,-0.00000000913,5,-5.670485322e-08,-5.926223899e-10,-0.
00000005670*4128b0f5
```

表3-101 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	TIMES header	报文头 Log header. For information about the log headers, see ASCII, or Binary.		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
2	clock status	时钟模型状态 Clock model status (not including current measurement data)	Enum	4	H
3	utc year	UTC 年 UTC year	Ulong	4	H+4
4	utc month	UTC 月 (0-12) UTC month (0-12) If UTC time is unknown, the value for month is 0.	Uchar	1	H+8
5	utc day	UTC 日 (0-31) UTC day (0-31) If UTC time is unknown, the value for day is 0.	Uchar	1	H+9
6	utc hour	UTC 小时 (0-23) UTC hour (0-23)	Uchar	1	H+10
7	utc min	UTC 分钟 (0-59) UTC minute (0-59)	Uchar	1	H+11
8	utc ms	UTC 毫秒 (0-60999) UTC millisecond (0-60999) Maximum of 60999 when leap second is applied.	Ulong	4	H+12

Field	Field type	Description	Format	Binary Bytes	Binary Offset
9	utc status	UTC 状态 0 = 非法 1 = 合法 2 = 警告, 由于缺乏年历, 闰秒值用默认值的 UTC status 0 = Invalid 1 = Valid 2 = Warning, leap second value is used as a default due to the lack of an almanac	Enum	4	H+16
10	System num	卫星系统数 Satellite system number	Ulong	4	H+20
11	Satellite system	卫星系统	Ulong	4	H+24
12	offset	相对于 GPS 时的接收机钟差, s. 正值表示接收机时钟早于 GPS 时, 以下为 GPS 时的计算公式: $\text{GPS 时} = \text{GPS 参考时间} - \text{钟差}$ Receiver clock offset in seconds from GPS system time. A positive offset implies that the receiver clock is ahead of GPS system time. To derive GPS system time, use the following formula: $\text{GPS system time} = \text{GPS reference time} - \text{offset.}$	Double	8	H+28

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		The GPS reference time can be obtained from the log header.			
13	offset std	接收机钟差的标准差, s Receiver clock offset standard deviation (s)	Double	8	H+36
14	utc offset	GPS 时到 UTC 时间的偏差, 通过历书参数计算, s UTC 时间= GPS 参考时间- 钟差+ UTC 偏差 The offset of GPS system time from UTC time, computed using almanac parameters. UTC time is GPS reference time plus the current UTC offset minus the receiver clock offset: UTC time = GPS reference time - offset + UTC offset	Double	8	H+44
15	Next satellite system offset = H + 20 + (#System num x 28)				
16	xxxx	32 位CRC 校验 32-bit CRC (ASCII and Binary only)	Ulong	4	H + 20 + (#System num x 28)
17	[CR][LF]	语句结束符 Sentence terminator (ASCII only)	-	-	-

表3-102 Clock Model Status

Clock Status (Binary)	Clock Status (ASCII)	Description
-----------------------	----------------------	-------------

Clock Status (Binary)	Clock Status (ASCII)	Description
0	VALID	The clock model is valid
1	CONVERGING	The clock model is near validity
2	ITERATING	The clock model is iterating towards validity
3	INVALID	The clock model is not valid

3.2.48. UVNAV 无人系统专用导航电文

适用产品：M720、M722

UVNAV(Unmanned Vehicle Navigation) 是面向无人系统专门优化的专用导航电文，包含无人系统控制锁需要的位置信息、速度信息、Heading 信息、卫星信息、状态信息等信息。

- **Message ID: 3063**
- **Log Type: Synch**

输出示例：

```
#UVNAVA,42.5,0,0,COM2,FINESTEERING,2374,439600000,29624735,00000000;63852
afc,0,31.15969324058,121.17808144944,36.0936,46.1148,0.0237,192.0890,NARR
OW_INT,0,0,0.012,0.010,0.3953,0.5923,0,0,0,0,0,0,0.0000,-0.0012,0.0
032,0.0027,110.159409,1,0,0,0,0,1752199582000000,50,0,0,0,180.000000,0.00
0000,0.000000,0.000000,0,0,1,0,0.0036,0.0046,1.000,0.0176,0.0116,0916,DOP
PLER_VELOCITY,SOL_COMPUTED,2c39,18,16,9,7,0,54,G9,G11,G19,J194,C7,C29,C38
,C41,G4,G5,G6,G12,G17,G20,G21,C8,C10,C27,C30,C32,C40,C1,C4,C59,R7,R8,R12,
R21,C3,C36,R6,E31,E33,J195,J196,J199,R22,E10,C60,R23,E12,E24,C5,C13,C20,C
2,E19,E16,G22,E4,G25,E25,E14,E11*f3d40669
```

表3-103 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	UVNAV header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
2	device ID	设备 ID Device ID	Hex	4	H+0
3	reserved	预留 Reserved	Ulong	4	H+4
4	lat	纬度，单位：度。 Latitude (degrees)	Double	8	H+8
5	lon	经度，单位：度。 Longitude (degrees)	Double	8	H+16
6	hgt	海拔高，单位：米。 Height above mean sea level (metres)	Double	8	H+24
7	hEll	大地高，单位：米。 Ellipsoidal height (metres)	Double	8	H+32
8	vel σ	速度标准差，单位：米/秒。 Velocity standard deviation (m/s)	Float	4	H+40
9	course σ	航向标准差，单位：度。 Course standard deviation (deg)	Float	4	H+44
10	pos type	定位类型，见 Position Type Position type, see Position Type	Enum	1	H+48
11	reserved	预留 Reserved	Uchar	1	H+49
12	reserved	预留 Reserved	Ushort	2	H+50
13	pos σ	水平位置标准差，单位：米。 Position standard deviation (m)	Float	4	H+52
14	hgt σ	高度标准差，单位：米。 Height standard deviation (m)	Float	4	H+56

Field	Field type	Description	Format	Binary Bytes	Binary Offset
15	hdop	水平精度因子 Horizontal dilution of precision.	Float	4	H+60
16	vdop	垂直精度因子 Vertical dilution of precision.	Float	4	H+64
17	noise	GPS 噪声 GPS noise	Long	4	H+68
18	AGC	自动增益控制监视器, 范围 0-79 Automatic gain control monitor, range 0 to 79.	Ushort	2	H+72
19	jamming status	干扰状态: 0:Unknown, 1:OK, 2:Warning, 3:Critical jamming status: 0:Unknown, 1:OK, 2:Warning, 3:Critical.	Uchar	1	H+74
20	reserved	预留 Reserved	Uchar	1	H+75
21	jamming indicator	是否发生干扰 jamming indicator.	Long	4	H+76
22	spoofing status	抗欺骗状态: 0:Unknown, 1:OK, 2:Warning, 3:Critical Spoofing status: 0:Unknown, 1:OK, 2:Warning, 3:Critical.	Uchar	1	H+80
23	reserved	预留 Reserved	Uchar	1	H+81
24	reserved	预留 Reserved	Uchar	1	H+82
25	reserved	预留	Uchar	1	H+83

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Reserved			
26	vel	速度,单位: 米/秒 Velocity (m/s)	Float	4	H+84
27	velN	N 方向速度,单位: 米/秒 N-direction velocity (m/s)	Float	4	H+88
28	velE	E 方向速度,单位: 米/秒 E-direction velocity (m/s)	Float	4	H+92
29	velD	垂直方向速度,正值表示高度增加(向上), 负值表示高度下降(向下), 单位: 米/秒 Vertical velocity, positive values indicate an increase in height (upward), negative values indicate a decrease in height (downward) (m/s)	Float	4	H+96
30	trk gnd	相对于真北的实际对地运动方向(相对地面轨迹), 单位: 度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Float	4	H+100
31	vel valid	为 TRUE 表示 NED 速度有效 True if NED velocity is valid	Uchar	1	H+104
32	reserved	预留 Reserved	Uchar	1	H+105
33	reserved	预留 Reserved	Uchar	1	H+106
34	reserved	预留 Reserved	Uchar	1	H+107

Field	Field type	Description	Format	Binary Bytes	Binary Offset
35	reserved	预留 Reserved	Long	4	H+108
36	UTC	UTC 时间, 单位: 微秒 UTC time (microsecond)	U64	8	H+112
36	solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+120
37	reserved	预留 Reserved	Uchar	1	H+121
38	reserved	预留 Reserved	Uchar	1	H+122
39	reserved	预留 Reserved	Uchar	1	H+123
40	heading	航向, 单位: 度。 Heading in degrees (0° to 359.999°)	Float	4	H+124
41	heading offset	航向偏移, 单位: 度。 Heading offset in degrees	Float	4	H+128
42	hdg std dev	航向标准差, 单位: 度。 Heading standard deviation in degrees	Float	4	H+132
43	reserved	预留 Reserved	Float	4	H+136
44	reserved	预留 Reserved	Uchar	1	H+140
45	RTCM crc status	RTCM CRC 状态, 1 为校验出错 RTCM CRC status, 1 is error;	Uchar	1	H+141
46	RTCM msg used	RTCM 报文使用状态, 1 为使用中 RTCM msg used status, 1 is used;	Uchar	1	H+142

Field	Field type	Description	Format	Binary Bytes	Binary Offset
47	heading type	航向类型 Heading type;	Uchar	1	H+143
48	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+144
49	lon σ	经度标准差, 单位: 米。 Longitude standard deviation (m)	Float	4	H+148
50	diff age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+152
51	verspd std	高程速度标准差, 单位: 米/秒。 Vertical speed standard deviation	Float	4	H+156
52	horspd std	水平速度标准差, 单位: 米/秒。 Horizontal speed standard deviation	Float	4	H+160
53	stn id	基准站 ID Base station ID	Ushort	2	H+164
54	vel type	速度类型, 见 Velocity Type Velocity type, see Velocity Type	Enum	1	H+166
55	pos sol stat	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+167
56	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+168
57	#solnSVs BDS	参与解算的 BDS 卫星数。 Number of satellites with BDS used in solution	Uchar	1	H+170
58	#solnSVs GPS	参与解算的 GPS 卫星数。 Number of satellites with GPS used in	Uchar	1	H+171

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		solution			
59	#solnSVs GAL	参与解算的 GAL 卫星数。 Number of satellites with GAL used in solution	Uchar	1	H+172
60	#solnSVs GLO	参与解算的 GLO 卫星数。 Number of satellites with GLO used in solution	Uchar	1	H+173
61	#solnSVs QZSS	参与解算的 QZSS 卫星数。 Number of satellites with QZSS used in solution	Uchar	1	H+174
62	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+175
63	satellite ID	跟踪卫星号，高 4bit 为系统号，低 12bit 为 PRN 号，系统号定义见 Satellite system, PRN 定义见 PRN Numbers; ASCII 格式：UShort 字段用代表系统的字符和 PRN 组合的字符串表示，比如 G1/C33; the PRN for GPS, Galileo, BeiDou, NavIC and QZSS, or the slot for GLONASS. The high 4 bits represent system id, and the low 12 bits represent PRN. System id see Satellite system, PRN see PRN Numbers ASCII format: The UShort field is represented by a string of characters representing the system and a combination of PRN, such as G1/C33;	Ushort	2	H+176

Field	Field type	Description	Format	Binary Bytes	Binary Offset
64		Next satellite offset = $H + 176 + (\#SVs \times 2)$			
65	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	$H + 176 + (\#SVs \times 2)$
66	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.2.49. VERSIONX 版本号信息

适用产品：M720、M722

- Message ID: 3019

输出示例：

```
#VERSIONXA,50.0,0,0,COM2,FINESTEERING,2381,378235000,6e54712f,00008000;2,
M722,0,0,0.0,1.0,0,0,00000002,250827-1-6e54712f-M,054430c4_7b28cc50,6182b
e09,DebugFR,50827,,00050009,ff1d8e968335b7b9,3250116008QB,00000007*189289
99
```

表3-104 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	Message header	VERSION header		H	0
2	Reserved		Enum	4	H
3	Product Series Type	产品型号 0 = UNKNOWN 72000 = M720 72002 = MS720 72200 = M722 72202 = MS722 76000 = B760 76002 = BS760	Enum	4	H+4
4	Reserved		ULong	4	H+8
5	Reserved		ULong	4	H+12
6	Version Number	版本号：	Version	4	H+16

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		ASCII 报文以主版本.子版本 格式输出，如“40.3”； Binary 报文以 Ulong 输出，主版本号(高 16 bits) + 子版本号(低 16 bits);	Number		
7	Aux Version Number	辅助版本号： ASCII 报文以辅助主版本.辅助子版本 格式输出，如“06.196”； Binary 报文以 Ulong 输出，辅助主版本号(高 16 bits) + 辅助子版本号(低 16bits);	Version Number	4	H+20
8			UShort	2	H+24
9			UShort	2	H+26
10			Hex	4	H+28
11			Char[32]	32	H+32
12			Hash[4]	32	H+64
13			Hash[4]	32	H+96
14			Enum	4	H+128
15			UShort	2	H+132
16			UShort	2	H+134
17			ULong	4	H+136
18			U64	8	H+140
19	SN	模组序列号	Char[16]	16	H+148
20	FeatureListMask	产品特定功能标识	Hex	4	H+164
21	xxxx	32-bit CRC(ASCII and binary only)	ULong	4	H+168
22	[CR][LF]	Sentence terminator(ASCII only)	-	-	-

3.3. GMF 从天线报文输出

3.3.1. BESTDOPS_1 最佳DOP 信息

适用产品：M720、M722

BESTDOPS 输出最佳 DOP。

精度衰减因子由接收机当前跟踪并在位置解中使用的卫星的几何分布算出。本指令每 60 秒，或者当卫星星座改变时更新一次。因此本指令输出的数据字段总数可变，并取决于被跟踪的卫星数。

- **Message ID: 3029**
- **Log Type: Synch**

输出示例：

```
#BESTDOPSA_1,59.5,0,0,COM2,COARSESTEERING,2381,390531000,3d7cea15,0000800
0;0.0000,0.0000,0.0000,0.0000,0.0000,0.0,169,E3632,S48,S48,S44,S46,S300,S
2358,J1324,S2100,J1324,S1587,S563,C812,S2100,C812,S2100,C812,S563,C812,S1
588,C812,S2098,E3128,S595,S1331,E3128,S330,S307,E3122,S595,S49,E3120,S330
,S307,E3122,S339,S2101,E3127,S1363,S822,J812,S56,E3120,S595,S49,E3120,S21
15,S48,C812,S306,S48,J812,S56,E3120,S1363,S822,J812,S56,E3120,S339,S2101,
E3128,S2115,S48,C812,S50,S2105,J1324,S1589,C812,S2357,E3125,S339,S819,E31
21,S1349,E3126,S851,S51,C812,S51,E3127,S1349,E3120,S1363,S1337,C812,S2356
,J1324,S2100,C812,S819,E3120,S851,S1840,J1324,S53,C812,S819,E3129,S595,S4
9,E3121,S1349,E3125,S339,S1587,E3123,S2131,S562,J2604,S56,E3120,S1363,E31
26,S1093,E3128,S1363,S1337,C812,S2356,J1324,S2100,C812,S306,S1329,C812,S2
100,J812,S56,E3120,S851,S1584,C812,S2100,J2604,S56,E3120,S1363,E3126,S109
3,E3128,S339,S1587,E3123,S2131,S562,J2604,S56,E3120,S1363,E3126,S1093,E31
28,S851,S2355,C812,S306,S304,J1324,S1589,C812,S306,S1329,C812,S2100,J812,
S56,E3120*ce16132d
```

表3-105 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTDOPS	报文头。		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
	header	Log header. For information about the log headers, see ASCII, or Binary.			
2	pdop	位置精度因子 - 假设 3-D 位置未知, 接收机钟差已知 Position dilution of precision - assumes 3D position is unknown and receiver clock offset is known	Float	4	H
3	gdop	几何精度因子 - 假设 3-D 位置和接收机钟差 (全部的 4 个参数) 未知 Geometric dilution of precision - assumes 3D position and receiver clock offset (all 4 parameters) are unknown	Float	4	H+4
4	hdop	水平精度因子 Horizontal dilution of precision.	Float	4	H+8
5	tdop	时间精度因子 - 假设 3-D 位置已知, 接收机钟差未知 GPS time dilution of precision – assumes 3D position is known and only the GPS receiver clock offset is unknown	Float	4	H+12
6	htdop	水平位置和时间精度因子 Horizontal position and time dilution of precision.	Float	4	H+16
7	elev mask	截止高度角, 单位: 度。 GPS elevation mask angle	Float	4	H+20
8	#sats	DOP 值计算用到的卫星总数 Number of satellites sats to follow	Ulong	4	H+24

Field	Field type	Description	Format	Binary Bytes	Binary Offset
9	sats	计算使用的卫星 ID, 高 4bit 为系统号, 低 12bit 为 PRN 号, 系统号定义见 Satellite system, PRN 定义见 PRN Numbers; ASCII 格式: UShort 字段用代表系统的字符和 PRN 组合的字符串表示, 比如 G1/C33; Satellites in use at time of calculation, The high 4 bits represent system id, and the low 12 bits represent PRN. System id see Satellite system, PRN see PRN Numbers ASCII format: The UShort field is represented by a string of characters representing the system and a combination of PRN, such as G1/C33;	Ushort	2	H+28
10	Next satellite offset = H+28+(#sats x 2)				
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+28+(#sats x 4)
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.3.2. BESTECEF_1 ECEF 坐标下的最佳位置和速度信息

适用产品: M720、M722

BESTECEF 输出 ECEF 坐标下的最佳位置和速度信息。

本指令包含接收机计算出的地心空间直角坐标系下最佳可用位置和速度信息。位置和速度的“status”字段表明了对应数据是否有效。

- **Message ID: 3028**

- Log Type: Synch

输出示例:

```
#BESTECEFA_1,39.0,0,0,COM2,FINESTEERING,2381,390590000,3d7cea15,00008000;
SOL_COMPUTED,NARROW_INT,0,-2828137.0865,4673843.5928,3281081.0919,0.0101,
0.0168,0.0096,SOL_COMPUTED,NARROW_INT,0,0.0010,-0.0023,0.0055,0.0235,0.02
59,0.0219,2.000,2.000,2.000,61,56,48,46,4828,37,1f,07,00,654*9267b41d
```

表3-106 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTECEF header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	pos sol status	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位状态, Position Type Position type, see Position Type	Enum	1	H+1
4	Reserved		Ushort	2	H+2
5	P-X	X 轴坐标, 单位: 米。 Position X-coordinate (m)	Double	8	H+4
6	P-Y	Y 轴坐标, 单位: 米。 Position Y-coordinate (m)	Double	8	H+12
7	P-Z	Z 轴坐标, 单位: 米。 Position Z-coordinate (m)	Double	8	H+20
8	P-X σ	X 轴坐标标准差, 单位: 米。 Standard deviation of P-X (m)	Float	4	H+28
9	P-Y σ	Y 轴坐标标准差, 单位: 米。	Float	4	H+32

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Standard deviation of P-Y (m)			
10	P-Z σ	z 轴坐标标准差, 单位: 米。 Standard deviation of P-Z (m)	Float	4	H+36
11	vel sol status	解算状态, Solution Status Solution status, see Solution Status	Enum	1	H+40
12	vel type	速度类型, 见 see Position or Velocity Type Velocity type, see Position or Velocity Type	Enum	1	H+41
13	Reserved		Ushort	2	H+42
14	V-X	x 轴速度, 单位: 米/秒。 Velocity vector along X-axis (m/s)	Double	8	H+44
15	V-Y	y 轴速度, 单位: 米/秒。 Velocity vector along Y-axis (m/s)	Double	8	H+52
16	V-Z	z 轴速度, 单位: 米/秒。 Velocity vector along Z-axis (m/s)	Double	8	H+60
17	V-X σ	x 轴速度标准差, 单位: 米/秒。 Standard deviation of V-X (m/s)	Float	4	H+68
18	V-Y σ	y 轴速度标准差, 单位: 米/秒。 Standard deviation of V-Y (m/s)	Float	4	H+72
19	V-Z σ	z 轴速度标准差, 单位: 米/秒。 Standard deviation of V-Z (m/s)	Float	4	H+76
20	V-latency	根据速度时标计算的延迟值, 以秒为单位。用历元时间减去延迟可得到更准确的速度结果。 A measure of the latency in the velocity	Float	4	H+80

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		time tag in seconds. It should be subtracted from the time to give improved results			
21	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+84
22	sol_age	解算时间，单位：秒。 Solution age in seconds	Float	4	H+88
23	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+92
24	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+93
25	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+94
26	#ggL1	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+95
27	ext sol stat	扩展解算状态 Extended Solution Status, see Extended Solution Status	Hex Ushort	2	H+96
28	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+98
29	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+99
30	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+100

Field	Field type	Description	Format	Binary Bytes	Binary Offset
31	Reserved	保留 Reserved	Hex	1	H+101
32	stn id	基准站 ID Base station ID	Ushort	2	H+102
33	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+104
34	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.3.3. BESTP_1 最佳位置信息

适用产品：M720、M722

BESTP 输出最佳位置信息。此外, BESTP 报告了一些状态指标, 包括差分龄期等。

该条报文包含了解算状态、定位类型、纬度、经度、海拔高、高程异常值、参考坐标系、纬度标准差、经度标准差、高程标准差、差分龄期、解算时间、跟踪到的卫星数、参与解算的卫星数、参与解算的 L1/E1/B1 卫星数、参与解算的多频信号卫星数、参与解算的信号、基站 ID 等信息。

- **Message ID: 3020**
- **Log Type: Synch**

输出示例:

```
#BESTPA_1,42.5,0,0,COM2,FINESTEERING,2381,390634000,3d7cea15,00008000;SOL
_COMPUTED,NARROW_INT,WGS84,0,31.15969339970,121.17808143539,46.1179,0.000
0,0.0071,0.0074,0.0213,1.000,1.000,61,56,48,48,4828,37,1f,07,00,654*53d24
a78
```

表3-107 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTP header	报文头。 Log header. For information about the log		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		headers, see ASCII, or Binary.			
2	pos sol stat	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型, 见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度, 单位: 度。 Latitude (degrees)	Double	8	H+4
7	lon	经度, 单位: 度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the ellipsoid (m) of the chosen datum When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差, 单位: 米。 Longitude standard deviation (m)	Float	4	H+36

Field	Field type	Description	Format	Binary Bytes	Binary Offset
12	hgt σ	高度标准差, 单位: 米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间, 单位: 秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61

Field	Field type	Description	Format	Binary Bytes	Binary Offset
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

差分龄期为 0 表示未使用差分校正。

不同的定位模式具有不同的最大记录速率，这些速率也由模型选项控制，最大速率: RTK 为 20Hz, PPP 为 20Hz, SPP 定位为 100Hz

表3-108 Solution Status

Binary	ASCII	Description
0	SOL_COMPUTED	已解出 Solution computed
1	INSUFFICIENT_OBS	观测数据不足 Insufficient observations
2	NO_CONVERGENCE	无法收敛 No convergence
3	SINGULARITY	参数矩阵奇异 Singularity at parameters matrix
4	COV_TRACE	协方差矩阵的迹超过最大值（迹>1000 米） Covariance trace exceeds maximum (trace > 1000 m)
5	VARIANCE	方差超过限制 Variance exceeds limits
6	PENDING	当输入 SETFIXPOS 指令时，接收机会计算自己的位置并确定固定位置是否有效 When a SETFIXPOS position command is entered, the receiver computes its own position and determines if the fixed position is

Binary	ASCII	Description
		valid
7	INVALID_FIX	使用 SETFIXPOS 指令输入的固定位置无效 The fixed position, entered using the SETFIXPOS position command, is not valid
8	RESIDUALS	残差过大 Residuals are too large
9	INTEGRITY_WARNING	残差过大导致位置不可靠 Large residuals make position unreliable
10~20	Reserved	

表3-109 Position or Velocity Type

Binary	ASCII	Description
0	NONE	无解 No solution
1	SINGLE	单点定位 Solution calculated using only data supplied by the GNSS satellites
2	SPPDIFF	伪距差分解 Solution calculated using pseudorange differential (DGPS, DGNSS) corrections
3	Reserved	
4	NARROW_INT	窄巷固定解 Multi-frequency RTK solution with carrier phase ambiguities resolved to narrow-lane integers
5	NARROW_FLOAT	窄巷浮点解 Multi-frequency RTK solution with unresolved, float carrier phase ambiguities
6	Reserved	
7	FIXEDPOS	位置由 SETFIXPOS 指令指定 Position has been fixed by the SETFIXPOS position command or by

Binary	ASCII	Description
		position averaging.
8	Reserved	
9	SBAS	SBAS 定位 Solution calculated using corrections from an SBAS satellite
10	PPP	收敛的 B2b、HAS、MADOCa PPP 定位解 Converged B2b or HAS or MADOCa PPP solution
11	PPP_FIXED	收敛的 PointSky、CLAS PPP 固定解 Converged PointSky or CLAS PPP solution
12	PPP_CONVERGING	收敛中的 PPP 定位解 Converging PPP solution
13	WIDE_INT	宽巷固定解 Multi-frequency RTK solution with carrier phase ambiguities resolved to wide-lane integers
14	L1_INT	L1 固定解 Single-frequency RTK solution with carrier phase ambiguities resolved to integers
15	L1_FLOAT	L1 浮点解 Single-frequency RTK solution with unresolved, float carrier phase ambiguities
16	DOPPLER_VELOCITY	速度由即时多普勒信息导出 Velocity computed using instantaneous Doppler
17-19	Reserved	
20	INS	纯惯导定位解 INS positioning solution
21	INS_SBAS	惯导与 SBAS 定位组合解

Binary	ASCII	Description
		INS position, where the last applied position update used a GNSS solution computed using corrections from an SBAS solution
22	INS_SPP	惯导与单点定位组合解 INS position, where the last applied position update used a single point GNSS(SINGLE) solution
23	INS_SPPDIFF	惯导与伪距差分定位组合解 INS position, where the last applied position update used a pseudorange differential GNSS (SPPDIFF) solution
24	INS_RTKFLOAT	惯导与载波相位差分浮点解的组合解 INS position, where the last applied position update used a floating ambiguity RTK (L1_FLOAT or NARROW_FLOAT) solution
25	INS_RTKFIXED	惯导与载波相位差分固定解的组合解 INS position, where the last applied position update used a fixed integer ambiguity RTK (L1_INT, WIDE_INT or NARROW_INT) solution
26	INS_PPP	惯导与 B2b、HAS、MADOCA PPP 定位解的组合解 INS position, where the last applied position update used B2b or HAS or MADOCA ppp converged solution
27	INS_PPPFIXED	惯导与 PointSky、CLAS PPP 固定解的组合解 INS position, where the last applied position update used PointSky or CLAS ppp converged solution

表3-110 Extended Solution Status

Nibble	Bit	Mask	Description	Value
NO	0	0x0001	伪距电离层改正	0 = 未知
	1	0x0002		1 = Klobuchar 广播星历改正
	2	0x0004		2 = SBAS 电离层格网改正
				3 = 多频改正
				4 = 伪距差分改正

				Pseudorange Iono Correction 0 = Unknown or default Klobuchar model 1 = Klobuchar Broadcast 2 = SBAS Broadcast 3 = Multi-frequency Computed 4 = SPPDiff Correction
	3	0x0008	云端同源使用标志	1: 使用, 0: 未使用
N1	4	0x0010	云端同源使用版本	0 = 2.0 版本
	5	0x0020		1 = 3.0 版本
	6	0x0040		2 = 4.0 版本
	7	0x0080		
N2	8	0x0100	云端同源龄期	0-61s
	9	0x0200		
	10	0x0400		
	11	0x0800		
N3	12	0x1000		
	13	0x2000		
	14	0x4000	双天线基线长异常	0: 正常 1: 异常
	15	0x8000	Reserved	

表3-111 BeiDou Signal-Used Mask

Bit	Mask	Description
0	0x01	BeiDou B1 used in Solution (B1I)
1	0x02	BeiDou B2 used in Solution (B2I)
2	0x04	BeiDou B3 used in Solution (B3I)
3	0x08	BeiDou B1C used in Solution
4	0x10	Beidou B2a used in Solution
5	0x20	Beidou B2b used in Solution

Bit	Mask	Description
6-7	0xC0	Reserved

表3-112 Galileo Signal-Used Mask

Bit	Mask	Description
0	0x01	Galileo E1 used in Solution
1	0x02	Galileo E5a used in Solution
2	0x04	Galileo E5b used in Solution
3	0x08	Galileo ALTBOC used in Solution
4	0x10	Galileo E6 used in Solution (E6B and E6C)
5-7	0xE0	Reserved

表3-113 GPS and GLONASS Signal-Used Mask

Bit	Mask	Description
0	0x01	使用 GPS L1 解算 GPS L1 used in Solution
1	0x02	使用 GPS L2 解算 GPS L2 used in Solution
2	0x04	使用 GPS L5 解算 GPS L5 used in Solution
3	0x08	Reserved
4	0x10	使用 GLONASS L1 解算 GLONASS L1 used in Solution
5	0x20	使用 GLONASS L2 解算 GLONASS L2 used in Solution
6	0x40	使用 GLONASS L3 解算 GLONASS L3 used in Solution
7	0x80	保留

Bit	Mask	Description
		Reserved

3.3.4. BESTSATINFO_1 参与定位的卫星信息

适用产品：M720、M722

BESTSATINFO 输出参与定位的卫星信息。

该消息包含参与定位的卫星信息。

- Message ID: 3031
- Log Type: Synch

输出示例：

```
#BESTSATINFOA_1,42.0,0,0,COM2,FINESTEERING,2381,390669000,3d7cea15,000080
00;62,C,1,00000007,GOOD,C,3,00000000,NOEPHEMERIS,C,59,00000005,GOOD,C,60,
00000005,GOOD,G,4,00000007,GOOD,G,18,00000007,GOOD,G,26,00000007,GOOD,G,2
7,00000007,GOOD,G,28,00000007,GOOD,J,196,00000007,GOOD,E,9,00000007,GOOD,
E,14,00000000,NOEPHEMERIS,C,2,00000007,GOOD,C,7,00000007,GOOD,C,10,000000
07,GOOD,C,11,00000007,GOOD,C,4,00000007,GOOD,G,16,00000003,GOOD,G,31,0000
0003,GOOD,J,194,00000007,GOOD,J,195,00000007,GOOD,J,200,00000005,GOOD,G,3
,00000007,GOOD,R,5-7,00000003,GOOD,R,7-4,00000001,GOOD,R,9-7,00000003,GOO
D,R,15-7,00000003,GOOD,R,16,00000003,GOOD,R,18+4,00000003,GOOD,E,4,000000
07,GOOD,E,6,00000007,GOOD,E,23,00000007,GOOD,E,31,00000007,GOOD,C,5,00000
006,GOOD,C,6,00000007,GOOD,C,9,00000007,GOOD,C,12,00000007,GOOD,C,13,0000
0000,ELEVATIONERROR,C,16,00000007,GOOD,C,23,0000001d,GOOD,C,24,0000001d,G
OOD,C,25,0000001d,GOOD,C,33,0000001d,GOOD,C,34,0000001d,GOOD,C,39,0000001
d,GOOD,C,41,0000001d,GOOD,C,43,0000001d,GOOD,C,44,0000001d,GOOD,C,56,0000
0000,NOEPHEMERIS,G,8,00000007,GOOD,C,14,00000002,GOOD,C,40,0000001d,GOOD,
G,9,00000007,GOOD,J,199,00000007,GOOD,E,19,00000007,GOOD,E,21,00000007,GO
OD,E,26,00000007,GOOD,C,32,00000019,GOOD,E,24,00000007,GOOD,R,17-1,000000
01,GOOD,E,33,00000000,ELEVATIONERROR,G,29,00000000,NOEPHEMERIS*ef7a4885
```

表3-114 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTSATINFO header	报文头。 Log header. For information about the log headers, see ASCII, or Binary		H	0
2	#entries	Number of records to follow	Ulong	4	H
3	System	GNSS 卫星系统列表。 see Satellite System 卫星系统使用与 GNSSOBS 报文一致的标识符定义	Enum	4	H+4
4	Satellite ID	卫 星 PRN 号 ◆ GPS: 1 到 32 , ◆ GLONASS: 1 到 24, ◆ Galileo 1 到 36, ◆ BDS 1 到 63, ◆ SBAS 120 到 158 及 183 到 192, ◆ QZSS 193 到 202 在二进制消息中,卫星 ID 字段由 1 个 Short 和 1 个 Ushort 组成。 • 最低的两个字节是系统标识符: (如 GPS 的 PRN, GLONASS 的通道号), 为 USHORT 类型; • 最高的两个字节是 GLONASS 的频率通道, 其他系统这两个字节值为零。 在 ASCII 消息中, 卫星 ID 字段是系统标识	Ulong	4	H+8

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		符。如果系统是 GLONASS，而频率通道不是零，则在系统标识符后加上频率通道数。例如，系统标识符是 13，频率通道-2，输出为 13-2,数据类型为 Short 类型。 In binary logs, the satellite ID field is 4 bytes. The 2 lowest-order bytes, interpreted as a USHORT, are the system identifier: for instance, the PRN for GPS, Galileo, BeiDou, NavIC and QZSS, or the slot for GLONASS. The 2 highest-order bytes are the frequency channel for GLONASS, interpreted as a SHORT and zero for all other systems. In ASCII and abbreviated ASCII logs, the satellite ID field is the system identifier. If the system is GLONASS and the frequency channel is not zero, then the signed channel is appended to the system identifier. For example, slot 13, frequency channel -2 is output as 13-2.			
5	Signal mask	See GPS Signal Mask, GLONASS Signal Mask, Galileo Signal Mask, BeiDou Signal Mask, QZSS Signal Mask and NavIC Signal MaskNavIC Signal Mask	Hex	4	H+12
6	Status	在二进制消息中，数值是“0”；在 ASCII 消息中为：“GOOD” Satellite status. See Observation Statuses	Ulong	4	H+16
7	Next satellite offset = H + 4 + (#entries x 16)				
8	xxxx	32-bit CRC 校验。	Hex	4	H+4 (#entries

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		32-bit CRC (ASCII and Binary only)			x 16)
9	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-115 Observation Statuses

Value	Name	Description
0	GOOD	观测数据正常 Observation is good
1	BADHEALTH	卫星在星历表或历书中被标记为健康状况不佳 Satellite is flagged as bad health in ephemeris or almanac
2	OLDEPHEMERIS	星历时间大于 3 小时 Ephemeris >3 hours old
6	ELEVATIONERROR	卫星位于高度截止角以下 Satellite was below the elevation cutoff
7	MISCLOSURE	观察值与预测值相差大 Observation was too far from predicted value
8	NODIFFCORR	无差分校正可用 No differential correction available
9	NOEPHEMERIS	无有效星历 No ephemeris available
10	INVALIDIODE	使用的 IODE 无效 IODE used is invalid
11	LOCKEDOUT	卫星已被锁定 Satellite has been locked out
12	LOWPOWER	卫星信号功率低 Satellite has low signal power
13	OBSL2	解算中未直接使用的 L2 观测值

Value	Name	Description
		An L2 observation not directly used in the solution
15	UNKNOWN	未使用观察结果，因为其类型未知 Observation was not used because it was of an unknown type
16	NOIONOCORR	电离层延迟校正不可用 No ionosphere delay correction was available
17	NOTUSED	解算中未使用的观测数据 Observation was not used in the solution
18	OBSL1	解算中未使用 L1 观测值 An L1 observation not directly used in the solution
19	OBSE1	解算中未使用 E1 观测值 An E1 observation not directly used in the solution
20	OBSL5	解算中未使用 L5 观测值 An L5 observation not directly used in the solution
21	OBSE5	解算中未使用 E5 观测值 An E5 observation not directly used in the solution
22	OBSB2	解算中未使用 B2 观测值 A B2 observation not directly used in the solution
23	OBSB1	解算中未使用 B1 观测值 A B1 observation not directly used in the solution
24	OBSB3	解算中未使用 B3 观测值 A B3 observation not directly used in the solution
25	NOSIGNALMATCH	信号类型不匹配 Signal type does not match
26	SUPPLEMENTARY	观测数据在解算中提供补偿 Observation contributes supplemental information to the solution

Value	Name	Description
99	NA	无有效观测数据 No observation available
100	BAD_INTEGRITY	观察数据异常，在解算中排除 Observation was an outlier and was eliminated from the solution
101	LOSSOFLOCK	此信号失锁 Lock was broken on this signal
102	NOAMBIGUITY	未匹配 RTK 模糊类型 No RTK ambiguity type resolved

表3-116 GPS Signal Mask

Bit	Mask	Description
0	0x01	解算中使用 GPS L1 频点 GPS L1 used in the solution
1	0x02	解算中使用 GPS L2 频点 GPS L2 used in the solution
2	0x04	解算中使用 GPS L5 频点 GPS L5 used in the solution

表3-117 GLONASS Signal Mask

Bit	Mask	Description
0	0x01	解算中使用 GLONASS L1 频点 GLONASS L1 used in the solution
1	0x02	解算中使用 GLONASS L2 频点 GLONASS L2 used in the solution
2	0x04	解算中使用 GLONASS L3 频点 GLONASS L3 used in the solution

表3-118 Galileo Signal Mask

Bit	Mask	Description
0	0x01	解算中使用 Galileo E1 频点 Galileo E1 used in the solution
1	0x02	解算中使用 Galileo E5A 频点 Galileo E5A used in the solution
2	0x04	解算中使用 Galileo E5B 频点 Galileo E5B used in the solution
3	0x08	解算中使用 Galileo ALTBOC 频点 Galileo ALTBOC used in the solution
4	0x10	解算中使用 Galileo E6 频点 Galileo E6 used in the solution

表 3-119 BeiDou Signal Mask

Bit	Mask	Description
0	0x01	解算中使用 BeiDou B1I 频点 BeiDou B1I used in the solution
1	0x02	解算中使用 BeiDou B2I 频点 BeiDou B2I used in the solution
2	0x04	解算中使用 BeiDou B3 频点 BeiDou B3 used in the solution
3	0x08	解算中使用 BeiDou B1C 频点 BeiDou B1C used in the solution
4	0x10	解算中使用 BeiDou B2a 频点 BeiDou B2a used in the solution
5	0x20	解算中使用 BeiDou B2b 频点

Bit	Mask	Description
		BeiDou B2b used in the solution

表3-120 QZSS Signal Mask

Bit	Mask	Description
0	0x01	解算中使用 QZSS L1 频点 QZSS L1 used in the solution
1	0x02	解算中使用 QZSS L2 频点 QZSS L2 used in the solution
2	0x04	解算中使用 QZSS L5 频点 QZSS L5 used in the solution
4	0x08	解算中使用 QZSS L6 频点 QZSS L6 used in the solution

表3-121 NavIC Signal Mask

Bit	Mask	Description
2	0x04	解算中使用 NavIC L5 频点 NavIC L5 used in the solution

3.3.5. BESTV_1 最佳可用速度

适用产品：M720、M722

BESTV 输出最佳可用速度。

本指令包含接收机计算出的最佳可用速度信息。此外，接收机还报告了速度状态指示符，对指示相应的数据是否有效非常有用。该速度测量有时会带来相关的延迟。

- Message ID: 3021
- Log Type: Synch

输出示例：

```
#BESTVA_1,42.5,0,0,COM2,FINESTEERING,2381,390703000,3d7cea15,00008000;SOL
_COMPUTED,DOPPLER_VELOCITY,0,0.0051,340.812356,-0.0029,0.000,1.000,0.000*
b40abc0d
```

表3-122 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	BESTV header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	vel sol stat	解算状态, 见 Vel Solution Status Solution status, see Vel Solution Status	Enum	1	H+0
3	vel type	速度类型, 见 Velocity Type Velocity type, see Velocity Type	Enum	1	H+1
4	reserved		Ushort	2	H+2
5	hor spd	对地水平速度, 单位: 米/秒。 Horizontal speed over ground, in metres per second	Double	8	H+4
6	trk gnd	相对于真北的实际对地运动方向 (相对地面轨迹), 单位: 度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Double	8	H+12
7	vert spd	垂直速度, 正值表示高度增加 (向上), 负值表示高度下降 (向下), 单位: 米/秒。 Vertical speed, in metres per second, where positive values indicate increasing altitude (up) and negative values indicate decreasing altitude (down)	Double	8	H+20
8	latency	根据速度时标计算的延迟值, 用历元时间减去延迟可得到更准确的速度结果, 单位: 秒。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results (s)	Float	4	H+28

Field	Field type	Description	Format	Binary Bytes	Binary Offset
9	age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+32
10	Reserved		Float	4	H+36
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.3.6. GNSSOBS_1 卫星观测值信息

适用产品：M720、M722

GNSSOBS 报文传输原始观测数据信息。

在精确相位处理中使用载波相位测量值时，如果在 ch-tr-status 字段中的 parity known 标志未被设置，测量值可能存在半个周期(1/2 cycle)的不确定性，此时的载波相位不应该使用。

伪距和载波相位的跟踪误差估计仅包含接收机跟踪环路的热噪声，没有考虑到可能的多路径误差或大气延迟。

如果对一个给定的 PRN 正在跟踪多个信号，则在 GNSSOBS 报文中出现具有相同 PRN 的每个信号的信息，用 signal ID 观测信号的类型 来区分。

- **Message ID: 3000**
- **Log Type: Sync**

输出示例：

```
#GNSSOBSA_1,41.5,0,0,COM2,FINESTEERING,2381,390731000,3d7cea15,00008000;1
84,31,G1C,0,5142,21233622.368,-111583430.926425,-1339.743,9,143,196.011,0
0017623,31,G2L,0,5226,21233628.594,-86948183.830508,-1044.077,9,140,195.0
00,00017633,31,G2W,0,4997,21233628.239,-86948157.579451,-1043.962,9,147,1
90.000,00017623,18,G1C,0,4374,24716365.889,-129885367.408831,1077.094,11,
171,202.011,00027613,18,G2L,0,4746,24716374.774,-101209362.428979,839.264
```

,10,155,199.011,00027623,18,G5Q,0,4822,24716387.443,-96992413.466633,804.241,10,153,202.011,00027623,3,G1C,0,4096,24738372.263,-130001069.447592,-2384.443,12,207,88.000,000375e3,3,G2L,0,4312,24738389.566,-101299509.993025,-1858.207,11,175,187.000,00037613,3,G5Q,0,4808,24738405.461,-97078779.327725,-1780.856,10,153,194.000,00037623,3,G2W,0,3771,24738393.848,0.000000,-1857.971,13,460,0.000,00034533,16,G1C,0,5065,20180838.534,-106051024.045430,80.116,9,145,196.011,00047633,16,G2W,0,4014,20180842.652,-82637191.831729,62.333,12,368,192.000,00047563,27,G1C,0,4948,20923553.032,-109954009.202253,2698.399,9,149,202.011,00057623,27,G2L,0,5164,20923563.241,-85678429.708168,2102.583,9,142,186.000,00057633,27,G5Q,0,5631,20923570.598,-82108591.324640,2014.931,8,127,202.011,00057633,27,G2W,0,5038,20923562.986,-85678491.461912,2102.607,9,146,194.000,00057623,28,G1C,0,4209,23137976.442,-121590892.830302,-2265.133,12,181,202.011,00067603,28,G2L,0,5000,23137986.488,-94746178.447897,-1765.333,9,147,199.011,00067623,28,G5Q,0,5398,23137998.971,-90798466.528437,-1691.832,8,134,202.011,00067633,26,G1C,0,5105,21128287.262,-111029894.408458,-1693.576,9,144,202.011,00077633,26,G2L,0,5187,21128296.605,-86516841.355054,-1319.709,9,141,199.011,00077633,26,G5Q,0,5604,21128303.016,-82911999.288041,-1264.636,8,128,202.011,00077633,26,G2W,0,5076,21128296.698,-86516841.095864,-1319.630,9,145,194.000,00077623,4,G1C,0,4801,21284667.500,-111851672.115987,1364.943,10,154,202.011,00087623,4,G2L,0,5112,21284674.549,-87157217.022370,1063.531,9,144,199.011,00087623,4,G5Q,0,5688,21284687.062,-83525683.949371,1019.253,8,125,202.011,00087633,4,G2W,0,3724,21284675.180,0.000000,1063.483,13,488,0.000,00084523,8,G1C,0,4124,23109113.135,-121439224.525697,3757.125,12,187,190.000,000a75f3,8,G2L,0,4627,23109129.291,-94627998.657438,2927.561,10,159,189.000,000a7623,8,G5Q,0,4814,23109136.658,-90685224.689581,2805.585,10,153,190.000,000a7623,8,G2W,0,4019,23109128.951,-94628032.399530,2927.567,12,271,182.000,000a75b3,9,G1C,0,4433,23771678.341,-124921014.243027,2993.336,11,167,184.000,000b7613,9,G2L,0,4342,23771689.009,-97341101.770511,2332.353,11,173,181.000,000b7603,9,G5Q,0,4938,23771697.940,-93285250.313171

,2235.354,9,149,184.000,000b7623,9,G2W,0,4136,23771688.881,-97341083.5065
45,2332.456,12,260,178.000,000b75b3,44,R1C5,0,4658,23069093.170,-12349065
7.747321,3042.827,10,158,200.011,00017613,46,R1C-2,0,3587,23051390.049,-1
23093132.216891,2044.018,14,376,86.000,00027580,46,R2C-2,0,3931,23051403.
588,-95739141.124905,1590.288,12,199,200.011,000275f3,55,R1C-3,0,3868,215
06590.353,-114803637.309116,2995.223,13,264,8.000,000475c3,55,R2C-3,0,461
6,21506605.465,-89291770.471422,2329.537,10,159,200.011,00047623,54,R1C4,
0,5179,21611259.641,-115646200.423743,-22.723,9,141,196.011,00057633,53,R
1C-1,0,4836,20491156.433,-109460068.065855,215.877,9,152,200.011,00067623
,53,R2C-1,0,4439,20491163.345,-85135637.932362,167.860,10,166,200.011,000
67613,52,R1C0,0,4785,21763316.161,-116296535.826773,-1776.449,10,154,200.
011,00087623,52,R2C0,0,4218,21763322.856,-90452887.440467,-1381.457,11,18
1,200.011,000875f3,42,R1C1,0,4530,21675771.319,-115869404.208265,-4004.70
3,10,162,200.011,00097613,42,R2C1,0,3653,21675782.210,-90120716.627405,-3
114.852,13,303,128.000,000975b3,56,R1C3,0,3508,24139796.182,0.000000,3982
.533,14,362,0.000,000a4590,56,R2C3,0,3264,24139808.113,0.000000,3097.461,
15,532,0.000,000a4530,9,E1C,0,4846,25201215.698,-132433276.255687,1621.89
4,10,152,201.011,00017623,9,E5Q,0,5294,25201234.615,-98895057.818545,1211
.279,9,138,200.011,00017633,9,E7Q,0,5116,25201226.174,-101474897.768808,1
242.747,9,143,201.011,00017633,14,E1C,0,5085,24978717.713,-131264042.1154
73,2596.965,9,144,201.011,00027623,14,E5Q,0,5330,24978734.496,-98021919.1
76409,1939.079,9,137,200.011,00027633,14,E7Q,0,5034,24978726.151,-1005789
80.643173,1989.773,9,146,201.011,00027623,23,E1C,0,5087,24189718.429,-127
117824.832928,-1381.883,9,144,194.000,00037633,23,E5Q,0,5404,24189736.793
, -94925722.428013,-1031.911,8,134,190.000,00037633,23,E7Q,0,5112,24189728
.426,-97402010.456617,-1058.588,9,144,194.000,00037623,31,E1C,0,5091,2400
4296.135,-126143425.415313,554.251,9,144,194.000,00047623,31,E5Q,0,5431,2
4004315.058,-94198088.785418,413.784,8,133,190.000,00047633,31,E7Q,0,5135
,24004306.571,-96655394.107864,424.543,9,143,194.000,00047633,4,E1C,0,497
3,23596148.801,-123998593.081159,-1035.217,9,148,194.000,00057623,4,E5Q,0

,5660,23596166.103,-92596422.898604,-773.120,8,126,190.000,00057633,4,E7Q
 ,0,5256,23596157.992,-95011944.688930,-793.265,9,139,194.000,00057633,6,E
 1C,0,5153,23743029.724,-124770461.878014,393.764,9,142,194.000,00067623,6
 ,E5Q,0,5473,23743047.876,-93172819.772744,294.026,8,132,190.000,00067633,
 6,E7Q,0,5285,23743040.001,-95603384.055567,301.655,9,138,194.000,00067633
 ,19,E1C,0,3799,26016009.748,-136715052.430946,-2961.514,13,249,102.000,00
 0775d3,19,E5Q,0,4550,26016032.746,-102092496.948837,-2211.570,10,162,170.
 000,00077613,19,E7Q,0,3873,26016023.708,-104755727.537540,-2269.804,13,26
 3,171.000,000775c3,26,E1C,0,4358,27729428.894,-145719123.781811,1842.227,
 11,172,172.000,00087603,26,E5Q,0,4530,27729450.010,-108816314.225812,1375
 .550,10,162,170.000,00087613,26,E7Q,0,4358,27729441.605,-111654966.633224
 ,1411.540,11,172,172.000,000875f3,21,E1C,0,4087,28236138.729,-148381907.7
 04049,-2198.979,12,208,138.000,000975e3,21,E5Q,0,4210,28236158.939,-11080
 4753.615996,-1642.150,12,199,170.000,000975e3,21,E7Q,0,4273,28236151.542,
 -113695274.344687,-1684.948,11,177,172.000,00097603,24,E1C,0,4068,2766722
 6.773,-145392254.921568,1951.409,12,209,144.000,000a75e3,24,E5Q,0,4563,27
 667253.470,-108572249.887111,1457.249,10,161,164.000,000a7623,24,E7Q,0,42
 69,27667243.213,-111404529.029631,1494.898,11,177,167.000,000a75f3,33,E1C
 ,0,3704,28253897.310,-148475239.752790,2316.236,14,299,93.000,000b75b3,33
 ,E5Q,0,4396,28253930.872,-110874498.120099,1729.619,11,169,159.000,000b76
 03,33,E7Q,0,4320,28253920.648,-113766832.923178,1774.707,11,174,159.000,0
 00b7603,59,C2I,0,5340,37193313.664,-193675343.393871,-64.578,9,136,203.01
 1,00017613,59,C6I,0,5252,37193300.938,-157377028.613390,-52.587,9,139,203
 .011,00017623,1,C2I,0,4910,37418852.859,-194849778.105203,-14.751,9,150,2
 05.011,00027603,1,C6I,0,4845,37418854.427,-158331420.107393,-11.882,10,15
 2,205.011,00027603,1,C7I,0,4851,37418853.337,-150670212.036588,-11.443,9,
 152,205.011,00027603,10,C2I,0,4416,38165555.023,-198738060.496152,870.267
 ,11,168,200.011,00037613,10,C6I,0,4618,38165552.996,-161490942.053849,707
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 ,673.061,10,156,200.011,00037613,12,C2I,0,5363,23241084.321,-121022417.13

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575938,479.888,9,142,194.000,00047633,12,C7I,0,5418,23241085.003,-9358221
8.888949,456.803,8,134,194.000,00047633,16,C2I,0,5184,36607658.930,-19062
5684.844502,509.254,9,141,194.000,00057633,16,C6I,0,4735,36607657.415,-15
4898978.745467,413.704,10,156,194.000,00057613,16,C7I,0,5034,36607661.670
,-147403881.750622,393.839,9,146,194.000,00057633,60,C2I,0,4992,38371980.
110,-199812964.195875,-62.883,9,147,203.011,00067603,60,C6I,0,4910,383719
66.555,-162364349.544495,-51.055,9,150,203.011,00067603,23,C2I,0,4690,253
49535.133,-132001685.386789,1919.525,10,157,194.000,00077613,23,C6I,0,479
9,25349530.227,-107262160.411295,1559.886,10,154,194.000,00077623,23,C1P,
0,4389,25349524.799,-133212649.703228,1937.190,11,170,187.000,000775f3,23
,C5P,0,4758,25349541.406,-99477044.807622,1446.607,10,155,195.000,0007762
3,24,C2I,0,5652,23353174.074,-121606100.239931,-1387.465,8,126,194.000,00
087633,24,C6I,0,5201,23353173.349,-98814919.914443,-1127.506,9,141,194.00
0,00087633,24,C1P,0,5025,23353164.010,-122721710.332510,-1400.158,9,146,1
95.000,00087623,24,C5P,0,5486,23353180.476,-91642895.675491,-1045.502,8,1
32,195.000,00087633,2,C2I,0,4465,38128381.069,-198544483.349337,-84.652,1
0,231,168.000,000975b3,2,C6I,0,4568,38128381.367,-161333659.300077,-68.98
0,10,209,201.011,000975c3,2,C7I,0,4664,38128379.354,-153527178.963624,-65
.599,10,158,201.011,000975f3,3,C2I,0,4919,36701538.575,-191114538.691064,
-27.730,9,150,203.011,000a7603,3,C6I,0,4780,36701536.953,-155296211.48764
1,-22.688,10,170,203.011,000a75e3,3,C7I,0,4869,36701537.182,-147781878.13
9900,-21.540,9,151,203.011,000a7603,6,C2I,0,4965,36066487.576,-187807660.
284264,484.574,9,148,194.000,000b7623,6,C6I,0,5038,36066482.873,-15260909
1.866704,393.709,9,146,194.000,000b7623,6,C7I,0,4788,36066483.673,-145224
785.693885,374.734,10,154,194.000,000b7623,7,C2I,0,4845,37175461.825,-193
582381.274353,884.729,10,152,200.011,000c7623,7,C6I,0,4869,37175456.236,-
157301523.071685,718.829,9,151,200.011,000c7623,7,C7I,0,5064,37175459.292
,-149690171.201570,683.940,9,145,200.011,000c7623,9,C2I,0,4968,37305578.4
76,-194259935.265488,587.782,9,148,194.000,000d7623,9,C6I,0,4603,37305573

.121, -157852089.780205, 477.745, 10, 160, 194.000, 000d7613, 9, C7I, 0, 5175, 37305
577.355, -150214099.136592, 454.587, 9, 142, 194.000, 000d7623, 11, C2I, 0, 5032, 24
115141.359, -125573869.955876, 2823.938, 9, 146, 200.011, 000e7623, 11, C6I, 0, 502
3, 24115139.889, -102039055.103710, 2294.490, 9, 146, 200.011, 000e7623, 11, C7I, 0
, 4870, 24115143.159, -97101698.779528, 2183.694, 10, 151, 200.011, 000e7623, 25, C
2I, 0, 5581, 22053368.023, -114837673.397230, 664.923, 8, 129, 194.000, 000f7633, 2
5, C6I, 0, 5408, 22053366.705, -93315011.616494, 540.339, 8, 134, 194.000, 000f7633
, 25, C1P, 0, 5139, 22053357.446, -115891169.725151, 670.892, 9, 143, 195.000, 000f7
623, 25, C5P, 0, 5431, 22053374.208, -86542176.207191, 501.223, 8, 133, 195.000, 000
f7633, 34, C2I, 0, 5362, 22810600.044, -118780780.441468, 1842.959, 8, 136, 194.000
, 00107633, 34, C6I, 0, 5303, 22810601.713, -96519122.032910, 1497.516, 9, 138, 194.
000, 00107623, 34, C1P, 0, 4938, 22810590.119, -119870460.061172, 1860.067, 9, 149,
195.000, 00107623, 34, C5P, 0, 5110, 22810610.571, -89513734.815769, 1388.880, 9, 1
44, 195.000, 00107633, 13, C2I, 0, 4101, 40831183.890, -212618688.954523, -1867.60
6, 12, 226, 194.000, 001175d3, 13, C6I, 0, 3776, 40831201.713, -172770176.693159, -1
518.053, 13, 292, 188.000, 001175b3, 13, C7I, 0, 3788, 40831208.059, -164410359.975
710, -1444.626, 13, 271, 50.000, 001175c3, 32, C2I, 0, 4549, 26273957.902, -13681539
1.071145, 1571.213, 10, 162, 170.000, 00127613, 32, C1P, 0, 4183, 26273948.013, -138
070529.853922, 1585.215, 12, 183, 175.000, 00127603, 32, C5P, 0, 4933, 26273973.176
, -103104715.801988, 1184.093, 9, 149, 175.000, 00127623, 56, C2I, 0, 5087, 37062339
.516, -192993322.498880, 740.617, 9, 144, 194.000, 00137623, 56, C6I, 0, 4852, 37062
342.330, -156822899.567154, 601.753, 9, 152, 194.000, 00137633, 33, C2I, 0, 4924, 24
759218.793, -128927747.700330, -2899.096, 9, 150, 194.000, 00147623, 33, C6I, 0, 49
57, 24759234.561, -104764419.645654, -2355.962, 9, 149, 194.000, 00147623, 33, C1P
, 0, 4302, 24759207.696, -130110526.742339, -2925.477, 11, 175, 194.000, 001475f3,
33, C5P, 0, 4978, 24759255.660, -97160637.776201, -2184.837, 9, 148, 194.000, 00147
623, 39, C2I, 0, 5526, 36045510.184, -187698429.101200, 405.768, 8, 130, 194.000, 00
157633, 39, C6I, 0, 5243, 36045506.318, -152520332.399723, 329.595, 9, 139, 194.000
, 00157633, 39, C1P, 0, 5157, 36045501.076, -189420382.779110, 409.431, 9, 142, 194.
000, 00157623, 39, C5P, 0, 5304, 36045514.494, -141450341.539383, 305.716, 9, 137, 1

94.000,00157633,40,C2I,0,5228,37017776.546,-192761275.541494,1110.734,9,1
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 9,139,188.000,00167633,40,C1P,0,4836,37017765.750,-194529644.296949,1120.
 910,10,152,188.000,00167613,40,C5P,0,5211,37017783.634,-145265735.632036,
 836.973,9,140,188.000,00167633,41,C2I,0,5166,23973431.732,-124835951.4361
 83,-985.324,9,142,194.000,00177633,41,C6I,0,5184,23973439.325,-101439468.
 040019,-800.672,9,141,194.000,00177623,41,C1P,0,4740,23973422.400,-125981
 176.074511,-994.295,10,156,195.000,00177623,41,C5P,0,5293,23973452.240,-9
 4076969.913443,-742.440,9,138,195.000,00177633,44,C2I,0,5437,23942931.778
 ,-124677124.432611,-958.767,8,133,194.000,00187633,44,C6I,0,5055,23942932
 .165,-101310381.830223,-779.236,9,145,194.000,00187623,44,C1P,0,5079,2394
 2920.290,-125820862.010267,-967.629,9,145,195.000,00187633,44,C5P,0,4567,
 23942939.583,-93957245.073338,-722.494,10,161,195.000,00187613,4,C2I,0,44
 74,38126753.043,-198536004.211518,-31.616,10,197,190.000,001975d3,4,C6I,0
 ,4124,38126747.176,-161326741.932072,-26.023,12,336,195.000,00197573,4,C7
 I,0,4575,38126751.076,-153520620.711440,-24.519,10,257,197.011,00197593,5
 ,C6I,0,4086,39774553.439,-168299151.547481,-39.329,12,303,174.000,001a759
 3,5,C7I,0,4460,39774555.019,-160155650.826838,-37.534,10,215,194.000,001a
 75c3,43,C2I,0,4417,25317798.698,-131836434.147048,3195.296,10,168,194.000
 ,001b7603,43,C6I,0,4553,25317807.676,-107127944.251351,2596.667,10,162,19
 4.000,001b7623,43,C1P,0,3798,25317786.618,-133045900.163585,3225.020,13,3
 11,123.000,001b75a3,43,C5P,0,4544,25317820.327,-99352582.754998,2408.178,
 10,162,194.000,001b7623,14,C7I,0,4739,25955233.117,-104510945.931553,-241
 0.435,10,156,188.000,001e7623,196,J1C,0,4370,36401794.862,-191292718.2565
 83,-133.976,11,171,202.011,00017603,196,J2L,0,5050,36401804.055,-14905929
 6.306105,-104.274,9,146,200.011,00017623,196,J5Q,0,5384,36401816.557,-142
 848544.550741,-99.863,8,135,202.011,00017633,199,J1C,0,4018,36821342.126,
 -193497458.843260,-3.830,12,232,155.000,000275d3,199,J2L,0,5432,36821350.
 021,-150777273.387163,-3.094,8,133,184.000,00027633,199,J5Q,0,5468,368213
 59.152,-144494921.957563,-2.872,8,132,184.000,00027633,195,J1C,0,4157,375

08404.274, -197107998.035026, 272.366, 11, 203, 196.011, 000375e3, 195, J2L, 0, 489
 2, 37508416.869, -153590701.857981, 212.135, 9, 151, 196.011, 00037623, 195, J5Q, 0
 , 4883, 37508427.336, -147191125.742746, 203.192, 9, 151, 196.011, 00037623, 194, J
 1C, 0, 5083, 38800782.898, -203899493.487894, -586.367, 9, 145, 196.011, 00047633,
 194, J2L, 0, 5649, 38800790.285, -158882749.594456, -456.932, 8, 126, 196.011, 0004
 7633, 194, J5Q, 0, 5559, 38800799.277, -152262670.296191, -437.908, 8, 129, 196.011
 , 00047633, 200, J1C, 0, 4492, 37652793.235, -197866760.307355, -1.305, 10, 163, 196
 .011, 00057613, 200, J5Q, 0, 5295, 37652811.744, -147757719.995515, -0.826, 9, 138,
 196.011, 00057633*cf0240f1

表3-123 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	GNSSOBS	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	# obs	对应的观测信息个数。 Number of observations with information to follow	Ulong	4	H
3	PRN/slot	卫星 PRN 号 Satellite PRN number of range measurement	Ushort	2	H+4
4	satellite system	卫星系统 为了减少 ASCII 报文时的编码长度，减少通讯带宽占用，将“satellite system”/“signal ID”/“glo freq”三个字段在映射成 ASCII 时合并成一个字符串，卫星系统名占用一个字符，信号名占用两个字符，GLONASS 频点如果有则加在后面，比	Uchar	1	H+6

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		如 GPS L1C/A, 对应字符串“G1C”, GLONSS L1C/A slot -3, 对应字符串“R1C-3”			
5	signal ID	Signal ID 频点, Follow RTCM 的对每个卫星系统的 ID 定义, 对应 ASCII follow RINEX 的定义, 固定两个字符表示	Uchar	1	H+7
6	glo freq	GLONASS 频率(-7~+6)	Char	1	H+8
7	reserved	保留	Uchar	1	H+9
8	C/No	载噪比。 Carrier to noise density ratio C/No = 10[log10(S/N0)] (dB-Hz) C/No * 128	Ushort	2	H+10
9	psr	码伪距观测量, 米。 Pseudorange measurement (m)	Double	8	H+12
10	adr	载波相位, 周。 Carrier phase, in cycles (accumulated Doppler range)	Double	8	H+20
11	dopp	瞬时多普勒, Hz Instantaneous carrier Doppler frequency (Hz)	Float	4	H+28
12	psr σ	码伪距标准差, 米。 Pseudorange measurement standard deviation (m) psr std * 128	Ushort	2	H+32
13	adr σ	载波相位标准差, 周。 Estimated carrier phase standard deviation	Ushort	2	H+34

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		(cycles) adt std * 8192			
14	locktime	连续跟踪时间，无周跳。 Number of seconds of continuous tracking (no cycle slipping)	Float	4	H+36
15	ch-tr-status	通道跟踪状态 Tracking status	Ulong	4	H+40
16...	下一个 OBS offset = H + 4 + (#obs x 40) Next PRN offset = H + 4 + (#obs x 40)				
variable	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+4+(#obs x 40)
variable	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-124 PRN Numbers

Command/Log	GPS PRN	SBAS PRN	SBAS QZSS L1S PRN	GLONASS Slot	Galileo SVID	QZSS PRN	BDS PRN	NavIC PRN
GNSOBS	1-32	120-158	183-192	38-61	1-36	193-202	1-63	1-14
GNSOBSOMP	1-32	120-158	183-192	38-61	1-36	193-202	1-63	1-14
SATINVIEW	1-32	120-158	183-191	1-24	1-36	193-202	1-63	1-14
BESTDOPS	G1-G32	S120-S158	S183-S191	R1-R24	E1-E36	J193-J202	C1-C63	I1-I14
UVNAV	G1-G32	S120-S158	S183-S191	R1-R24	E1-E36	J193-J202	C1-C63	I1-I14

表3-125 Satellite system

Satellite system	System ID	ASCII name	Comments
GPS	0	'G'	
GLONASS	1	'R'	

Satellite system	System ID	ASCII name	Comments
GALLIEO	2	'E'	
SBAS	3	'S'	
QZSS	4	'J'	
BDS	5	'C'	
NavIC	6	'I'	
Other	7	'L'	Lband, not defined in RINEX

表3-126 Channel Tracking Status

Nibble	Bit	Mask	Description	Range Value
N0	0	0x00000001	基带跟踪状态 BB Tracking state	0-23, see Tracking State
	1	0x00000002		
	2	0x00000004		
	3	0x00000008	保留 Reserved	
N1	4	0x00000010	PLD	
	5	0x00000020		
	6	0x00000040		
	7	0x00000080		
N2	8	0x00000100	保留 Reserved	
	9	0x00000200		
	10	0x00000400		
	11	0x00000800	保留 Reserved	
	12	0x00001000	伪距平滑标志 PSR Smooth Flag	0 = 未平滑, 1 = 平滑 0 = Not smoothed, 1 = smoothed
	13	0x00002000	载波有效标志 ADR Valid Flag	0 = 无效, 1 = 有效 0 = invalid, 1 = valid

	14	0x00004000	载波相位偏差版本号 PhaseShiftVer	0 = 版本0，1 = 版本1 0 = version0, 1 = version1 always set to 1 since 202411
	15	0x00008000	保留 Reserved	
N4	16	0x00010000	频道号 SV channel number	1 ~ n
	17	0x00020000		
	18	0x00040000		
	19	0x00080000		
N5	20	0x00100000		
	21	0x00200000		
	22	0x00400000		
	23	0x00800000		
N6	24	0x01000000	保留 Reserved	
	25	0x02000000		
	26	0x04000000		
	27	0x08000000		
N7	28	0x10000000		
	29	0x20000000		
	30	0x40000000		
	31	0x80000000		

表3-127 Tracking State

State	Description
0	0x20
1	0x21，环路快速收敛
2	0x22，高动态跟踪
3	0x23，FINE TRACK

State	Description
4	0x24, 高灵敏度 PLL 跟踪
5	0x25, 高灵敏度 FLL 跟踪
6	0x26
7	0x27

表3-128 SignalID Mapping of RTCM and Range

下表为 CHC 扩展的 RTCM SignalID 与 扩展的 Range SignalType 之间的转换关系:

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
GPS	1					Reserved
	2	L1	C/A	1C	0 - L1C/A	
	3	L1	P	1P	4 - L1P	
	4	L1	Z-tracking or similar	1W	8 - L1W	
	5-7					Reserved
	8	L2	C/A	2C	1 - L2C/A	
	9	L2	P	2P	5 - L2P	
	10	L2	Z-tracking or similar	2W	9 - L2P(Y),semi-code less	
	11-14					Reserved
	15	L2	L2C(M)	2S	17 - L2C(M)	
	16	L2	L2C(L)	2L	13 - L2C(L)	
	17	L2	L2C(M+L)	2X	21 - L2CX	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
	18-21					Reserved
	22	L5	I	5I	6 - L5I	
	23	L5	Q	5Q	14 - L5Q	
	24	L5	I+Q	5X	10 - L5X	
	25-29					Reserved
	30	L1	L1C-D	1S	12 - L1CD	
	31	L1	L1C-P	1L	16 - L1CP	
	32	L1	L1C-(D+P)	1X	20 - L1CX	
GLONASS	1					Reserved
	2	G1	C/A	1C	0 - L1C/A	
	3	G1	P	1P	4 - L1P	
	4-7					Reserved
	8	G2	C/A	2C	1 - L2C/A	
	9	G2	P	2P	5 - L2P	
	10-14					Reserved
	15	G2a	D	6A	13 - L2SCI	
	16	G2a	P	6B	17 - L2OCP	
	17	G2a	D+P	6X	21 - L2OCX	
	18-21					Reserved
	22	G3	I	3I	2 - L3I	
	23	G3	Q	3Q	6 - L3Q	
	24	G3	I+Q	3X	10 - L3X	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
	25-29					Reserved
	30	G1a	D	4A	12 - L1OCD	
	31	G1a	P	4B	16 - L1OCP	
	32	G1a	D+P	4X	20 - L1OCX	
Galileo	1					Reserved
	2	E1	C no data	1C	2 - E1C	
	3	E1	A	1A	0 - E1A	
	4	E1	B I/NAV OS/CS/ SoL	1B	1 - E1B	
	5	E1	B+C	1X	3 - E1X	
	6	E1	A+B+C	1Z	4 - E1Z	
	7					Reserved
	8	E6	C	6C	7 - E6C	
	9	E6	A	6A	5 - E6A	
	10	E6	B	6B	6 - E6B	
	11	E6	B+C	6X	8 - E6X	
	12	E6	A+B+C	6Z	9 - E6Z	
	13					Reserved
	14	E5B	I	7I	16 - E5bI	
	15	E5B	Q	7Q	17 - E5bQ	
	16	E5B	I+Q	7X	18 - E5bX	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
	17					Reserved
	18	E5(A+B)	I	8I	19 - E5AltBOC-I	
	19	E5(A+B)	Q	8Q	20 - E5AltBOC-Q	
	20	E5(A+B)	I+Q	8X	21 - E5AltBOC-X	
	21					Reserved
	22	E5A	I	5I	11 - E5aI	
	23	E5A	Q	5Q	12 - E5aQ	
	24	E5A	I+Q	5X	13 - E5aX	
	25-32					Reserved
SBAS	1					Reserved
	2	L1	C/A	1C	0 - L1C/A	
	3-21					Reserved
	22	L5	I	5I	6 - L5I	
	23	L5	Q	5Q	14 - L5Q	
	24	L5	I+Q	5X	10 - L5X	
	25-32					Reserved
QZSS	1					Reserved
	2	L1	C/A	1C	0 - L1C/A	
	3-8					Reserved
	9	L6	D	6S	26 - L6D	
	10	L6	P	6L	27 - L6P	
	11	L6	D+P	6X	28 - L6X	
	12	L6	E	6E	29 - L6E	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
	13	L6	D+E	6Z	30 - L6Z	
	14					Reserved
	15	L2	L2C(M)	2S	17 - L2CM	
	16	L2	L2C(L)	2L	13 - L2CL	
	17	L2	L2C(M+L)	2X	21 - L2CX	
	18-21					Reserved
	22	L5	I	5I	6 - L5I	
	23	L5	Q	5Q	14 - L5Q	
	24	L5	I+Q	5X	10 - L5X	
	25-29					Reserved
	30	L1	L1C(D)	1S	12 - L1CD	
	31	L1	L1C(P)	1L	16 - L1CP	
	32	L1	L1C(D+P)	1X	20 - L1CX	
BDS	1					Reserved
	2	B1	I	2I	0 or 4 B1I	
	3	B1	Q	2Q	20 - B1Q	
	4	B1	I+Q	2X	B1X ¹⁾	
	5-7					Reserved
	8	B3	I	6I	2 or 6 B3I	
	9	B3	Q	6Q	22 - B3Q	
	10	B3	I+Q	6X	B3X ²⁾	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
	11-13					Reserved
	14	B2	I	7I	1 or 5 B2I	
	15	B2	Q	7Q	21 - B2Q	
	16	B2	I+Q	7X	B2X ³⁾	
	17-21					Reserved
	22	B2a	D	5D	12 - B2aD	
	23	B2a	P	5P	9 - B2aP	
	24	B2a	D+P	5X	15 - B2aX	
	25	B2b	I	7D	11 - B2bI	
	26	B2b	Q	7P	8 - B2bP	
	27	B2b	I+Q	7Z	14 - B2bX	
	28-29					Reserved
	30	B1C	D	1D	10 - B1CD	
	31	B1C	P	1P	7 - B1CP	
	32	B1C	D+P	1X	13 - B1CX	
NIC	1-21					Reserved
	22	L5	A	5A	0 - L5A	
	23-32					Reserved
Other		XW1L1	P	1P	0 - XW1L1L	
		XW1L1	D	1D	4 - XW1L1H	
		XW1L1	D+P	1X		
		XW1B2	P	2P	1 - XW1B2L	
		XW1B2	D	2D	5 - XW1B2H	

GNSS System	Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	RangeSignalType	Comments/Notes
		XW1B2	D+P	2X		
		L-Band			19 - L-Band	

3.3.7. GNSSOBSCMP_1 压缩格式的卫星观测值信息

适用产品：M720、M722

压缩格式的 GNSSOBS 报文。

双天线的数据，从天线的观测值的报文头为 GNSSOBSCMP_1, GNSSOBSCMPA_1, GNSSOBSCMPB_1; 命令请求分别为 GNSSOBSCMP_1, GNSSOBSCMPA_1, GNSSOBSCMPB_1

- Message ID: 3058
- Log Type: Sync

输出示例：

```
#GNSSOBSCMPA_1,42.0,0,0,COM2,FINESTEERING,2381,390791000,3d7cea15,0000800
0;185,2376010075B1FA2F21E0210A66FA22D8711F0020801E1000,337601005CDDFB0F54
E0210A853B50D0711FE01F801EF000,237601006ADDFB0F52E0210AC75550D0811F401F60
1E9000,13760200871C04B08DA6C70B50D214C39212C020A01D1000,237602003E340360D
8A6C70B6C5F6DF882126020001EF000,237602002412030039A7C70B53BABDB88212C0200
01E6001,0376030085ACF69F8C25CF0BA80F28BEA2038012601D1000,03760300D2BBF8AF
2A26CF0BE90297F4A203E01EA01DF000,137603004B09F95F9126CF0B6B6610B59203C01F
E01D6001,F3440300A1BBF8EF4C26CF0B0000000012030000201D9000,337604006E3900E
0C0629F099B3EDAAD81100020601E1000,837504009D2C0080E3629F0934FB1A933210801
F601D9000,23760500BF6B0A90C267F60936F7B0F4811BC020601E1000,33760500E01E08
300F68F6094A9E90E6711BC01E801EF000,3376050034C807504D68F60926E2F49C611BC0
20001F6001,23760500D41E08400E68F6098B6090E6811BC01F601E9000,23760600F316F
7EFEFA00B0B4EBC96BE821CC020201E1000,237606008D0EF90F3CA10B0B968AAAD8811C6
020401EF000,337606009558F92FA1A10B0B6597F894711CC020A01E6001,337607005158
F96F0F81150AAB5943E0711AC020801E1000,33760700D4D0FA1F5A81150A8D5CA5D6711A
```

6020A01EF000, 33760700FB07FB8F8D81150Aafb3B38D611AC020001F6001, 337607006FD0FAEF5A81150AC75CA5D6811AC01F601E9000, 33760800573005906B59240AEBFE82D68104C020401E1000, 23760800200B04F0A359240A00090CCF81046020601EF000, 33760800E4DF03E0075A240AED766B866104C020001F6001, 03450800E10A04F0A159240A00000000F2040000201D9000, 13760A00F6A10E0042B5FF0A43406BC69208401FA01D1000, 23760A00D3660BA0C1B5FF0AFE7BC4DE8208201F401EF000, 23760A0036ED0A60FCB5FF0AFF65D19A8208401F201E6001, F3750A00FB660BF0BDB5FF0A445AC4DEA208401E801D9000, 23760B005D9C0BA0B8A8510BC75B96909209801EC01D1000, 03760B005D0C09E008A9510BEE0CD2B49209201EA01DF000, 23760B0098AB081050A9510BDA859EF28209801E201E6001, 83750B00330C095007A9510B341FD2B43209C01D601D9000, 137601002DC00BC072E9FB0A92B473A6922C8020E0311100, 9075020092F907109616FB0A57429EAB522E4002E0141100, C3750200AC3306601717FB0A3A5B97CCF22E802020157100, 137604000E960B10313A3D0A41D1F6AA92378008C0111100, 03760400F30209308E3A3D0A8151A3AFA237802080117100, 237605002BB7FF9FA3284E0A78CD559B71360020802E1100, 13760600EFD000401912C509EF60F7F982358020001A1100, 0376060042A200F05112C509B89F15ED92358020C0197100, 137608003A0EF9FF3317630AFE22D48F92348020C01D1100, F37508007F99FAFF6917630ACC72889AB2348020601D7100, 03760900614AF07F94775B0A52814C94822A802000221100, C3750900DEC7F32FBB775B0AEC87029E022AC00320217100, D3750A002F9A0FF0E04A7D0B793542D1E238C00020291100, 70450A0093220C201D4B7D0B00000000A3380000A0287100, 23760100E94006601013020CFA2AB49C8209A020201E1200, 3376010076AB04D0A613020C9F31159C71098020A01E6201, 33760100ABCA04406413020C070DBFF48109A020601EE200, 23760200A5230A10A28AE50B5F4C72AF810EA020601E1200, 337602003B920790288BE50BE33F14AA710E8020A01E6201, 23760200CAC40790E38AE50BD8941B83810EA020601EE200, 33760300D195FA7F94C68A0B8F9311EB8117C01F601E1200, 33760300B5F4FB8F27C78A0B7B229AD67117401FC01E6201, 33760300A4D9FBBFE1C68A0B89D0CAB07117C01F601EE200, 23760400351E0210B572710B9FB2B4FB811FC01F601E1200, 337604000B9501B04973710B65A306E3711F401FC01E6201, 33760400659F01500A73710BAF4A8ABD811FC01F601EE200, 237605006ADBFBEFA0D9410B61D6F89A8104C01F401E1200, 33760500ECE7FC7F2ADA410BEEC65FFA6104401FE01E6201, 3376050035D3FC4FE7D9410B075C7FD57104C01F801EE200, 337606004C6F0180FAC9510B038780907106C01F801E1200, 337606003312011087CA5

10BC73D8EF27106401FC01E6201, 337606006619010049CA510B0A9679CD7106C01F801EE
200, C37507004267F42F54EA6B0C82D62DD702134014201D1200, 037607003457F71F0FEB
6B0C71332AE89213C01CC01D6201, F3750700201DF7BFC1EA6B0CDD6A79BFA213E01C601D
E200, 03760800E82607106861360DCD162FD2A21A001D801D1200, 137608005F5705E00E6
2360DC0AFD984921AC01CC01D6201, 03760800317B0510C861360DE19591D9921A001DC01
DE200, E3750900186FF7DF04DB790D8E62DCA5B215C018801D1200, F37509009F9AF90FA2
DB790DDD82C0E3A215C01C801D6201, 03760900246FF93F71DB790DD4629BB7A215001DA0
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8A22E0D496BA6889218001CE01D6201, 03760A00BFD005B09CA22E0D0EB877DD9218601CC
01DE200, 93750B004F15098054BA750D7C7692A842212013201D1200, 13760B00C6C806B0
58BB750DD29FC6E59221601BE01D6201, 13760B0079F506F005BB750D5509AFB99221601B
A01DE200, 13760100B7BFFFFFF514ABC11828BB0F4713BE020A01E1500, 23760100E5CBFFA
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20C01D1500, 13760300E8BE02302FA93112B3367DE0920A8020E01D7500, 23760300C19C0
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01F0DF3C140BA4A674EC710CC01FA01ED500, 33760500F7FE0190AFFD7311EEF4BFA37110
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7F0730F4B9130C103492A38217C01F201E1500, 2376070090170660CCB9130C80ACBC9C81
17C01F401E7500, 03760700A8900730A0B9130C31001C919217E01EA01DE501, 137607003
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7118E01FC01E6501, F375090069ABFFEF7F762E1230B35FAA9202801CC01D1500, D375090
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B18103E020401ED500, 33760B007FE60150FAF631113C14BACE8106C01F401E1500, 23760
B004E8B0130D5F63111E92ABBE78106C01F601E7500, 23760B0015780160DEF6311117A46
3D88206C01F201ED500, 23760C00FD6E0370BFC7B8118B72F9F682078020201E1500, 2376
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60D0028E10140D712C9110E92CE978209C01F001E7500, 33760D00D0C901F0FA12C911A00
9558C7109C01F801ED500, 23760E0078FD0AE045C77B0B4EAC7986810B8020401E1500, 33
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4E57B8710B8020A01ED500, 33760F00879302B05E21830A7D8052A87119C01FE01E1500, 3
3760F00C01702405221830A9E519EF07119C01FC01E7500, 23760F00859902200A21830A9
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, 23761000275B05603FECDD0AE66264AB8122E01F601E6501, E375110054BBF87FE9E57A1
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01, 23761200938B0400E314850CAF46D2DB8120601D401E6501, 3376130055DF0220792CA
B11DCBCD4FF8138C01F601E1500, 23761300555502408D2CAB11F20B9FA78238C01F201E7
500, 23761400D5A6F42FA372D20BFEC10FCE8121C01F401E1500, 237614005BC7F65F2273
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842F116E36EB917127C01FA01E6501, 33761600004F04700537A511D5A5B5837128001F80
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01E6501, 33761700540AFC2FB3D96F0B48443D8E7129C01F801E1500, 3376170039C8FC6F
ECD96F0B11B169F37129C01F801E7500, 237617000A01FCBF66D96F0B5B95C1FC8229E01F

```

001EE501,337617001D04FDEF56DA6F0BB444CFE37129E01FA01E6501,33761800FB22FCA
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F7F5C302E12455770D9B2042020E01DD500,83751A0017D0FF1F215CF71201C49BA722056
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BC01FE01D6501,13761E00699AF6BF6ABE640CDBB815C3820E001F201ED500,2376010096
76FF1F8AC45B11F7FEFA9892C4C020C01D1400,23760100E594FF4FD3C45B119053709D71
C48020801EF400,337601007E99FF5F35C55B11D51B36FC71C4C020A01E6401,F3750200C
4FCFFDF53CC8E1185A375F7B2C7E01A601D1400,3376020039FDFF5F93CC8E11749051837
1C7801EC01EF400,3376020081FDFF7FDCCC8E117A072EE371C7801EC01E6401,F3750300
251101F07448E211ACB59EC0A2C30020601D1400,23760300C0D40000D748E211190E96D8
82C30020201EF400,2376030065CB00C02A49E2119F533ABA81C30020401E6401,3376040
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761C20020001FF400,337604001C4CFE6FC93D8112C68341EC71C20020E01E6401,137605
0012FFFF1F354AF41139B1CAB492C80020E01D1400,3376050093FFFFAFC84AF411633365
B171C80020A01E6401*9596587e

```

表3-129 报文格式说明

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
1	GNSSOBSCMP	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	#obs	对应的观测信息个数。 Number of satellite observations with information to follow	Ulong	4	H

Field	Field Type	Description	Format	Binary Bytes	Binary Offset
3	1st gnssobs record	GNSSOBS 压缩信息。 Compressed gnssobs log in format of Table GNSSOBS Record Format (GNSSOBSCMP only)	Hex	24	H+4
4	Next gnssobscomp offset = H+4 (#obs x 24)				
5	xxxx	32 位 CRC 校验。 32-bit CRC (ASCII and Binary only)	Ulong	4	H+4+(#obs x 24)
6	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

表3-130 GNSSOBS Record Format (GNSSOBSCMP only)

Data	Description	Bits first to last	Length(bits)	Scale Factor	Units
Channel Tracking Status	Channel tracking status word	0-31	32	see Table Channel Tracking Status	-
Doppler Frequency	Instantaneous carrier Doppler frequency	32-59	28	1/256	Hz
Pseudorange (PSR)	Pseudorange measurement	60-95	36	1/128	m
ADR	ADR (Accumulated Doppler Range) is calculated as follows: ADR_ROLLS = (GNSSOBSCMP_PSR / WAVELENGTH + GNSSOBSCMP_ADR) / MAX_VALUE Round to the closest integer	96-127	32	1/256	cycles

Data	Description	Bits first to last	Length(bits)	Scale Factor	Units
	IF ($\text{ADR_ROLLS} \leq 0$) ADR_ROLLS = $\text{ADR_ROLLS} - 0.5$ ELSE ADR_ROLLS = $\text{ADR_ROLLS} + 0.5$ At this point integerise ADR_ROLLS CORRECTED_ADR = GNSOBSCMP_ADR - (MAX_VALUE*ADR_ROLLS) where ADR has units of cycles WAVELENGTH = 0.1902936727984 for GPS L1 WAVELENGTH = 0.2442102134246 for GPS L2 MAX_VALUE = 8388608 GLONASS satellites emit L1 and L2 carrier waves at a satellite-specific frequency				
StdDev-PSR	Pseudorange measurement standard deviation	128-131	4	See Table StdDev-PSR Values	m
StdDev-ADR	ADR measurement standard deviation	132-135	4	$(n+1)/512$	cycles
PRN/Slot	Refer to Table PRN Numbers	136-143	8	1	-
Lock Time	Number of seconds of continuous tracking (no cycle slipping) This field is constrained to a	144-164	21	1/32	s

Data	Description	Bits first to last	Length(bits)	Scale Factor	Units
	maximum value of 2,097,151 which represents a lock time of 65535.96875 s (2097151/32).				
C/No	Carrier to noise density ratio The C/No is constrained to a value between 20-51 dB-Hz. Thus, if it is reported that C/No = 20 dB-Hz, the actual value could be less. Likewise, if it is reported that C/No = 51, the true value could be greater.	165-169	5	(20+n)	dB-Hz
GLONASS Frequency number	GLONASS Frequency number	170-175	6	n+7	1
Satellite system	Satellite system, see Satellite system	176-179	4	1	1
Signal type	Signal type, see GNSSOBS Signal Type	180-185	6	n-1	1
Reserved		186-191	6		

表3-131 StdDev-PSR Values

Code	StdDev-PSR(m)
0	0.050
1	0.075
2	0.113
3	0.169
4	0.253
5	0.380

Code	StdDev-PSR(m)
6	0.570
7	0.854
8	1.281
9	2.375
10	4.750
11	9.500
12	19.000
13	38.000
14	76.000
15	152.000

3.3.8. PPPP_1 PPP 定位的位置信息

适用产品：M720、M722

PPPP 输出 PPP position 定位的位置信息。

本指令包含接收机 PPP 定位的位置及定位精度、状态等信息。

- **Message ID: 3026**
- **Log Type: Synch**

输出示例：

```
#PPPPA_1,41.0,0,0,COM2,FINESTEERING,2381,390866000,3d7cea15,00008000;INSU
FFICIENT_OBS,SINGLE,WGS84,0,31.15967845825,121.17808628794,31.1559,10.021
1,0.9226,1.0991,3.0758,0.000,0.000,62,56,15,17,4928,37,00,00,00,400*a683e
b6d
```

表3-132 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	PPPP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
2	sol stat	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型, 见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度, 单位: 度。 Latitude (degrees)	Double	8	H+4
7	lon	经度, 单位: 度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the WGS84 ellipsoid (m) When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差, 单位: 米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差, 单位: 米。 Height standard deviation (m)	Float	4	H+40

Field	Field type	Description	Format	Binary Bytes	Binary Offset
13	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间，单位：秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61
24	stn id	基准站 ID Base station ID	Ushort	2	H+62

Field	Field type	Description	Format	Binary Bytes	Binary Offset
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.3.9. RTKP_1 RTK 位置信息

适用产品：M720、M722

RTKP 输出 RTK 位置信息。

本指令包含接收机 RTK 定位的位置及定位精度、状态等信息。

- **Message ID: 3024**
- **Log Type: Synch**

输出示例：

```
#RTKPA_1,42.0,0,0,COM2,FINESTEERING,2381,390953000,3d7cea15,00008000;SOL_
COMPUTED,NARROW_INT,WGS84,0,31.15969337754,121.17808140525,36.0891,10.021
1,0.0022,0.0023,0.0060,1.000,0.000,62,54,46,48,4728,37,17,07,00,654*ff9e6
7c7
```

表3-133 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	RTKP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态，见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型，见 Position Type Position type, see Position Type	Enum	1	H+1

Field	Field type	Description	Format	Binary Bytes	Binary Offset
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度，单位：度。 Latitude (degrees)	Double	8	H+4
7	lon	经度，单位：度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高，单位：米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值，单位：米。 Undulation - the relationship between the geoid and the WGS84 ellipsoid (m) When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.	Float	4	H+28
10	lat σ	纬度标准差，单位：米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差，单位：米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差，单位：米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间，单位：秒。 Solution age in seconds	Float	4	H+48

Field	Field type	Description	Format	Binary Bytes	Binary Offset
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	#solnMultiSVs	参与解算的多频信号卫星数。 Number of satellites with multi-frequency signals used in solution	Uchar	1	H+54
18	#solnL1SVs	参与解算的 L1/E1/B1 卫星数。 Number of satellites with L1/E1/B1 signals used in solution	Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。	-	-	-

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		Sentence terminator (ASCII only)			

3.3.10. RTKV_1 RTK 速度信息

适用产品：M720、M722

RTKV 输出 RTK 速度。

本指令包含接收机计算出的 RTK 速度信息。此外，接收机还报告了速度状态指示符，对指示相应的数据是否有效非常有用。该速度测量有时会带来相关的延迟。

- **Message ID: 3025**
- **Log Type: Synch**

输出示例：

```
#RTKVA_1,39.0,0,0,COM2,FINESTEERING,2381,390987000,3d7cea15,00008000;SOL_
COMPUTED,NARROW_INT,0,0.0071,324.538826,-0.0088,0.000,2.000,0.000*a824725
3
```

表3-134 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	RTKV header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态，见 Vel Solution Status Solution status, see Vel Solution Status	Enum	1	H+0
3	vel type	速度类型，见 Velocity Type Velocity type, see Velocity Type	Enum	1	H+1
4	reserved		Ushort	2	H+2
5	hor spd	对地水平速度，单位：米/秒。 Horizontal speed over ground, in metres per second	Double	8	H+4

Field	Field type	Description	Format	Binary Bytes	Binary Offset
6	trk gnd	相对于真北的实际对地运动方向（相对地面轨迹），单位：度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Double	8	H+12
7	vert spd	垂直速度，正值表示高度增加（向上），负值表示高度下降（向下），单位：米/秒。 Vertical speed, in metres per second, where positive values indicate increasing altitude (up) and negative values indicate decreasing altitude (down)	Double	8	H+20
8	latency	根据速度时标计算的延迟值，用历元时间减去延迟可得到更准确的速度结果，单位：秒。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results (s)	Float	4	H+28
9	age	差分龄期，单位：秒。 Differential age in seconds	Float	4	H+32
10	Reserved		Float	4	H+36
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
11	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.3.11. SPPP_1 伪距位置信息

适用产品：M720、M722

SPPP 输出伪距位置信息。

本指令包含接收机伪距定位的位置及定位精度、状态等信息。

- Message ID: 3022
- Log Type: Synch

输出示例:

```
#SPPPA_1,39.5,0,0,COM2,FINESTEERING,2381,391029000,3d7cea15,00008000;SOL_
COMPUTED,SINGLE,WGS84,0,31.15968983320,121.17809298747,35.5882,10.0211,1.
6530,1.9009,6.4674,0.000,0.000,61,54,0,0,4628,37,17,07,00,0*2ab816a1
```

表3-135 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	SPPP header	报文头。 Log header. For information about the log headers, see ASCII, or Binary.		H	0
2	sol stat	解算状态, 见 Position Solution Status Solution status, see Position Solution Status	Enum	1	H+0
3	pos type	定位类型, 见 Position Type Position type, see Position Type	Enum	1	H+1
4	datum id#	坐标系 ID Datum ID number 0 = WGS84 1 = USER	Enum	1	H+2
5	Reserved		Uchar	1	H+3
6	lat	纬度, 单位: 度。 Latitude (degrees)	Double	8	H+4
7	lon	经度, 单位: 度。 Longitude (degrees)	Double	8	H+12
8	hgt	海拔高, 单位: 米。 Height above mean sea level (metres)	Double	8	H+20
9	undulation	高程异常值, 单位: 米。 Undulation - the relationship between the geoid and the WGS84 ellipsoid (m)	Float	4	H+28

Field	Field type	Description	Format	Binary Bytes	Binary Offset
		When using a datum other than WGS84, the undulation value also includes the vertical shift due to differences between the datum in use and WGS84.			
10	lat σ	纬度标准差, 单位: 米。 Latitude standard deviation (m)	Float	4	H+32
11	lon σ	经度标准差, 单位: 米。 Longitude standard deviation (m)	Float	4	H+36
12	hgt σ	高度标准差, 单位: 米。 Height standard deviation (m)	Float	4	H+40
13	diff_age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+44
14	sol_age	解算时间, 单位: 秒。 Solution age in seconds	Float	4	H+48
15	#SVs	跟踪到的卫星数。 Number of satellites tracked	Uchar	1	H+52
16	#solnSVs	参与解算的卫星数。 Number of satellites used in solution	Uchar	1	H+53
17	Reserved		Uchar	1	H+54
18	Reserved		Uchar	1	H+55
19	ext sol stat	扩展解算状态 Extended solution status	Hex Ushort	2	H+56
20	GPS and GLONASS sig mask	GPS 和 GLONASS 信号标志 GPS and GLONASS signals used mask	Hex	1	H+58
21	BeiDou sig mask	BeiDou 信号标志 BeiDou signals used mask	Hex	1	H+59

Field	Field type	Description	Format	Binary Bytes	Binary Offset
22	Galileo sig mask	Galileo 信号标志 Galileo signals used mask	Hex	1	H+60
23	Reserved	保留 Reserved	Hex	1	H+61
24	stn id	基准站 ID Base station ID	Ushort	2	H+62
25	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+64
26	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

3.3.12. SPPV_1 伪距速度信息

适用产品：M720、M722

SPPV 输出伪距速度。

本指令包含接收机计算出的伪距速度信息。此外，接收机还报告了速度状态指示符，对指示相应的数据是否有效非常有用。该速度测量有时会带来相关的延迟。

- **Message ID: 3023**
- **Log Type: Synch**

输出示例：

```
#SPPVA_1,37.0,0,0,COM2,FINESTEERING,2381,391065000,3d7cea15,00008000;SOL_
COMPUTED,DOPPLER_VELOCITY,0,0.0028,115.634324,0.0096,0.000,0.000,0.000*4b
f82ed5
```

表3-136 报文格式说明

Field	Field type	Description	Format	Binary Bytes	Binary Offset
1	SPPV	报文头。		H	0

Field	Field type	Description	Format	Binary Bytes	Binary Offset
	header	Log header. For information about the log headers, see ASCII, or Binary.			
2	sol stat	解算状态, 见 Vel Solution Status Solution status, see Vel Solution Status	Enum	1	H+0
3	vel type	速度类型, 见 Velocity Type Velocity type, see Velocity Type	Enum	1	H+1
4	reserved		Ushort	2	H+2
5	hor spd	对地水平速度, 单位: 米/秒。 Horizontal speed over ground, in metres per second	Double	8	H+4
6	trk gnd	相对于真北的实际对地运动方向 (相对地面轨迹), 单位: 度。 Actual direction of motion over ground (track over ground) with respect to True North, in degrees	Double	8	H+12
7	vert spd	垂直速度, 正值表示高度增加 (向上), 负值表示高度下降 (向下), 单位: 米/秒。 Vertical speed, in metres per second, where positive values indicate increasing altitude (up) and negative values indicate decreasing altitude (down)	Double	8	H+20
8	latency	根据速度时标计算的延迟值, 用历元时间减去延迟可得到更准确的速度结果, 单位: 秒。 A measure of the latency in the velocity time tag in seconds. It should be subtracted from the time to give improved results (s)	Float	4	H+28
9	age	差分龄期, 单位: 秒。 Differential age in seconds	Float	4	H+32

Field	Field type	Description	Format	Binary Bytes	Binary Offset
10	Reserved		Float	4	H+36
11	xxxx	32-bit CRC 校验。 32-bit CRC (ASCII and Binary only)	Hex	4	H+40
12	[CR][LF]	语句结束符。 Sentence terminator (ASCII only)	-	-	-

4. 常用功能配置

4.1. RTCM 快捷输出配置指令

此指令可用于对 RTCM 数据进行快速配置。

指令格式：

```
>OUTMSG,RTCMSet*CS
```

指令输入示例：

```
>OUTMSG,MSM4,1*41
```

表4-1 RTCM 快捷输出配置指令参数说明

参数项	参数值	参数描述
RTCMSet	MSM4	RTCM1074/RTCM1084/RTCM1094/RTCM1114/RTCM1124/RTCM1134
	MSM5	RTCM1075/RTCM1085/RTCM1095/RTCM1115/RTCM1125/RTCM1135
	MSM7	RTCM1077/RTCM1087/RTCM1097/RTCM1117/RTCM1127/RTCM1137
	RTCMEPH	RTCM1019/RTCM1020/RTCM1041/RTCM1042/RTCM1044/RTCM1045/RTCM1046

4.2. 基准站模式

RTK 基准站（固定基站）即将接收机天线安装在固定位置，在整个使用过程天线不移动。同时将已知测站的精密坐标和接收到的卫星信息直接或经过处理后实时发送给流动站接收（待定位点），流动站接收机在接收卫星观测值的同时也接收到基准站信息，进行 RTK 定位解算实现 RTK 高精度定位，达到 cm 或者 mm 级定位精度。

在已知精密坐标时输入接收机中的指令示例如表固定坐标配置指令示例。

表4-2 固定坐标配置指令示例

指令示例	指令描述
>SETFIXPOS,ENABLE,31.159682963,121.178077581,47.824*74	手动输入固定坐标
>OUTMSG,COM1,RTCM1005,5*72	输出基站天线参考点坐标
>OUTMSG,COM1,RTCM1033,5*77	输出接收机和天线说明
>OUTMSG,COM1,MSM4,1*10	输出差分数据
>SAVECONFIG*0B	保存当前端口所有配置

自主优化设置基准站：即在将架设基准站的点没有精确坐标。自主优化设置基准站坐标时输入接收机中的指令如表自主优化设置坐标配置指令示例。

表4-3 自主优化设置坐标配置指令示例

指令示例	指令描述
>CONFIGAUTOBASE,ENABLE,60,0.1,0.1*1D	自动平滑坐标，并固定坐标

>OUTMSG,COM1,RTCM1005,5*72	输出基准站天线参考点坐标
>OUTMSG,COM1,RTCM1033,5*77	输出接收机和天线说明
>OUTMSG,COM1,MSM4,1*10	输出差分数据
>SAVECONFIG*0B	保存当前端口所有配置

4.3. 移动站模式

RTK 流动站（移动站）是实时接收基准站的差分改正数信息，同时接收卫星信号进行 RTK 定位解算，实现 RTK 高精度定位。接收机可自适应识别 RTCM 数据输入的端口和格式。

表4-4 流动站的常用指令示例

指令示例	指令描述
>OUTMSG,COM1,GGA,1*3B	输出 GGA 数据
>OUTMSG,COM1,MSM4,1*10	输出原始观测值数据
>OUTMSG,COM1,RTCM1005,5*72	输出原始观测值数据
>SAVECONFIG*0B	保存当前端口所有配置

5. 附录

5.1. AntennaIndex

表5-1 AntennaIndex

序号	参数名	说明
0	MASTER	主天线
1	SLAVE	辅天线

5.2. AntennaType

表5-2 AntennaType

序号	参数名	序号	参数名	序号	参数名
1	NONE	103	JAVTRIUMPH_2A+G	205	SPP135000.00
2	3S-02-TSADM	104	JAVTRIUMPH_2A+P	206	SPP571212238+GP
3	3S-02-TSATE	105	JAVTRIUMPH_3A	207	SPPSP85
4	AERAT1675_120	106	JAVTRIUMPH_LSA	208	SPPSP85UHF
5	AERAT1675_542E	107	JNSCR_C146-22-1	209	STHCR3-G3
6	AERAT2775_43	108	JNSMARANT_GGD	210	STXS10SX017A
7	AOAD/M_B	109	JPLD/M_R	211	STXS700A
8	AOAD/M_T	110	JPLD/M_RA_SOP	212	STXS8PX003A
9	AOAD/M_T_RFI_T	111	JPSLEGANT_E	213	STXS800
10	AOAD/M_TA_NGS	112	JPSODYSSEY_I	214	STXS800A
11	APSAPS-3	113	JPSREGANT_DD_E	215	STXS900A
12	ARFAS13DFS	114	JPSREGANT_DD_E1	216	STXS990A
13	ARFAS1FS	115	JPSREGANT_DD_E2	217	STXS9I
14	ASH700228A	116	JPSREGANT_SD_E	218	STXS9PX001A
15	ASH700228B	117	JPSREGANT_SD_E1	219	STXS9SA7224V3.0
16	ASH700228C	118	JPSREGANT_SD_E2	220	STXS900
17	ASH700228D	119	LEIAR10	221	STXSA1000
18	ASH700228E	120	LEIAR20	222	STXSA1200
19	ASH700699.L1	121	LEIAR25	223	STXSA1500
20	ASH700700.A	122	LEIAR25.R3	224	STXSA1800
21	ASH700700.B	123	LEIAR25.R4	225	TIAPENG2100B
22	ASH700700.C	124	LEIAS05	226	TIAPENG2100R
23	ASH700718A	125	LEIAS10	227	TIAPENG3100R1
24	ASH700718B	126	LEIAS11	228	TIAPENG3100R2

25	ASH700829.2	127	LEIAT202+GP	229	TIAPENG6J2
26	ASH700829.3	128	LEIAT202-GP	230	TIAPENG7N
27	ASH700829.A	129	LEIAT302+GP	231	TOP700779A
28	ASH700829.A1	130	LEIAT302-GP	232	TOP72110
29	ASH700936A_M	131	LEIAT303	233	TPSCR.G3
30	ASH700936B_M	132	LEIAT502	234	TPSCR.G5
31	ASH700936C_M	133	LEIAT503	235	TPSCR.G5C
32	ASH700936D_M	134	LEIAT504	236	TPSCR3_GGD
33	ASH700936E	135	LEIAT504GG	237	TPSCR4
34	ASH700936E_C	136	LEIATX1230	238	TPSG3_A1
35	ASH700936F_C	137	LEIATX1230+GNSS	239	TPSG5_A1
36	ASH701008.01B	138	LEIATX1230GG	240	TPSHIPER_GD
37	ASH701023.A	139	LEIAX1202	241	TPSHIPER_GGD
38	ASH701073.1	140	LEIAX1202GG	242	TPSHIPER_HR
39	ASH701073.3	141	LEIAX1203+GNSS	243	TPSHIPER_HR+PS
40	ASH701933A_M	142	LEICGA100	244	TPSHIPER_LITE
41	ASH701933B_M	143	LEICGA60	245	TPSHIPER_PLUS
42	ASH701933C_M	144	LEIFLX100	246	TPSHIPER_VR
43	ASH701941.1	145	LEIGG02PLUS	247	TPSLEGANT_G
44	ASH701941.2	146	LEIGG03	248	TPSLEGANT2
45	ASH701941.A	147	LEIGG04	249	TPSLEGANT3_UHF
46	ASH701941.B	148	LEIGG04PLUS	250	TPSODYSSEY_I
47	ASH701945B.99	149	LEIGS08	251	TPSPG_A1
48	ASH701945B_M	150	LEIGS08PLUS	252	TPSPG_A1+GP
49	ASH701945C_M	151	LEIGS09	253	TPSPN.A5
50	ASH701945D_M	152	LEIGS12	254	TRM105000.10
51	ASH701945E_M	153	LEIGS14	255	TRM115000.00
52	ASH701945G_M	154	LEIGS15	256	TRM115000.00+S
53	ASH701946.2	155	LEIGS15.R2	257	TRM115000.10
54	ASH701946.3	156	LEIGS16	258	TRM14177.00
55	ASH701975.01A	157	LEIGS18	259	TRM14532.00
56	ASH701975.01AGP	158	LEIICG60	260	TRM14532.10
57	CHAPS9017	159	LEIICG70	261	TRM159800.00
58	CHCC220GR	160	LEIMNA950GG	262	TRM159900.00

59	CHCC220GR2	161	LEISR299_INT	263	TRM22020.00+GP
60	CHCI80	162	LEISR399_INT	264	TRM22020.00-GP
61	CHCX91+S	163	LEISR399_INTA	265	TRM23903.00
62	CNTAT340	164	MAC4647942	266	TRM27947.00+GP
63	CNTAT350	165	MPL_WAAS_2224NW	267	TRM27947.00-GP
64	CNTAT500	166	MPL_WAAS_2225NW	268	TRM29659.00
65	CNTAT600	167	MPLL1/L2_SURV	269	TRM33429.00+GP
66	CNTT30	168	MVECR152GNSSA	270	TRM33429.00-GP
67	CNTT300	169	MVEGA152GNSSA	271	TRM33429.20+GP
68	CNTT300PLUS	170	NAVAN2004T	272	TRM39105.00
69	EML_REACH_RS2	171	NAVAN2008T	273	TRM41249.00
70	EML_REACH_RS2+	172	NAX3G+C	274	TRM41249USCG
71	EML_REACH_RX	173	NOV_WAAS_600	275	TRM4800
72	ESVUA92	174	NOV501	276	TRM55970.00
73	FOIA90	175	NOV501+CR	277	TRM55971.00
74	GINCYF90	176	NOV502	278	TRM57970.00
75	GING20M	177	NOV502+CR	279	TRM57971.00
76	GING30	178	NOV503+CR	280	TRM5800
77	GMXZENITH06	179	NOV531	281	TRM59800.00
78	GMXZENITH10	180	NOV531+CR	282	TRM59800.00C
79	GMXZENITH15	181	NOV533+CR	283	TRM59800.80
80	GMXZENITH16	182	NOV600	284	TRM59800.99
81	GMXZENITH20	183	NOV702	285	TRM59900.00
82	GMXZENITH25	184	NOV702GG	286	TRMR10
83	GMXZENITH25PRO	185	NOV703GGG.R2	287	TRMR10-2
84	GMXZENITH35	186	NOV750.R4	288	TRMR12
85	GMXZENITH40	187	NOV750.R5	289	TRMR12I
86	GMXZENITH60	188	NOV850	290	TRMR2
87	HEMS631	189	RNG80971.00	291	TRMR4-3
88	HGGCYH8372	190	SEN67157596+CR	292	TRMR6-4
89	HITAT45101CP	191	SEPALTUS_NR3	293	TRMR780
90	HXCCGX601A	192	SEPCHOKE_B3E6	294	TRMR8-4
91	HXCCGX611A	193	SEPCHOKE_MC	295	TRMR8_GNSS
92	IGAIG8	194	SEPPOLANT_X_MF	296	TRMR8_GNSS3

93	ITT3750323	195	SEPVC6150L	297	TRMR8S
94	JAV_GRANT-G3T	196	SJTTL111	298	TRMSPS985
95	JAV_GRANT-G3T+G	197	SLGAT45101CP	299	TRMSPS986
96	JAV_RINGANT_G3T	198	SOK_RADIAN_IS	300	TRSAX4E02
97	JAVGRANT_G5T+GP	199	SOK502	301	TWIVC6050
98	JAVRINGANT_DM	200	SOK600	302	TWIVC6150
99	JAVRINGANT_G5T	201	SOK702	303	TWIVP6000
100	JAVTRIUMPH_1M	202	SOKGCX3	304	TWIVP6050_CONE
101	JAVTRIUMPH_1MR	203	SOKGRX3	305	TWIVSP6037L
102	JAVTRIUMPH_2A	204	SOKSA500		
306	MASTER_ANT_USER1	说明: Userdefinedmainantennatype1			
307	SLAVE_ANT_USER1	说明: Userdefinedslaveantennatype1			

5.3. RadomeType

表5-3 RadomeType

序号	参数值	说明	序号	参数值	说明	序号	参数值	说明
1	0	NONE	14	13	JVDM	27	26	SCIS
2	1	ARFC	15	14	JVGR	28	27	SCIT
3	2	ARFS	16	15	JVSD	29	28	SLGZ
4	3	CHCD	17	16	LEIC	30	29	SNOW
5	4	CNTS	18	17	LEIM	31	30	SPKE
6	5	CONE	19	18	LEIS	32	31	STHC
7	6	DUTD	20	19	LEIT	33	32	STXG
8	7	HGGS	21	20	MMAC	34	33	STXR
9	8	HITZ	22	21	NEVE	35	34	STXS
10	9	HXCM	23	22	NOVC	36	35	TCWD
11	10	HXCS	24	23	NOVS	37	36	TPSD
12	11	JAVC	25	24	OLGA	38	37	TPSH
13	12	JAVD	26	25	PFAN			